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Intelligent Drug Sales Forecasting and Healthcare -Insights with Artificial Intelligence

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Declaration

I, Bope Ranasinghage Gihan Lakmal, hereby declare that the thesis entitled Intelligent Drug Sales Forecasting and Healthcare -Insights with Artificial Intelligence", submitted to the Faculty of Computing and Technology, University of Kelaniya, in fulfilment of the requirements for the award of the degree of **Bachelor of Science Honours in Computer Science**, is the result of my own independent work carried out under the guidance and supervision of **Prof N G J Dias**. This work has not been submitted, in whole or in part, to any conference, journal, book, university, or other institution for the purpose of obtaining any degree, diploma, or other academic qualification. I confirm that all sources of information and ideas from other works have been properly acknowledged in accordance with accepted academic conventions.

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Abstract

The pharmaceutical sector depends heavily on accurate demand forecasting to maintain proper stock levels, reduce medicine shortages, avoid overstocking, and support better healthcare service delivery. In Sri Lanka, predicting drug sales for areas is important because medicine demand can vary according to region, seasonal patterns, patient behaviour, disease trends, and healthcare access. Traditional forecasting methods are often limited when handling complex sales patterns, especially when the data contains non-linear trends, sudden changes, and differences between drug categories. Therefore, this research project focuses on developing an intelligent machine learning-based system for predicting drug sales data for areas in Sri Lanka.

The main objective of this project is to design and implement a drug sales prediction system that can forecast future pharmaceutical sales using historical sales data. The system uses multiple forecasting models, including statistical, machine learning, deep learning, and advanced artificial intelligence techniques. The implemented models include XGBoost, LightGBM, SARIMAX, Prophet, LSTM, GRU, Transformer, Temporal Fusion Transformer, N-BEATS, and Informer-based forecasting models. In addition to basic forecasting, the project also includes advanced components such as ensemble forecasting, meta-learning, neural architecture search, federated learning, causal inference, uncertainty estimation, and explainability. These components help improve prediction performance, support comparison between models, and provide more meaningful interpretation of the forecasting results.

The system was implemented as a microservice-based application with separate services for forecasting, model training, explainability, advanced AI functions, API gateway operations, and frontend interaction. A web-based user interface was developed to allow users to select drug categories, choose forecasting models, enter prediction dates, view forecast results, and access explanation outputs. The backend services expose REST API endpoints for forecasting, training, explainability, meta-learning, neural architecture search, federated learning, and causal analysis. The project also includes Docker-based deployment support, Kubernetes configuration, monitoring endpoints, and service health checks, making the implementation suitable for research demonstration and future deployment.

The dataset used in this research consists of historical pharmaceutical sales records categorized into multiple drug groups from C1 to C8. These categories represent different pharmaceutical product types, and the models were trained to identify time-series patterns within each category. The system evaluates forecasting behaviour using common performance measures such as Mean Absolute Error, Root Mean Squared Error, Mean Absolute Percentage Error, and related model comparison metrics. The inclusion of explainability techniques, such as SHAP-based and fallback feature importance analysis, helps identify how lagged sales values and temporal features influence the prediction results. Causal analysis was also included to examine possible relationships between temporal factors and drug sales behaviour.

The final implementation demonstrates that machine learning and deep learning methods can be effectively applied to pharmaceutical sales forecasting in a Sri Lankan context.

The system provides a practical platform for analysing historical drug sales, forecasting future demand, comparing multiple models, and supporting data-driven decision-making. This research can assist pharmacies, healthcare planners, distributors, and decision-makers in improving inventory management, reducing medicine wastage, and planning pharmaceutical supply more effectively. Overall, the project contributes a complete AI-based forecasting framework for predicting drug sales data for specific areas in Sri Lanka, combining predictive accuracy, explainability, scalability, and practical usability.

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Chapter 1: Introduction

The healthcare industry relies heavily on medicine availability. Medicines are not ordinary commercial products because their availability directly affects patient treatment, recovery, and public health. because its availability impacts the wellbeing of patients. Pharmacy, hospitals, medicine distributors, and the healthcare administration have the responsibility of ensuring the right medicines are available in sufficient quantities when required. When this process is not managed properly, it can create serious consequences. A shortage of essential medicines can delay treatment, increase patient risk, and reduce the quality of healthcare services. On the other hand, excessive stock can lead to expired medicines, financial loss, storage problems, and unnecessary wastage. Therefore, medicine stock management is both a healthcare issue and an inventory planning issue.

Drug sales forecasting plays a significant role since it could facilitate better medicine inventory planning. Using the sales history data, one is able to analyse the demands and forecast future sales of pharmacies and health care institutions. Earlier studies in the area of pharmaceutical drug sales forecasting have shown that time-series forecasting is applicable in categorical medicine demand forecasting and planning decisions [6]. These forecasted data is useful to make decisions regarding the amount of stock to buy, times to restock medicines, and medicine categories that may need more focus. There are several factors in the pharmaceutical domain that cause the shifts in the demands of medicines. These include seasonal illness, regional illnesses, doctor's prescriptions, consumers' demand, economy, and supply chain limitation. Therefore, manual stock estimation alone is not enough for reliable decision making.

The above challenges become prominent for Sri Lanka in the wake of recent national and international events that affect the country. The recent coronavirus outbreak caused health service delivery issues through the limitations in transportation, import of goods, and access to health facilities. This problem was exacerbated further by economic problems that arose in 2022. Lack of foreign exchange hindered the import of essential goods such as medicines and medical supplies [11]. These events demonstrated the importance of intelligent, data-driven systems that can support pharmaceutical demand forecasting and reduce the risk of shortages.

The demand for medicines can also vary from one area to another. A pharmacy located near a hospital may have different sales behaviour compared to a pharmacy in a residential or rural area.

Similarly, areas with higher population density, different income levels, and different disease patterns may show different sales trends. For example, respiratory medicines may have higher demand during periods of increased cough, asthma, or breathing related illnesses, while antihistamines may increase during allergy related seasons. Pain relief and anti-inflammatory medicines may also show repeated demand patterns because they are commonly used by many patients. Therefore, forecasting drug sales for specific areas is useful for understanding local demand behaviour.

Traditional forecasting methods and manual estimation techniques are often limited when dealing with real world pharmaceutical sales data. Drug sales data is usually time-dependent, meaning that previous values of sales affect future sales values. Sales may also contain trends, seasonality, sudden spikes, and irregular fluctuations. Time series forecasting studies show that sequential models would be helpful when historical values and time-dependency affect future results [8]. A single forecasting method may not perform well for every drug category because each category can have different demand behaviour. Thus, this study will incorporate different forecasting methods, including statistical models, machine learning models, deep learning models, and ensemble forecasting methods. Machine learning methods such as XGBoost have also been explored in recent drug sales prediction research, showing their usefulness for structured forecasting problems [10]. This allows the system to compare different models and identify suitable forecasting behaviour for different drug categories.

The dataset in this research is organized into eight drug sales categories, labelled C1 to C8. These categories represent common pharmaceutical groups such as anti-inflammatory drugs, analgesics, anxiolytics, sedatives, respiratory medicines, and antihistamines. The dataset was prepared in a category-wise time-series format so that each category could be analysed separately. Original data was collected from two pharmacies in Kurunegala. However, because pharmaceutical sales data is sensitive business information, it was difficult to collect a large amount of complete real-world data. Some pharmacy owners were not willing to share full records due to concerns related to privacy, tax, and business confidentiality. Therefore, synthetic data was also used to strengthen the dataset. The final dataset is a hybrid dataset that combines original pharmacy sales data and synthetic data.

This research does not focus only on producing a prediction value. It aims to develop a complete intelligent forecasting system that can be used practically. Artificial intelligence has been widely applied in healthcare-related prediction and decision-support tasks, including disease prediction, image-based diagnosis, and medical analytics [1]-[5].

These studies show that AI-based approaches can identify meaningful patterns from healthcare data and support more informed decision-making. In the same direction, this project applies AI methods to pharmaceutical sales forecasting. The system includes a web-based interface where a user can select a drug category, choose a forecasting model, enter a future date, and view the predicted sales result. The backend is designed using a microservice-based architecture with separate services for forecasting, training, explainability, advanced AI functions, API gateway handling, and frontend interaction. This architecture makes the system easier to maintain, test, deploy, and extend in the future.

Another essential aspect of this research is explainability. In systems applied to healthcare, users should be able to understand the rationale behind a prediction. A forecast value without an explanation may not be sufficient for pharmacy decision-making. Previous research has demonstrated that SHAP-based explainability can identify the input features influencing neural-network-based sales forecasts and provide more transparent support for business decision-making [9]. For example, a pharmacy owner may want to know whether a prediction has been influenced primarily by recent sales, older sales patterns, weekly changes, monthly trends, or other time-series features. To address this need, the system includes explainability methods such as SHAP-based analysis and fallback feature-importance techniques. These methods make the forecasting output more transparent and understandable.

This research project also explores advanced artificial intelligence concepts such as meta-learning, neural architecture search, federated learning, and causal inference. Meta-learning is useful for studying how knowledge from one drug category can support another category. Neural architecture search supports automating model improvements. Federated learning is important for privacy-preserving forecasting, especially when multiple pharmacies may want to train models without directly sharing raw data. Causal inference helps examine possible relationships between time-related factors and sales behaviour. These advanced modules increase the research value of the system and show possible future directions for pharmaceutical forecasting.

Overall, this research presents an AI based drug sales prediction system for specific areas in Sri Lanka. It combines time-series forecasting, machine learning, deep learning, explainability, advanced AI modules, web application development, API-based access, and deployment preparation. The purpose of the system is to support better pharmacy stock planning, reduce medicine shortages, minimize wastage, and promote data-based decision-making in the pharmaceutical industry. This study demonstrates how artificial intelligence can be applied to a practical healthcare problem and how forecasting systems can support both business efficiency and public health outcomes.

1.1. BACKGROUND OF THE STUDY

In the healthcare sector, the availability of medicines plays a major role in maintaining public health and ensuring proper patient care. It is required that pharmacies, hospitals, drug distributors, and healthcare providers manage their medicines inventory wisely as either drug shortage or excessive stock can lead to significant issues. In case medicines required by the patient are not in stock, there is a risk of delay in treatment, increased health dangers, and high costs. However, overstocking leads to losses due to the expiration of medicines and inefficient inventory management. For this reason, the accurate prediction of medicines sales became the necessity for effective pharmaceutical supply chain management. It was shown by previous research that forecasting techniques can help in pharmaceutical demand forecasting and decision making [6].

In Sri Lanka, the need for reliable medicine demand forecasting became more important after recent national and global challenges. The COVID-19 pandemic of 2019 and its aftermath resulted in certain difficulties with medicine supply chains related to import restrictions, transportation issues, and patient's changing buying behaviour. Afterward, the financial crisis of 2022 caused additional problems such as import restrictions, foreign currency shortages, medicine price increase, and pharmaceuticals supply delays. All these circumstances demonstrated that better planning procedures should be used in order to ensure the constant availability of the essential medicines in uncertain times.

There are several factors that can affect the medicine demands, among them location, seasonal patterns of diseases, patients' behaviour, healthcare services accessibility, weather conditions, and regional specifics of treatments. For example, respiratory-related medicines may show higher demand during rainy periods or seasons with increased breathing difficulties, while antihistamines may increase during allergy-related periods. Moreover, sales dynamics in different locations could vary because of population, income level, presence of hospitals nearby, prescriptive practices, and customer's buying habits. That is why manual estimations are not sufficient enough to provide proper stock planning of drugs. A data-driven forecasting system is required to identify patterns in historical drug sales and predict future demand more accurately. Drug sales data is time-dependent, which makes time series forecasting the appropriate method of forecasting. because they can learn patterns from past observations and support future demand prediction [8].

1.2. PROBLEM STATEMENT

The main problem addressed in this research is the difficulty of accurately predicting future pharmaceutical drug sales for specific areas based on the data in Sri Lanka. The vast majority of pharmacies depend on manual experience, rough estimation, or past purchase histories to estimate their future stocks levels. Although such an approach is effective only under stable conditions, it becomes less reliable when demand changes due to seasonal patterns, disease outbreaks, economic problems, medicine price changes, or supply chain restrictions. Previous research has shown that pharmaceutical drug sales prediction can support better categorical demand forecasting and planning decisions [6].

If future demand is underestimated, pharmacies may not have enough stock to serve patients. This can directly affect patient treatment, especially for medicines used for pain relief, respiratory conditions, allergies, anxiety, and other common health needs. And overestimation of future demand, can result in pharmacies purchasing more stock than required, which can lead to expired medicines, financial loss, and storage issues. Therefore, poor forecasting affects both business performance and healthcare service quality.

Another problem is that many existing forecasting approaches are limited to single models or simple prediction methods. They may not compare statistical, machine learning, and deep learning models together. Time-series forecasting research shows that different forecasting methods can behave differently depending on the structure, temporal dependencies, and complexity of the data [8]. Some systems also do not provide explainability, which means users cannot understand why a prediction was made. In a healthcare-related environment, this is a limitation because pharmacy owners and healthcare planners need understandable results before making inventory decisions. Explainable AI has been identified as important for improving transparency and supporting decision-making in forecasting-based business systems [9]. Therefore, this research focuses on developing a complete AI-based drug sales forecasting system that supports prediction, model comparison, explainability, and practical web/API-based usage.

1.3. RESEARCH GAP

Despite numerous research conducted on sales forecasting, healthcare analytics, and pharmaceutical demand prediction, several gaps still remain when applying these approaches to real-world drug sales forecasting in Sri Lanka. Most of the existing forecasting models are designed to forecast sales in general, but only a limited number of studies focus specifically on pharmaceutical drug sales prediction. Pharmaceutical drug sales forecasting differs from sales forecasting of regular products because medicine demand in such case is directly related to the state of public health, disease prevalence, seasonal illnesses, healthcare accessibility, supply chain constraints, and treatment of patients. This means that forecasting methodology that has been developed to be used with general sales data will be not appropriate for use with pharmaceutical sales data.

Another significant research gap is associated with pharmaceutical drug sales forecasting in Sri Lanka is the lack of pharmaceutical forecasting systems specific for Sri Lanka. In the last years, Sri Lanka was exposed to several disruptions, such as COVID-19 related import disruptions, global shipping delays, and the 2022 financial crisis. This resulted in medicine shortages, delays in importing medicines, change in prices, and uncertainty. However, most existing forecasting studies do not consider the practical challenges faced by Sri Lankan pharmacies, distributors, and healthcare authorities. As a result, there is a need for a forecasting system that is designed around local pharmaceutical sales data and can support category-wise demand prediction in the Sri Lankan context.

Most existing approaches also focus on a single forecasting model or a limited set of techniques. rely on a single prediction method or only a few forecasting techniques. These include either traditional statistical techniques such as ARIMA, SARIMAX, or Prophet, or machine learning models such as XGBoost and deep learning models such as LSTM. However, using only one model reduces the accuracy of forecasts due to the different nature of drug categories. Some categories may follow seasonal patterns, some may show non-linear demand changes, and others may depend strongly on recent sales trends. Therefore, there is a gap in systems and a need for a drug sales forecasting system that compares multiple statistical, machine learning, deep learning and ensemble models in the forecasting process.

Another typical limitation in many forecasting systems is the lack of explainability. In making healthcare-related decisions, users require the cause of the recommendation before putting faith in it. To overcome this problem, the proposed system has been integrated with explainability capabilities using SHAP-based feature importance analysis. This enables users to see what previous sales patterns have contributed the most to the forecasting outcome. Many machine learning and deep learning models can produce accurate outputs, but they often behave like black-box systems. This creates difficulty for pharmacists, distributors, and healthcare planners who need to justify purchasing and stock management decisions. Existing drug sales forecasting studies often focus mainly on prediction accuracy and provide limited explanation about which historical features or lagged sales values influenced the forecast. Therefore, there is a need for explainable forecasting systems that provide both prediction results and interpretable support.

Furthermore, most research implementations remain as isolated experiments, notebooks, or standalone scripts. They may show forecasting results, but they are not always developed into complete usable systems. A practical healthcare forecasting solution should provide API access, web-based interaction, model training support, health monitoring, and deployment readiness. Thus, there is a gap between research models and real-world deployable healthcare analytics systems since many of existing studies do not incorporate microservices, frontend dashboard, API gateway, Docker deployment, Kubernetes configuration, and monitoring endpoints.

Finally, there is a gap concerning the usage of advanced AI approaches such as meta-learning, neural architecture search, federated learning, and causal inference in the problem of pharmaceutical sales prediction. Meta-learning allows adaptation of model predictions between different drug categories; neural architecture helps in finding suitable architectures for a model, federated learning can be applied to preserve patients' privacy while the prediction is made, and causal inference can assist in understanding the possible influencing factors on drug sales. However, the mentioned approaches are rarely used in drug sales forecasting.

This research addresses these gaps by developing a complete AI-based drug sales prediction system that combines multiple forecasting models, explainability, advanced AI modules, API-based access, and microservice-based deployment for pharmaceutical sales forecasting in Sri Lanka.

1.4. AIMS AND OBJECTIVES

The aim of this research is to develop an AI-based drug sales forecasting system for predicting future category-wise pharmaceutical demand in specific areas of Sri Lanka.

The objectives of this research are:

- To collect and prepare historical drug sales data under category-wise groups from C1 to C8.
- To preprocess and format the dataset into a suitable time-series structure for forecasting.
- To implement statistical forecasting models such as SARIMAX and Prophet.
- To implement machine learning models such as XGBoost and LightGBM using lag-based features.
- To implement deep learning models such as LSTM, GRU, Transformer, TFT, N-BEATS, and Informer for sequential sales forecasting.
- To compare model performance using evaluation metrics such as MAE, RMSE, and MAPE.

- To include explainability methods such as SHAP and fallback feature importance analysis.
- To develop a web-based interface and API-based backend for practical forecasting access.
- To design the system using a microservice-based architecture with deployment support through Docker and Kubernetes.

1.5. SCOPE OF THE STUDY

The scope of this research is limited to predicting pharmaceutical drug sales using historical category-wise sales data. The dataset is organized into eight drug categories, labelled C1 to C8. These categories include anti-inflammatory drugs, analgesics, anxiolytics, sedatives, respiratory medicines, and antihistamines. The research focuses on forecasting future sales values for these categories rather than predicting individual patient prescriptions or medical diagnoses.

Code	Category	Common Pharmacy Medicines
C1 - M01AB	Anti-inflammatory acetic acid derivatives	Diclofenac, Indomethacin, Aceclofenac, Etodolac
C2 - M01AE	Anti-inflammatory propionic acid derivatives	Ibuprofen, Naproxen, Ketoprofen, Flurbiprofen
C3 - N02BA	Salicylic acid analgesics	Aspirin, Disprin, Ecosprin, Ascard
C4 - N02BE	Other analgesics / fever and pain relief	Paracetamol, Panadol, Calpol, Fevadol
C5 - N05B	Anxiolytics	Diazepam, Lorazepam, Alprazolam, Clonazepam
C6 - N05C	Hypnotics and sedatives	Zolpidem, Zopiclone, Nitrazepam, Midazolam
C7 - R03	Medicines for obstructive airway diseases	Salbutamol inhaler, Ventolin, Beclometasone inhaler, Budesonide inhaler, Montelukast
C8 - R06	Antihistamines	Cetirizine, Loratadine, Fexofenadine, Chlorpheniramine, Levocetirizine

Figure 1:1: Drug Category Classification Used in the Study

The data set used in this project is a hybrid data set. Original sales data was collected from two pharmacies in Kurunegala, but the available real data was limited because pharmacy owners were not willing to share complete sensitive business records due to tax, privacy, and commercial concerns. Thus, both real and synthetic data were used in order to support the model and training and testing of models. The final dataset combines original pharmacy sales data and synthetic data.

This research covers data preprocessing, feature engineering, model training, forecasting, model evaluation, explainability, API development, web application design, and deployment preparation. However, the system is not intended to replace doctors, pharmacists, or healthcare authorities. It is designed as a decision-support tool that can help users understand future sales trends and make better stock planning decisions.

1.6. SIGNIFICANCE OF THE STUDY

This research is significant because it involves the use of artificial intelligence technology for a real-world healthcare problem faced by pharmacists in Sri Lanka. There have been many cases in which the use of artificial intelligence technology has shown promising results in healthcare prediction and decision making such as disease prediction and medical image-based diagnosis [1]-[5]. Precise forecasting of drugs sales can help to maintain stock, prevent any shortages of medicine, overstock and waste of medicine. Previously there have been works done in categorical drug sales forecasting where time-series prediction has proven its effectiveness in pharmaceutical demand planning and inventory related decision making [6]. This is especially important in countries where medicine imports, supply chains, and prices can be affected by economic and political conditions.

The system can support pharmacists get a forecast based on data analysis rather than relying on manual estimates. This will be valuable also for the distributors and health care administrators to have forecasts of future demands of medicines of various categories. Since the system includes multiple forecasting models, users can compare model behaviour and choose a suitable forecasting approach based on accuracy and practical needs. Time-series forecasting research shows that different forecasting models may perform differently depending on the structure and temporal behaviour of the dataset [8]. Therefore, comparing statistical, machine learning, and deep learning models is useful for identifying the most suitable forecasting approach for each drug category.

Another important significance of this research is explainability. By including SHAP-based analysis and fallback feature importance methods, the system helps users understand the reason behind forecast outputs. Explainable AI has been recognized as important for improving transparency and supporting decision-making in forecasting-based business systems [9]. This improves trust and makes the system more useful for real decision-making. The microservice-based architecture, API gateway, frontend dashboard, Docker support, Kubernetes configuration, and monitoring features also make the project more practical as a deployable software solution.

1.7. STRUCTURE OF THE THESIS

This thesis is composed of six chapters. The first chapter covers the background of the research, research problem, research gap, aim, objectives, scope, and importance of the study. Chapter 2 reviews related literature on artificial intelligence in healthcare, drug sales forecasting, time-series forecasting, machine learning, deep learning, explainable AI, and advanced AI techniques. The third chapter will cover the methodologies employed in the research, which include datasets and data pre-processing, feature engineering, forecasting models, methods for explainability, architecture of the system, design of the API, web application design, deployment, and evaluation method.

The fourth chapter will focus on the result and analysis of the implemented system. This chapter will focus on the performance of statistical models, machine learning models, deep learning models, ensemble forecast, explainability output, advanced AI components, web application testing, API endpoint testing, and deployment testing. Chapter 5 will discuss the results in relation to the research objectives, which include model performance, practicality, strengths, weaknesses, and future improvement. Finally, chapter 6 will be the conclusion of the research.

Chapter 2: Literature Review

The use of artificial intelligence and machine learning in healthcare has increased rapidly during recent years because these technologies can support prediction, diagnosis, planning, and decision-making. In healthcare environments, accurate prediction is highly important because decisions are often connected with patient safety, treatment availability, resource allocation, and cost management. Many recent studies show that machine learning and deep learning models can identify hidden patterns from medical data more effectively than traditional manual analysis. For example, Abbas et al. [1] proposed an explainable artificial intelligence-based model for ocular disease prediction, showing how AI can be used not only to make predictions but also to improve transparency in healthcare decision making. Similarly, Khan et al. [2], Nazir et al. [3], Gamage et al. [4], and Wijesinghe et al. [5] applied deep learning techniques for medical image-based disease recognition. Although these studies mainly focus on disease identification, they show the broader importance of AI in healthcare and support the idea that intelligent systems can improve medical decision-making when properly designed and evaluated.

In the pharmaceutical domain, one of the most important challenges is predicting future drug demand accurately and in advance. Drug sales forecasting helps pharmacies, hospitals, distributors, importers, and healthcare authorities maintain proper stock levels and avoid both shortages and overstocking. This requirement became more important during recent crisis periods in Sri Lanka. During the COVID-19 pandemic that began in 2019 and continued into the following years, international transportation restrictions, lockdowns, shipping delays, and interruptions in global supply chains affected the availability of many essential goods, including medicines. Since Sri Lanka depends significantly on imported pharmaceutical products and raw materials, any delay in importation can directly affect medicine availability in local pharmacies and hospitals. Later, the 2022 financial and economic crisis created further pressure on the healthcare supply chain. Foreign currency shortages, import restrictions, increased prices, fuel shortages, and delays in supplier payments made it difficult to import medicines on time. As a result, the country experienced shortages of several pharmaceutical products, while healthcare providers and patients faced uncertainty regarding medicine availability.

These situations show that traditional manual stock estimation methods are not sufficient during unstable economic and public health conditions. If future demand is underestimated, essential medicines may become unavailable at the exact time patients need them, which can delay treatment, increase health risks, and reduce public trust in the healthcare system. This is especially serious for drugs required for chronic illnesses, pain management, respiratory conditions, infections, and emergency treatment. On the other hand, if future demand is overestimated, pharmacies and suppliers may purchase excessive stock, which can lead to expired medicines, storage issues, unnecessary financial burden, and wastage of limited resources. During periods of import restrictions and economic instability, these mistakes become even more costly because medicine procurement decisions must be made carefully with limited foreign exchange and restricted supply channels. Therefore, drug sales prediction is not only a business or inventory management problem, but also a healthcare planning and national resource management problem.

Time-series forecasting is commonly used for sales prediction because sales data usually depends on time-based patterns. Historical sales values often contain trends, seasonality, sudden changes, and repeating cycles. In pharmaceutical sales, these patterns may be influenced by disease outbreaks, seasonal illness, climate changes, holidays, healthcare access, and patient behaviour. A time-series forecasting system attempts to learn from previous sales values and estimate future demand. In this type of problem, the order of data is important because the past directly influences the future. Therefore, random data splitting is usually not suitable for time-series forecasting. Instead, historical records must be used carefully so that the model learns from earlier periods and predicts later periods. This makes drug sales forecasting different from ordinary classification or regression problems.

Traditional statistical forecasting methods such as ARIMA, SARIMAX, and Prophet are widely used for time-series data because they can model trend and seasonal behaviour. SARIMAX is useful when the data contains autoregressive patterns, moving average behaviour, and seasonal effects. Prophet is also popular because it is designed to handle trend changes and seasonality in a relatively flexible manner. These models are useful as baseline approaches because they provide interpretable forecasting behaviour and are suitable for many standard time-series problems. However, pharmaceutical sales data can be more complex because medicine demand may change due to non-linear patterns, irregular demand spikes, sudden shortages, and category-specific differences. Therefore, traditional statistical methods alone may not always be sufficient.

Ekanayake et al. [6] directly addressed the problem of predicting medical drug sales in a specific area using categorical drug sales data and time-series forecasting. Their study is highly relevant to this research because it focuses on pharmaceutical sales prediction in a Sri Lankan context. The study demonstrates that historical categorical drug sales data can be used to forecast future sales and support decision-making in medicine supply management.

This provides a strong foundation for the present project, which extends the idea further by implementing a complete forecasting system with multiple models, web-based interaction, explainability, advanced AI components, and microservice-based deployment.

Machine learning models such as XGBoost and LightGBM have become popular for forecasting and regression tasks because they can capture non-linear relationships and handle structured data effectively. XGBoost is especially useful in sales prediction because it builds an ensemble of decision trees and improves performance through gradient boosting. Xiong [10] proposed an AI-driven model for drug sales prediction using XGBoost and enhanced optimization techniques, showing that boosting-based models can be effective for pharmaceutical demand forecasting. This supports the use of XGBoost in the current research, especially for category-wise drug sales prediction using lag-based features generated from historical sales records. LightGBM is another gradient boosting framework that is suitable for efficient model training and structured forecasting tasks [7]. These models are useful because they can learn relationships between previous sales values and future sales outputs without requiring strict statistical assumptions.

Deep learning methods are also widely used in time-series forecasting because they can learn complex temporal patterns from sequential data. Recurrent Neural Networks, especially Long Short-Term Memory networks and Gated Recurrent Units, are commonly used when the prediction depends on previous time steps. Hewamalage, Bergmeir, and Bandara [8] reviewed recurrent neural networks for time-series forecasting and discussed their strengths, limitations, and future directions. Their work shows that RNN-based models are useful for capturing temporal dependencies but also require careful design, preprocessing, and evaluation. In the current project, LSTM and GRU models are included because drug sales data contains sequential behaviour where previous sales values can influence future demand.

Transformer-based models have also become important in modern forecasting because they use attention mechanisms to learn relationships across time steps. Unlike traditional recurrent models, Transformers can capture long-range dependencies more effectively and can process sequences in a more flexible way. In this project, Transformer-based approaches such as Transformer, Temporal Fusion Transformer, N-BEATS, and Informer models are considered to improve forecasting ability and compare deep learning performance against traditional and machine learning models. These models are useful when drug sales patterns are affected by both short-term and long-term changes. By including multiple deep learning models, the system does not depend on a single architecture and can compare how different sequence learning methods behave for pharmaceutical sales data.

Another important area in the literature is explainable artificial intelligence. In healthcare-related systems, prediction accuracy alone is not enough. Users should also be able to understand how and why the system produces a particular output. Explainability is especially important in medical and pharmaceutical applications because decision-makers must trust the system before using its recommendations. Chen et al. [9] proposed an explainable artificial intelligence framework for sales forecasting by combining a WaveNet neural network, multiple regression, and SHAP-based interpretation. Their study highlights the importance of transparency in forecasting systems, particularly when predictions support operational decisions. This directly supports the explainability component of the present research, where SHAP-based analysis and fallback feature importance methods explain the contribution of lagged sales features and temporal patterns..

In this project, explainability is connected directly to the forecasting workflow. When the system predicts future sales for a selected drug category, it is also important to identify which past sales values or temporal patterns influenced the output. For example, lagged sales values may show whether recent demand has a strong effect on the predicted result. This type of explanation can help pharmacists, distributors, and healthcare planners understand whether the forecast is based on stable historical behaviour or recent changes in demand. In real healthcare decision-making, this is valuable because users may be reluctant to rely on a black-box prediction without any supporting explanation. Therefore, explainability improves both technical transparency and practical usability.

In addition to forecasting and explainability, this project also investigates several advanced artificial intelligence techniques that can strengthen pharmaceutical sales prediction beyond a single-model forecasting approach. The system mainly focuses on category-wise drug sales forecasting using historical time-series data from C1 to C8 drug categories, but it extends the prediction pipeline with meta-learning, neural architecture search, federated learning, and causal inference modules. Meta-learning is included because drug categories may not always have equal sales behaviour, data volume, or forecasting difficulty. In a real pharmaceutical environment, some drug groups may have long and stable sales histories, while others may have limited or irregular records due to changes in demand, supplier availability, seasonal illnesses, or import restrictions. A meta-learning approach allows the system to learn transferable patterns across multiple drug categories and then adapt this learned knowledge to a target category with fewer samples or different sales behaviour.

Neural Architecture Search is another advanced component included in this project to reduce the manual effort involved in selecting suitable neural network structures for drug sales prediction. Since time-series forecasting performance can depend heavily on model structure, number of layers, hidden units, activation behaviour, and training configuration, manual experimentation can become time-consuming and inconsistent. The NAS component explores different architecture configurations and attempts to identify better-performing structures for the given drug category. This supports automated model optimization and makes the system more research-oriented because it does not depend only on fixed predefined models.

Federated learning is also included as a future-ready design for pharmaceutical forecasting because real sales data may belong to different pharmacies, hospitals, distributors, or regional branches. In practice, these organizations may not be willing or legally allowed to share raw sales records due to privacy, commercial sensitivity, or operational restrictions. Federated learning provides a privacy-preserving training mechanism where multiple clients can train local models using their own data, and only model updates are aggregated to build a global model. This makes the system suitable for a distributed healthcare environment where drug sales prediction can be improved using data from many locations without centralizing sensitive records.

Causal inference is used to move the system beyond correlation-based forecasting. Most forecasting models can identify patterns between historical inputs and future sales, but they do not directly explain whether a particular temporal factor may influence changes in drug demand. The causal analysis module in this project examines relationships between variables such as sales lag values, weekly trends, seasonal indicators, and sales behaviour. This helps the system provide more meaningful analytical insight rather than only producing a numerical forecast. For example, if lagged sales values or seasonal indicators show a strong relationship with future demand, decision-makers can better understand the reasons behind predicted changes. This is important in pharmaceutical planning because demand may be influenced by healthcare events, seasonal diseases, regional behaviour, and supply conditions.

A major difference between this research and many existing forecasting studies is that the present project is not limited to model training alone. It is implemented as a complete system with a microservice-based architecture. The system includes separate services for forecasting, model training, explainability, advanced AI processing, frontend interaction, and API gateway routing. This design improves maintainability because each service has a separate responsibility. For example, the forecasting service handles prediction requests, the training service handles model training, the explainability service provides interpretation outputs, and the advanced AI service handles NAS, federated learning, and causal inference functions. The API gateway provides a single access point for backend functions, while the frontend service allows users to interact with the system through a web interface.

This system-level design is important because a practical healthcare analytics system must be usable, testable, and deployable. A forecasting model that works only in a notebook or script may be useful for experimentation, but it may not be suitable for real users. In contrast, a web-based and API-based system allows users to select drug categories, choose models, enter prediction dates, and view forecast results through an interface. The project also includes Docker support, Kubernetes configuration, monitoring endpoints, and service health checks. These features show that the system is designed not only for academic experimentation but also for future deployment and scaling.

Overall, the reviewed literature shows that artificial intelligence has strong potential in healthcare prediction, pharmaceutical sales forecasting, and decision support. Previous studies have demonstrated the effectiveness of deep learning in medical applications [1]-[5], the relevance of time-series forecasting for drug sales prediction [6], the usefulness of recurrent neural networks in forecasting [8], the value of SHAP-based explainability [9], and the strength of XGBoost-based models for drug sales prediction [10]. However, many existing approaches focus only on a single model, a limited experiment, or a standalone forecasting pipeline. The present research addresses this limitation by developing a complete AI-based drug sales prediction system for specific areas in Sri Lanka. The implemented system combines statistical forecasting models, machine learning models, deep learning models, ensemble forecasting, explainability, meta-learning, neural architecture search, federated learning, causal inference, API-based access, and web-based visualization. It is also designed using a microservice architecture with separate services for forecasting, training, explainability, advanced AI operations, frontend interaction, and API gateway routing. This makes the project not only a forecasting experiment but also a complete deployable healthcare analytics platform that can support real-world pharmaceutical demand planning, inventory management, and decision-making in the Sri Lankan healthcare context.

Chapter 3: Methodology

In this research, a drug sales prediction system is designed that is based on AI and can forecast future drug sales for particular categories in certain regions of Sri Lanka. It is created by using time-series sales data where the drugs are denoted by categories C1 to C8. The proposed methodology follows a complete forecasting pipeline, starting from data collection and preprocessing, then moving into feature engineering, model training, prediction generation, explainability, evaluation, and system deployment.

The proposed framework incorporates a range of forecasting techniques rather than relying on one forecasting technique. Statistical models, machine learning models, deep learning models, and more sophisticated AI-based models are applied to study different patterns in sales. Statistical models like SARIMAX and Prophet are utilized for the detection of trend and seasonality patterns. Machine learning models like XGBoost and LightGBM are employed to find out non-linear dependencies between the lags in the sales data. Deep learning models like LSTM, GRU, Transformer, TFT, N-BEATS, and Informer are employed to detect sequential and long-term behaviour of time-series. In addition, the framework includes advanced modules like meta-learning, neural architecture search, federated learning, and causal inference.

The system is implemented as a microservice-based application with separate services for forecasting, model training, explainability, advanced AI processing, frontend interaction, and API gateway routing. This structure allows the system to be modular, testable, scalable, and easier to maintain. A web-based interface is also developed so that users can select a drug category, choose a forecasting model, enter a prediction date, and view the forecast output with visual results.

Moreover, the methodological framework incorporates the concept of explainability for the purpose of providing an interpretable output of predictions. Explainability analysis is performed through the use of SHAP-based analysis and fallback approaches of assessing feature importance, with the focus being on how historical sales values and time-dependent features impact the prediction process. It is crucial since forecasting in the pharmaceutical industry plays a key role in decision-making in the area of healthcare and inventory management. The overall system methodology is represented in the block diagram below in figure 3:1.

3.1 SYSTEM WORKFLOW

Workflow of the system in this study can be explained as an end-to-end prediction pipeline for pharmaceutical drug sales predictions. This starts by collecting the data set that comprises historical drug sales data. In this case, the sales data are classified according to their respective pharmaceutical categories ranging from C1 to C8. These raw sales records are treated as time-series data because each sales value is connected to a specific date or time period. After collecting the data, preprocessing is carried out to prepare the dataset for model training. This includes reading the category-wise CSV files, parsing date values, arranging records in chronological order, handling missing or inconsistent values, and preparing the sales column as the prediction target.

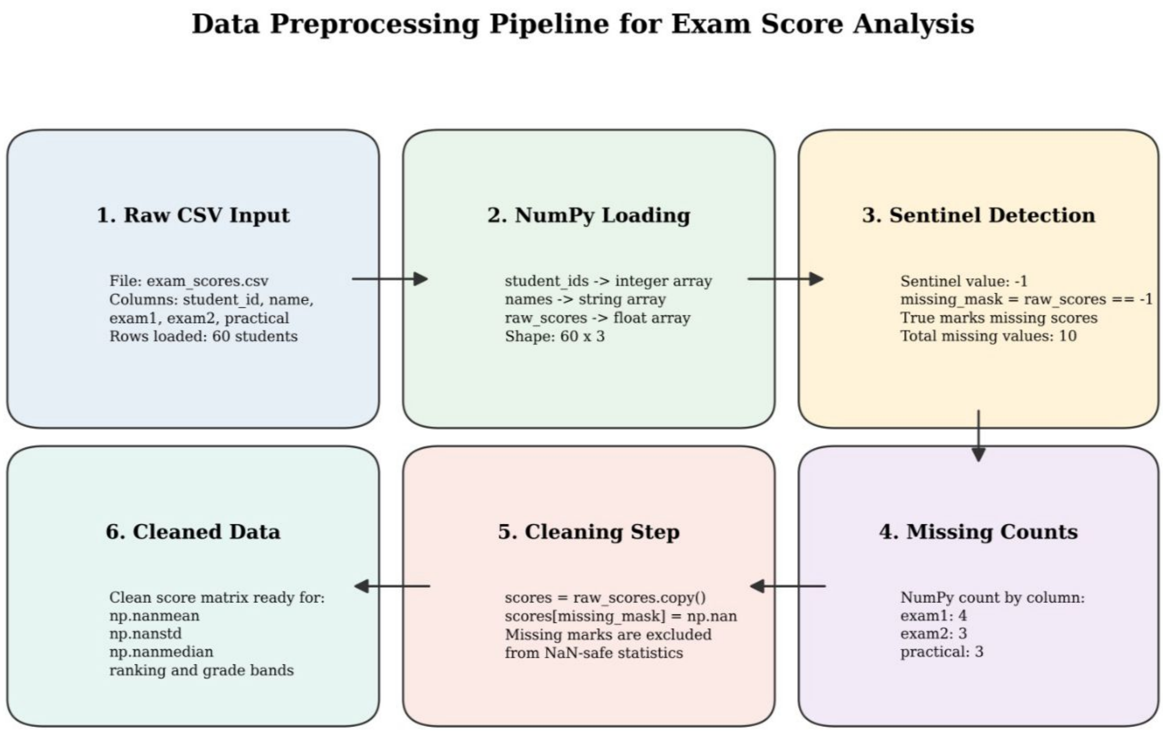

Figure 3:1:Data Preprocessing Pipeline

After preprocessing, feature engineering is performed to convert the raw time-series data into a suitable format for forecasting models. Since future drug sales are highly influenced by previous sales behaviour, lag features are generated from past sales values. These lag features allow machine learning models such as XGBoost and LightGBM to learn the relationship between previous demand and future sales. For deep learning models, the sales data is transformed into sequential input windows so that models such as LSTM, GRU, Transformer, TFT, N-BEATS, and Informer can learn temporal patterns from historical sequences. Additional time-based patterns such as trend and seasonality are also considered where applicable.

Once the data preprocessing phase is completed, the training of the forecasting models takes place. Training of different forecasting models takes place for each drug category in order to capture the sales behavior of each category individually. SARIMAX and Prophet are some of the statistical models used for modeling trend and seasonality in the time series. Lag features based machine learning models such as XGBoost and LightGBM are used for modeling drug sales. Normalized sequential data-based deep learning models such as LSTM, GRU, and Transformer-based are also used. The trained models and associated scalers files are stored as model artifacts so that the models can be reused again at the time of forecasting without retraining.

During the forecasting stage, the user selects a drug category, forecasting model, and prediction date through the web interface. This request is sent to the backend API, where the relevant trained model is loaded and used to generate the predicted drug sales value. If the selected date belongs to the historical data period, the system can return the closest available actual value. If the date is in the future, the selected model performs forecasting based on the learned historical sales patterns. The forecast result is returned with details such as predicted sales value, selected category, model used, prediction date, and a visualization output.

After generating the forecast, the explainability phase is executed as part of the process that explains the forecast. SHAP-based explainability or fallback feature importance methods are used to identify which previous sales values or lag features contributed most to the prediction. This is important because pharmaceutical forecasting supports decision-making, and users need to understand the reason behind the predicted demand. The explainability output helps users interpret whether the forecast is mainly influenced by recent sales behaviour, seasonal changes, or other time-based patterns.

Finally, the output delivery is provided in both the web application and REST API formats. Web UI provides users with visual interaction, including input selection and forecast visualization, forecast charting, and explanation outputs. The API layer allows forecasting, training, explainability, and advanced AI functions to be accessed programmatically. The complete workflow is supported by a microservice architecture, where separate services handle forecasting, training, explainability, advanced AI operations, frontend interaction, and API gateway routing. This workflow makes the system modular, scalable, and suitable for future deployment in pharmaceutical demand planning environments.

3.2 DATASET DESCRIPTION

Code	Category	Common Pharmacy Medicines
C1 - M01AB	Anti-inflammatory acetic acid derivatives	Diclofenac, Indomethacin, Aceclofenac, Etodolac
C2 - M01AE	Anti-inflammatory propionic acid derivatives	Ibuprofen, Naproxen, Ketoprofen, Flurbiprofen
C3 - N02BA	Salicylic acid analgesics	Aspirin, Disprin, Ecosprin, Ascard
C4 - N02BE	Other analgesics / fever and pain relief	Paracetamol, Panadol, Calpol, Fevadol
C5 - N05B	Anxiolytics	Diazepam, Lorazepam, Alprazolam, Clonazepam
C6 - N05C	Hypnotics and sedatives	Zolpidem, Zopiclone, Nitrazepam, Midazolam
C7 - R03	Medicines for obstructive airway diseases	Salbutamol inhaler, Ventolin, Beclometasone inhaler, Budesonide inhaler, Montelukast
C8 - R06	Antihistamines	Cetirizine, Loratadine, Fexofenadine, Chlorpheniramine, Levocetirizine

Figure 3:2:Drug Category Classification Used in the Study

The used dataset in the current research was generated by combining the real pharmaceutical sales data and synthetic data. At the early stage of the project, the goal was to gather the original records of the drug sales from the pharmacies in Sri Lanka. Data was collected from two pharmacies situated in the Kurunegala region. Records provided the initial insight about the patterns of pharmaceutical sales over time and about different drug group demands. The gathered data was associated with sales of various categories of medicines, which were used to build the dataset.

However, during the data gathering process, it was hard to gather enough sales data for real pharmacies. This is because many pharmacy owners did not want to provide their sales data due to its sensitivity. Owners feared privacy issues, tax issues, confidentiality, and misuse of the business information. Due to such constraints, the amount of real data gathered was insufficient to train and test several forecasting models. The machine learning and deep learning models need some amount of historical data to be able to learn the sales patterns. This was the reason behind the use of synthetic data generation for the dataset.

Thus, it can be said that the final dataset employed in this study is actually a hybrid dataset that contains both the real dataset and the artificial dataset. The actual data set that was collected from the pharmacies of Kurunegala was employed to analyse the actual sales behaviour, while the artificial data set was created to increase the size of the data set and train various models. The artificial data was created such that it resembles the characteristics of the real data set in terms of time series analysis, namely trend, variability, and category-wise behaviour.

The final dataset is organized into eight drug categories, named C1 to C8. Each category represents a different group of pharmaceutical products. These categories were created manually by grouping medicines according to their usage and sales behaviour. Category-wise organization is important because different drug groups may have different demand patterns. For example, some drugs may show seasonal demand, while others may have more stable or irregular sales behaviour. By separating the dataset into C1 to C8 categories, the forecasting system can train and predict sales for each drug group individually.

The data has been organized in a time series format. This means each row consists of the date field and a sales field which holds sales information for a particular drug category. The date field indicates the period during which the sale was made whereas the sales field indicates the total sales quantity for that particular category. Since the data is time-dependent, the order of the records is important. Earlier records are used to learn historical patterns, and later records are used for testing and forecasting.

For implementation, the dataset is stored as category-wise CSV files. Separate CSV files are maintained for each drug category, such as C1.csv, C2.csv, C3.csv, and so on up to C8.csv. This structure makes it easier to load, preprocess, train, and forecast each category independently. The model training pipeline reads the relevant category file, prepares the time-series data, creates lag features or sequential input windows, and trains the selected forecasting model. This category-wise file structure also supports future expansion, where additional drug categories or area-specific datasets can be added to the system.

Overall, the dataset used in this research represents a practical solution to the data availability challenges in pharmaceutical sales forecasting. Although collecting large-scale real pharmacy data was difficult due to privacy and business sensitivity, the hybrid dataset allowed the research to proceed while preserving realistic sales behaviour. The combination of original Kurunegala pharmacy data and synthetic time-series data provided a suitable foundation for developing, training, and evaluating the proposed drug sales prediction system.

3.3 DATA PREPROCESSING

Data preprocessing becomes a significant part of the research process in order to prepare the data for analysis by forecasting models. The initial data set was formed by pharmaceutical sales record taken from pharmacies and later increased with the use of artificial data. In order to use the records in further model training, it needed to organize them in such a way that each group of drugs could be pre-processed independently. That is why the pre-processed data were categorized into groups C1 to C8.

The first preprocessing step was date parsing. Each sales record was connected with a specific date or time period, so the date column had to be converted into a proper date-time format. This allowed the system to recognize the chronological order of the records and treat the dataset as time-series data. Date parsing is especially important in forecasting because the order of data points directly affects model training. After converting the date values, the records were sorted according to time so that earlier sales values appeared before later sales values.

The next step was missing value handling. In real-world pharmacy data, missing values can occur due to incomplete records, manual entry mistakes, unavailable sales logs, or data collection limitations. Missing values can negatively affect model training and may produce inaccurate forecasts if they are not handled properly. Therefore, the dataset was checked for missing or invalid values before training. Where possible, missing values were corrected or filled using suitable methods such as forward filling, backward filling, or interpolation based on nearby sales records. If any records were highly inconsistent or unusable, they were removed from the dataset.

After handling missing values, the dataset was formatted category-wise. Each drug category from C1 to C8 was stored in a separate CSV file. This made it easier to train and evaluate models independently for each category. Category-wise formatting is useful because different drug categories can have different demand patterns. For example, one category may show seasonal demand, while another may show stable or irregular sales behaviour. By separating the data, the model can learn the unique sales behaviour of each category instead of mixing all drug groups into one general pattern.

Data cleaning was also carried out to improve dataset quality. This included removing duplicate records, checking incorrect date formats, ensuring numerical sales values, and identifying unusual sales entries. Since the dataset contains both original and synthetic records, cleaning helped maintain consistency between both data sources. Sales values were checked to make sure they were suitable for forecasting and did not contain invalid text values or incorrect formats. This step helped reduce noise and prepared the dataset for reliable feature creation.

Finally, the pre-processed data was prepared for model input. For machine learning models such as XGBoost and LightGBM, lag features were created from previous sales values so that the models could learn how past demand influences future demand. For deep learning models such as LSTM, GRU, and Transformer-based models, the sales data was converted into sequential input windows. These input sequences allow the models to learn temporal patterns from historical sales behaviour. For models that require scaling, the data was normalized before training. After preprocessing, the cleaned and formatted data became suitable for statistical forecasting, machine learning, deep learning, and advanced AI-based forecasting models.

3.4 TIME-SERIES FEATURE ENGINEERING

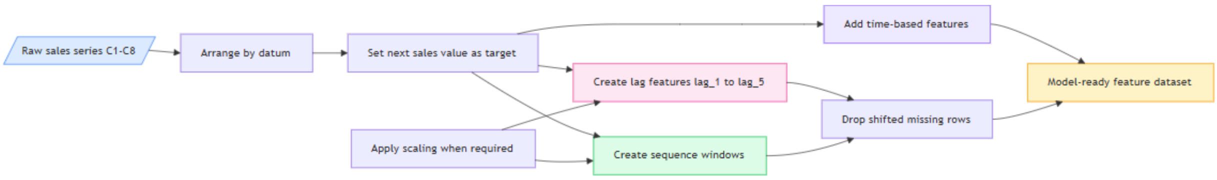

Figure 3:3:Time-Series Feature Engineering Process

In this project, feature engineering was performed to convert the category-wise drug sales records into a suitable format for different forecasting models. Since the dataset is time-series based, the most important information comes from previous sales values and time-related patterns. Drug sales usually do not change randomly; the demand for a future date is often influenced by the sales behaviour of previous days or weeks. Therefore, past sales values, time order, trend behaviour, and seasonal patterns were considered during feature preparation.

The main feature engineering method used in the project is lag feature generation. For machine learning models such as XGBoost and LightGBM, previous sales values are converted into lag features. For example, if the system predicts the next sales value for category C1, it uses a fixed number of previous sales records as input. In the implementation, lag-based forecasting is used so that the model can learn the relationship between recent historical sales and the next expected sales value. These lag features help the model identify whether sales are increasing, decreasing, or remaining stable.

For deep learning models such as LSTM, GRU, Transformer, TFT, N-BEATS, and Informer, the sales records are prepared as sequential windows. Instead of using only one previous value, a sequence of past sales values is given to the model. This allows the model to learn temporal behaviour from the data. LSTM and GRU models are useful for learning short-term and medium-term dependencies, while Transformer-based models can capture longer relationships in the sales sequence. This is important because pharmaceutical sales may follow repeated demand behaviour across time.

prophet is also used in this project to model trend and seasonality in the category-wise drug sales data. For Prophet, the date column is formatted as ds, and the sales value is formatted as y, which is the required input structure for Prophet forecasting. This allows Prophet to analyse historical sales movement over time and generate future sales predictions. Prophet is useful in this research because pharmaceutical sales may contain seasonal patterns, gradual trend changes, and repeated demand behaviour across weekly or monthly periods. By including Prophet, the system can compare traditional time-series forecasting behaviour with machine learning and deep learning models.

Trend-based features are also important in this project. A trend shows whether the sales of a drug category are generally increasing, decreasing, or moving steadily over time. In pharmaceutical sales, a trend may happen due to changes in patient demand, medicine availability, seasonal illnesses, or supply limitations. By considering the sales movement across time, the forecasting models can better understand long-term behaviour instead of only depending on a single previous value.

Seasonal and time-related behaviour is also considered in the advanced analysis part of the project. The causal inference module creates temporal features such as week, month, quarter, year, lag values, and seasonal indicators. These features help analyse whether drug sales are affected by time-based patterns. For example, some medicines may have higher demand during rainy seasons, respiratory illness periods, or allergy-related periods. Monthly and weekly patterns are useful because pharmaceutical demand can change depending on seasonal disease behaviour and patient purchasing habits.

Past sales values are important because they represent the most direct signal of future demand. If a drug category has shown high sales in recent periods, there is a possibility that demand may continue unless there is a sudden external change. Similarly, if sales have decreased over several previous periods, the future demand may also be lower. By using lag features, sequential windows, Prophet-based date formatting, trend information, and seasonal time-based variables, the system prepares the dataset in a way that allows different forecasting models to learn meaningful drug sales patterns from the C1-C8 category-wise data.

3.5 DATA NORMALIZATION AND SCALING

Data normalization and scaling were applied in this project to prepare the drug sales time-series data for different forecasting models. Since the sales values can vary between drug categories, using raw values directly may affect the training stability of some models, especially deep learning models. Models such as LSTM, GRU, Transformer, TFT, N-BEATS, and Informer learn from sequential numerical inputs, so scaling helps keep the input values within a suitable range and improves training efficiency.

In the implementation, MinMaxScaler was used for models such as LSTM, GRU, Transformer, and LightGBM. This scaler transforms the sales values into a normalized range, allowing the models to learn patterns without being affected by large differences in numerical scale. For other advanced deep learning models such as TFT, N-BEATS, and Informer, StandardScaler was used to standardize the data based on mean and standard deviation. This is useful for models that perform better when the input data is cantered and scaled.

After training, the scaler objects were saved separately with the trained models. This is important because the same scaling process must be used again during forecasting. When the system loads a trained model to predict future drug sales, it also loads the corresponding scaler file. The input data is first transformed using the saved scaler, and after prediction, the forecasted value is converted back into the original sales scale using inverse transformation. This ensures that the final output shown to the user is in the actual sales value format, not in the normalized scale. Therefore, normalization and scaling are important preprocessing steps in this project because they support stable model training, improve compatibility with deep learning architectures, and ensure that predictions can be correctly transformed back into meaningful pharmaceutical sales values.

3.6 TIME-SERIES TRAIN-TEST SPLITTING

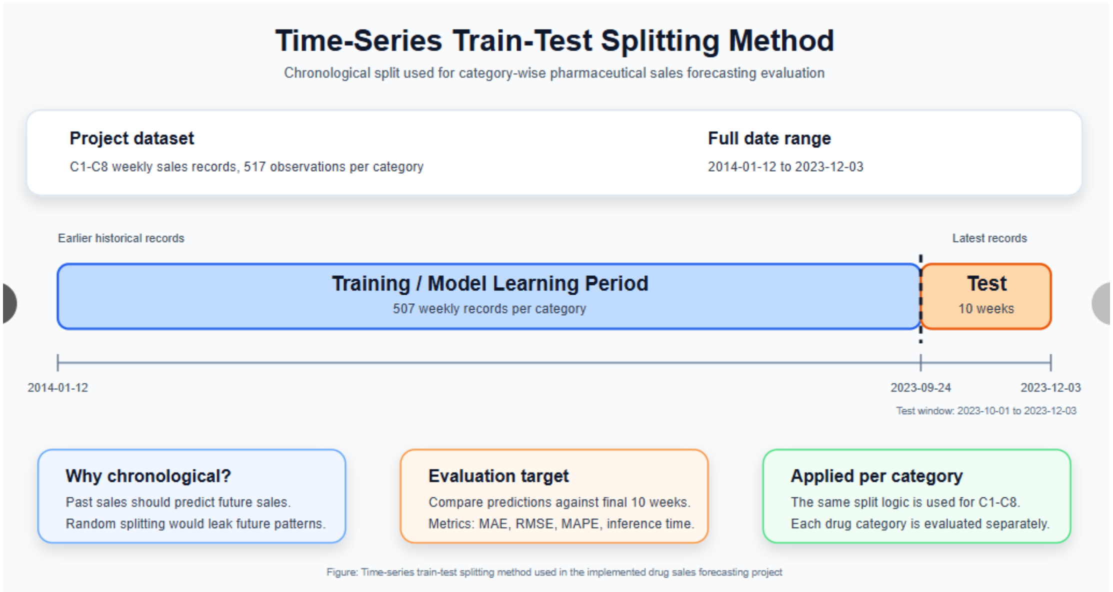

Figure 3:4: Time-Series Train-Test Splitting Method

In this project, the drug sales dataset is handled as time-series data because each sales value is connected to a specific date or time period. Therefore, the order of the records is very important. Unlike normal machine learning datasets, time-series data should not be randomly shuffled before training and testing. If random splitting is used, future sales values may appear in the training set while earlier values appear in the testing set. This can cause data leakage and give unrealistic model performance because the model would indirectly learn information from the future.

To avoid this issue, the dataset was split while preserving chronological order. Earlier sales records were used for model training, and later sales records were used for testing and evaluation. This method better represents the real-world forecasting situation, where a model is trained using past sales data and then used to predict future demand. For example, the system learns from previous historical sales values of a drug category and then forecasts sales for upcoming time periods.

This chronological splitting approach was applied category-wise for the C1 to C8 drug sales datasets. Each category was treated as an individual time-series forecasting problem. By using earlier data for training and later data for testing, the models could be evaluated based on how well they predict unseen future sales behaviour. This is important because pharmaceutical sales forecasting is mainly concerned with future demand planning, not random record prediction.

Maintaining the correct time order also helps models learn realistic trend and seasonal patterns. If the data were shuffled, models such as SARIMAX, Prophet, LSTM, GRU, Transformer, and other forecasting models would lose the natural sequence of sales behaviour. Therefore, time-series train-test splitting was used to preserve the temporal structure of the dataset and ensure that evaluation results are closer to real forecasting conditions.

3.7 FORECASTING MODEL DEVELOPMENT

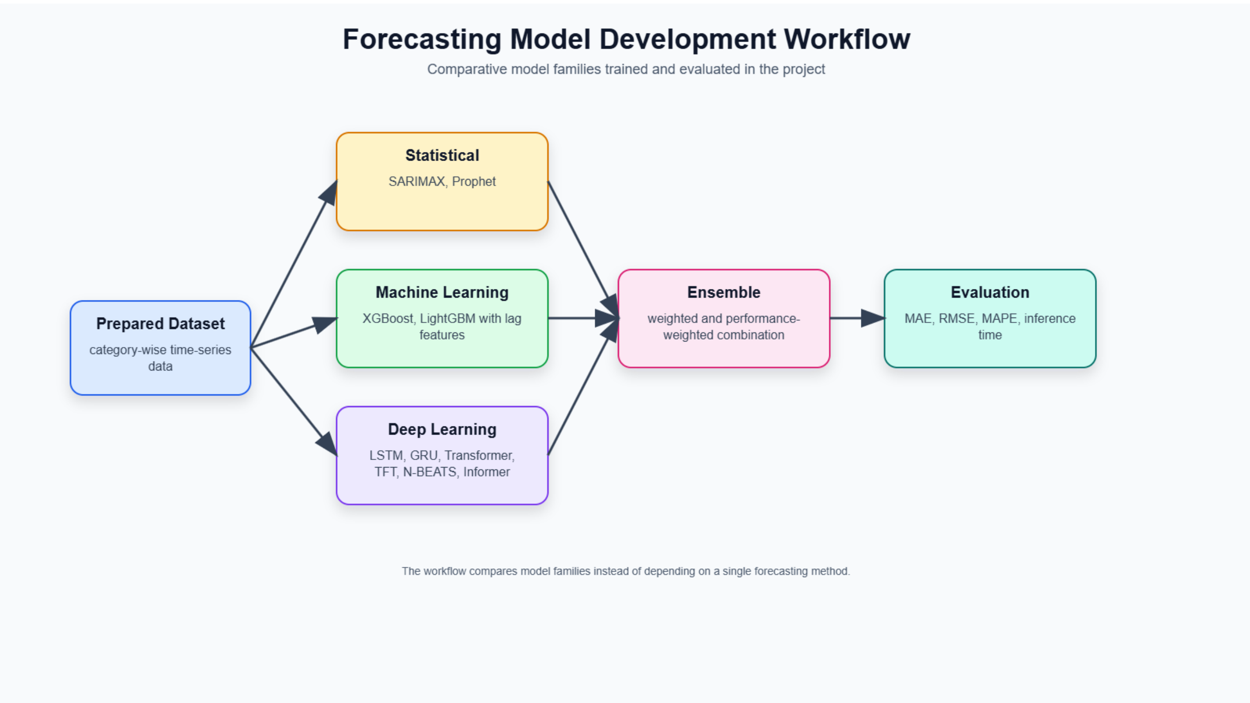

Figure 3:5: Forecasting Model Development Workflow

Forecasting model development is one of the main stages of this research. The purpose of this stage is to train and implement different forecasting models that can predict future pharmaceutical drug sales using historical category-wise sales data. Since drug demand can behave differently across categories, the system does not depend on a single forecasting method. Instead, multiple model types are developed and compared, including statistical models, machine learning models, deep learning models, and ensemble-based approaches.

The dataset is divided into drug categories from C1 to C8, and each category is treated as a separate forecasting problem. This category-wise approach allows the models to learn the individual sales behaviour of each drug group. Some categories may have stable sales patterns, while others may show seasonal, irregular, or trend-based changes. By training models separately for each category, the system can produce more focused predictions instead of using one generalized model for all drug categories.

The model development process begins by preparing the data according to the requirements of each model. For statistical forecasting models such as SARIMAX and Prophet, the data is prepared in a time-series format with date and sales values. For machine learning models such as XGBoost and LightGBM, lag features are created from previous sales values so that the models can learn how past demand affects future sales. For deep learning models such as LSTM, GRU, Transformer, TFT, N-BEATS, and Informer, the data is converted into sequential input windows and scaled using appropriate normalization or standardization techniques.

After preparing the input data, models are trained and saved as artifacts so that they can be reused during forecasting. The system stores trained models in separate folders according to model type and category. This allows the forecasting service to load the required model when a user selects a category, model type, and prediction date from the web interface or API request. The use of saved model artifacts avoids unnecessary retraining during prediction and makes the forecasting process faster and more practical.

The project includes a wide range of forecasting models because each model has different strengths. Statistical models are useful for understanding trend and seasonality. Machine learning models are effective for learning non-linear relationships from lag features. Deep learning models can capture sequential and long-term dependencies from historical sales patterns. Ensemble forecasting is used to improve reliability by combining or comparing multiple model outputs. This multi-model development strategy helps the system identify which approaches are most suitable for pharmaceutical drug sales prediction.

Overall, forecasting model development in this project is designed to support accurate, flexible, and category-wise drug sales prediction. By implementing multiple forecasting techniques and saving trained models for later use, the system becomes capable of comparing model performance, generating future sales forecasts, and supporting better pharmaceutical demand planning.

3.7.1 Statistical Models

Statistical forecasting models were included in this project to provide traditional time-series forecasting support and to compare their performance with machine learning and deep learning methods. These models are useful because they are designed to analyse historical patterns such as trend, seasonality, and temporal dependency.

The first statistical model used in this research is SARIMAX. SARIMAX is an extension of ARIMA that can model seasonal behaviour in time-series data. It uses past values, moving average components, and seasonal patterns to predict future sales. In the context of pharmaceutical drug sales, SARIMAX is useful because some drug categories may show repeating demand behaviour over time. For example, certain medicines may show higher sales during particular seasons or disease periods. SARIMAX helps capture such structured time-based patterns from the historical sales records.

The second statistical model used in the project is Prophet. Prophet is a time-series forecasting model designed to handle trend and seasonality in historical data. In this project, the date column is prepared as ds, and the sales value is prepared as y, which is the required input format for Prophet. Prophet is useful because it can model long-term trend changes and repeated seasonal behaviour. This makes it suitable for analysing category-wise drug sales patterns where demand may change across weekly or monthly periods.

3.7.2 Machine Learning Models

Machine learning models were used to learn non-linear relationships between previous sales values and future demand. Unlike statistical models, machine learning models do not require strict assumptions about the structure of the time series. Instead, they learn patterns directly from input features created from the historical sales data.

The first machine learning model used in this project is XGBoost. XGBoost is a gradient boosting algorithm that builds multiple decision trees and combines them to improve prediction accuracy. In this research, lag features are created from previous sales values and used as inputs to the XGBoost model. This allows the model to learn how recent sales behaviour affects future sales. XGBoost is useful for drug sales forecasting because it can capture non-linear patterns and handle structured tabular features effectively.

The second machine learning model used is LightGBM. LightGBM is another gradient boosting framework that is known for efficient training and good performance on structured data. Similar to XGBoost, LightGBM uses lag-based features generated from the historical sales records. It is included in the system to compare another boosting-based approach with XGBoost and other forecasting models. Since pharmaceutical sales patterns may vary across drug categories, using both XGBoost and LightGBM helps evaluate which machine learning method performs better for different category-wise sales behaviours.

3.7.3 Deep Learning Models

Deep learning models were included because drug sales data is sequential and may contain complex temporal patterns. These models can learn relationships across multiple time steps and are useful when future sales depend on both recent and earlier historical behaviour. Before training deep learning models, the sales data is usually normalized or standardized and converted into sequential input windows.

The LSTM model is used because it is designed to learn long-term dependencies in sequence data. It can remember useful information from previous time steps and use it to predict future sales. This is important when a drug category has demand patterns that continue over several weeks or months.

The GRU model is also used as a recurrent neural network approach. GRU is similar to LSTM but has a simpler structure, which can make training faster while still capturing temporal dependencies. It is useful for comparing whether a lighter recurrent model can perform well for drug sales forecasting.

The Transformer model is included because it uses attention mechanisms to learn relationships between different points in a sequence. Unlike recurrent models, Transformers can focus on important time steps directly. This is useful when sales behaviour depends on both recent and older sales patterns.

The Temporal Fusion Transformer (TFT) is included as an advanced forecasting model designed for time-series prediction. TFT can capture temporal relationships and is useful for complex forecasting tasks where multiple time-related patterns may influence the output.

The N-BEATS model is another deep learning forecasting architecture used in this project. It is designed specifically for time-series forecasting and can learn trend and pattern representations from historical sales data.

The Informer model is included because it is designed for long-sequence time-series forecasting. It is useful when the model needs to capture longer historical dependencies while maintaining efficient computation.

By including these deep learning models, the project can compare traditional recurrent models, attention-based models, and specialized time-series forecasting architectures.

3.7.4 Ensemble Forecasting

Ensemble forecasting is used in this project to improve prediction reliability by combining or comparing outputs from multiple forecasting models. Since each model has different strengths, a single model may not always perform best for every drug category. For example, SARIMAX or Prophet may perform better when the sales pattern has clear seasonality, while XGBoost or LightGBM may perform better when lag-based non-linear relationships are stronger. Deep learning models may be more useful when the sales pattern contains complex sequential behaviour.

The ensemble approach helps reduce dependence on one model by considering multiple forecasting outputs. In the system, model predictions can be compared using evaluation metrics such as MAE, RMSE, and MAPE. The best-performing model can then be identified for each category, or ensemble outputs can be used to provide more stable forecasting behaviour. This is useful in pharmaceutical sales prediction because different drug categories from C1 to C8 may not follow the same demand pattern.

Overall, ensemble forecasting supports better decision-making by allowing the system to evaluate different models and reduce forecasting uncertainty. Instead of assuming one model is always best, the system compares statistical, machine learning, and deep learning models to identify the most suitable forecasting approach for category-wise drug sales prediction.

3.8 MODEL TRAINING PROCESS

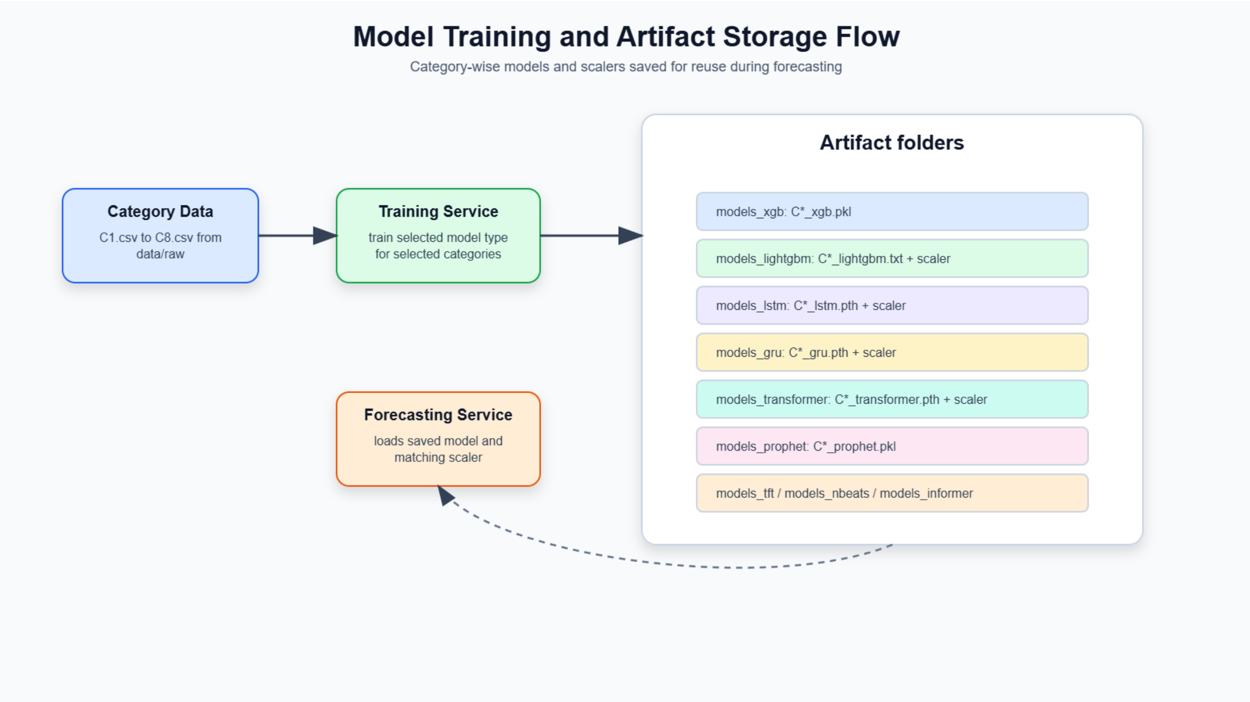

Figure 3:6: Model Training and Artifact Storage Flow

The model training process in this project is designed to train forecasting models separately for each pharmaceutical drug category. The dataset is organized into eight categories, from C1 to C8, and each category is treated as an individual time-series forecasting problem. This category-wise training approach is important because different drug groups may have different sales patterns. Some categories may show stable demand, while others may have seasonal changes, sudden increases, or irregular sales behaviour. By training models separately for each category, the system can learn the unique behaviour of each drug group more accurately.

During training, the relevant category-wise CSV file is loaded from the dataset. The sales data is then pre-processed and transformed according to the selected model type. For machine learning models such as XGBoost and LightGBM, lag features are generated from previous sales values. For deep learning models such as LSTM, GRU, Transformer, TFT, N-BEATS, and Informer, the sales data is converted into sequential input windows and normalized or standardized using suitable scalers. Statistical models such as Prophet and SARIMAX use the time-series structure directly to learn trend and seasonal patterns.

After each model is trained, the trained model is saved as a model artifact. These saved artifacts allow the system to reuse trained models during forecasting without retraining them every time a prediction request is made. The project stores trained models in separate folders based on model type. For example, XGBoost models are stored in the models_xgb folder, LightGBM models are stored in the models_lightgbm folder, LSTM models are stored in the models_lstm folder, GRU models are stored in the models_gru folder, Transformer models are stored in the models_transformer folder, and similar folders are used for other models such as Prophet, TFT, N-BEATS, and Informer. This folder structure makes the system organized and allows the forecasting service to load the correct model based on the selected category and model type.

For models that require scaling, the scaler files are also saved together with the trained model artifacts. This is important because the same scaler used during training must also be used during forecasting. During prediction, the input data is transformed using the saved scaler, and the predicted output is converted back into the original sales scale using inverse transformation. This ensures that the final forecast value shown to the user is meaningful and understandable.

The project also includes a dedicated training service as part of the microservice architecture. This service provides an API endpoint for running model training. Through the training endpoint, a request can specify the model type and the required drug categories. The backend then loads the relevant data, trains the selected model, and saves the generated model artifacts into the correct model folder. This design makes the training process more modular and allows training to be triggered separately from forecasting. As a result, the system can support future retraining when new sales data becomes available.

Overall, the model training process ensures that each drug category has its own trained forecasting models, saved artifacts, and reusable scaler files. This supports efficient forecasting, organized model management, and future system expansion.

3.9 FORECASTING PROCESS

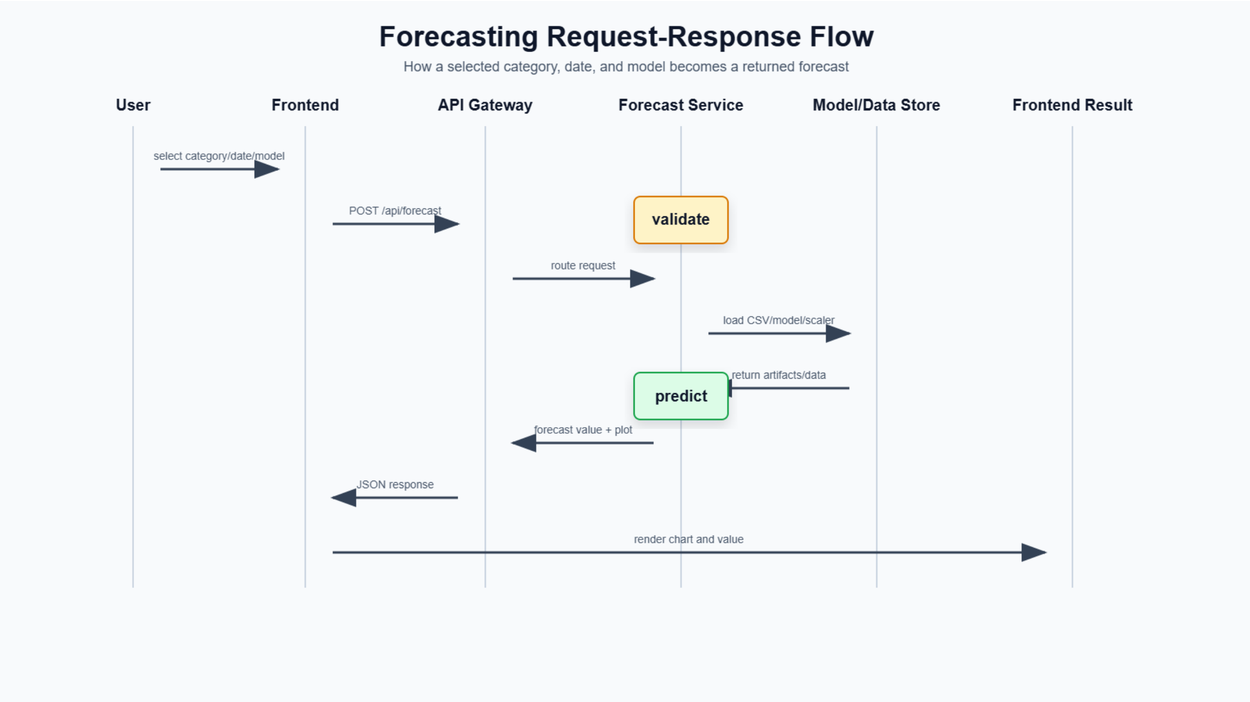

Figure 3:7: Forecasting Request-Response Flow

The forecasting process in this project begins from the user interface, where the user selects the required drug category, prediction date, and forecasting model. The drug category is selected from the available C1 to C8 categories, and the model can be selected from the implemented forecasting methods such as SARIMAX, Prophet, XGBoost, LightGBM, LSTM, GRU, Transformer, or ensemble-based forecasting. The user also enters the date for which the future drug sales value should be predicted. After these inputs are provided, the frontend sends the request to the backend forecasting API.

When the backend receives the forecasting request, it first checks the selected drug category, date, and model type. The system then loads the relevant category-wise sales dataset and identifies whether the requested date belongs to the historical data period or a future period. If the selected date is within the historical data range, the system can return the closest available actual sales value from the dataset. If the selected date is in the future, the system continues with the forecasting process and loads the trained model related to the selected category and model type.

For future forecasting, the backend uses the saved trained model artifacts from the model storage folders. Depending on the model type, the system also loads the corresponding scaler file if scaling was used during training. For example, deep learning models such as LSTM, GRU, Transformer, TFT, N-BEATS, and Informer require normalized or standardized input data. Therefore, the same scaler used during training is applied again during prediction. For machine learning models such as XGBoost and LightGBM, lag features are generated from the most recent historical sales values before prediction.

After the correct model and input format are prepared, the forecasting model predicts the future sales value for the selected category and date. The predicted output is then converted back into the original sales scale if scaling was applied. This ensures that the final forecast value is meaningful to the user. The system also records which model was used to generate the prediction so that the output can clearly show the selected forecasting approach.

Finally, the forecast result is returned to the frontend through the API response. The response includes the predicted sales value, drug category, input date, closest reference date, model used, and plot information. A chart is generated to visualize the historical sales pattern together with the forecasted value. This allows the user to understand the prediction in a visual form rather than only reading a numerical output. Through this process, the system provides a complete forecasting workflow from user input to model prediction and visual result display.

3.10 EXPLAIN ABILITY METHOD

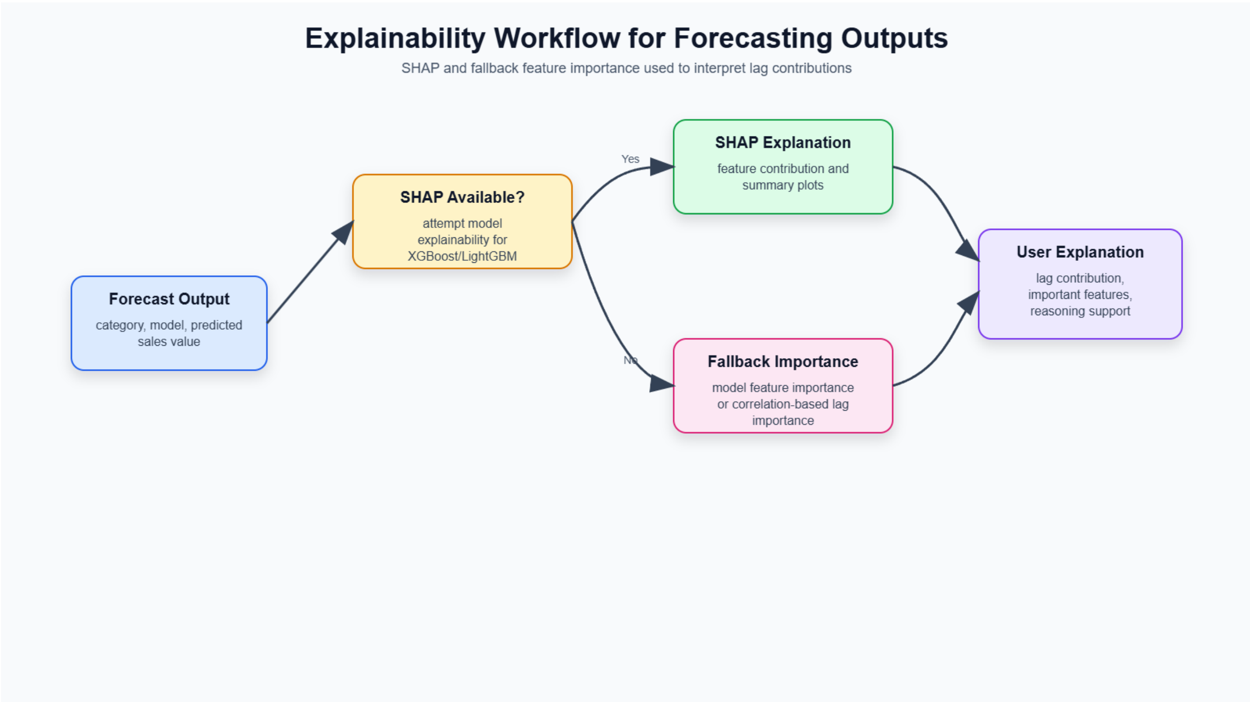

Figure 3:8: Explainability Workflow for Forecasting Outputs

Explainability is included in this project to make the forecasting results more understandable and useful for pharmaceutical decision-making. In drug sales prediction, simply showing a predicted value is not always enough. Pharmacists, distributors, and healthcare planners need to understand why the system predicted a certain sales value before using it for stock planning or purchasing decisions. Therefore, the system includes explainability support to identify which historical sales patterns contributed to the prediction.

The main explainability technique used in this project is SHAP-based analysis. SHAP helps estimate the contribution of each input feature to the model output. In the context of this project, the most important input features are usually lag features created from previous sales values. These lag values represent recent sales behaviour, and SHAP can show which past sales periods had a stronger influence on the final forecast. This allows users to understand whether the prediction is mainly affected by very recent sales, earlier demand behaviour, or a combination of multiple lagged values.

However, because explainability tools can sometimes face compatibility or runtime limitations depending on the model and environment, the project also includes fallback feature importance methods. If SHAP is not available or cannot generate a result for a selected model, the system uses model-based feature importance or historical correlation-based importance to estimate the contribution of lag features. This ensures that the explainability endpoint can still provide useful output even when full SHAP analysis is not possible.

Lag contribution is especially important in pharmaceutical sales forecasting because previous demand is one of the strongest indicators of future demand. For example, if a drug category has shown increasing sales in recent weeks, the forecast may also increase. By showing the importance of previous sales values, the system gives users more confidence in the prediction. This is useful for pharmacy decisions such as stock ordering, inventory balancing, and identifying categories that may need closer monitoring.

Overall, the explainability method improves transparency and trust in the forecasting system. It helps convert the system from a black-box prediction tool into a decision-support platform that provides both forecast values and reasoning behind those values.

3.11 ADVANCED AI MODULES

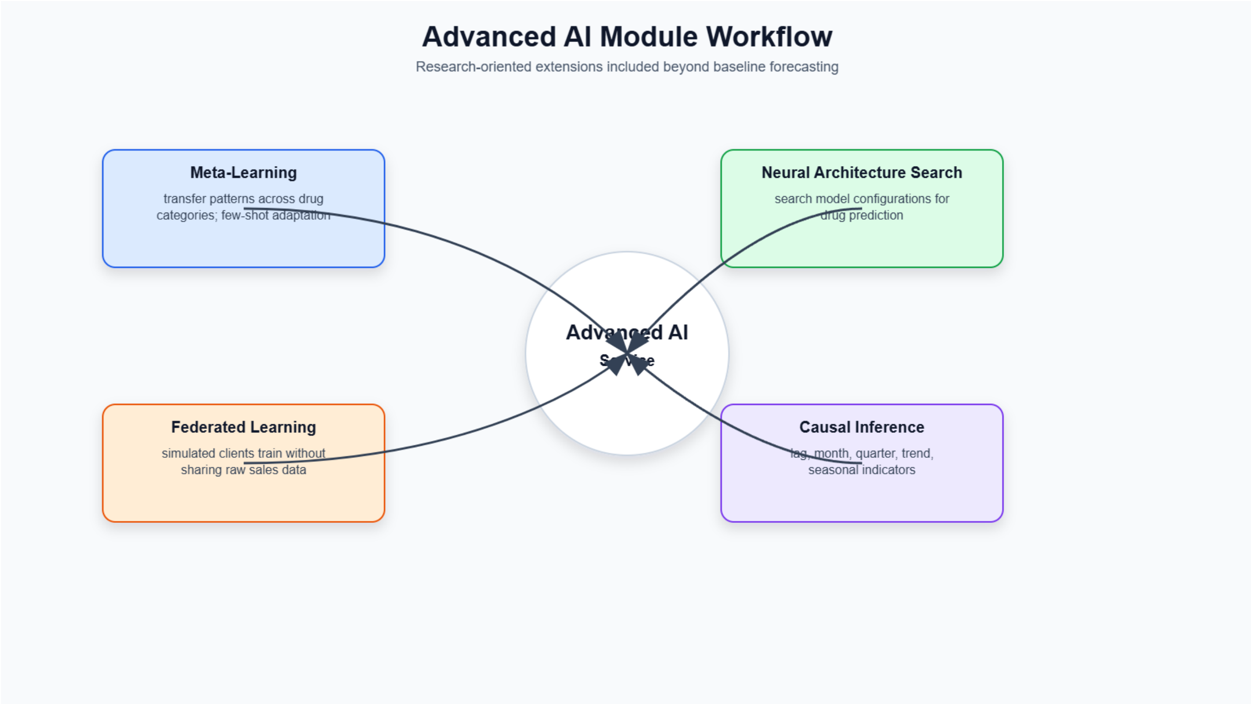

Figure 3:9: Advanced AI Module Workflow

3.11.1 Meta-Learning

Meta-learning is included in this project to explore how knowledge learned from one drug category can support forecasting in another category. Since the dataset is divided into C1 to C8 drug categories, each category may have different sales behavior and data availability. Some categories may have stronger historical patterns, while others may show irregular or limited sales behavior. Meta-learning helps the system learn transferable patterns across categories and adapt to a target category more efficiently. This is useful for future expansion when new drug categories or area-specific datasets are added to the system.

3.11.2 Neural Architecture Search

Neural Architecture Search is used to support automated model structure exploration. In deep learning forecasting, model performance can depend on the number of layers, hidden units, activation functions, and other architectural settings. Manually testing many architectures can be time-consuming. The NAS module explores possible neural network configurations and attempts to identify better model structures for drug sales prediction. This makes the system more flexible and research-oriented because it does not depend only on manually selected architectures.

3.11.3 Federated Learning

Federated learning is included as a privacy-preserving learning approach. In a real pharmaceutical environment, sales data may be stored separately by different pharmacies, hospitals, distributors, or regional branches. These organizations may not want to share raw sales records because of privacy, tax, or business confidentiality concerns. Federated learning allows local clients to train models using their own data and share only model updates instead of raw data. These updates can then be aggregated into a global model. This approach is suitable for future pharmacy-based forecasting systems where multiple locations contribute to model improvement without directly exposing sensitive sales data.

3.11.4 Causal Inference

Causal inference is used to analyse possible relationships between time-based factors and drug sales behaviour. Forecasting models can predict future values, but they do not always explain whether a particular factor may influence changes in demand. The causal inference module creates features such as lagged sales values, week, month, quarter, year, trend, and seasonal indicators. These features are used to explore possible causal relationships and understand how temporal patterns may affect drug sales. This helps the system provide deeper analytical insight in addition to numerical forecasting.

3.12 SYSTEM ARCHITECTURE

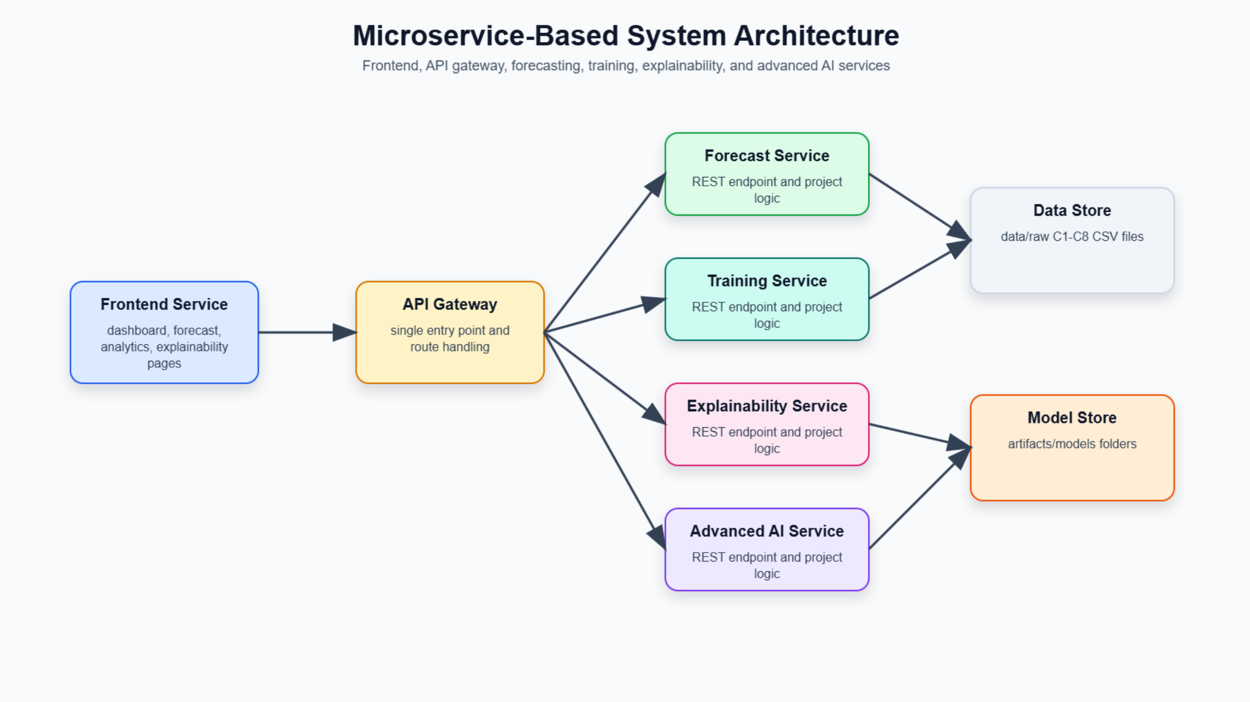

Figure 3:10: Microservice-Based System Architecture

The proposed system is implemented using a microservice-based architecture. This architecture separates the system into different services, where each service has a specific responsibility. This improves maintainability, scalability, and testing because each component can be developed and updated independently.

The frontend service provides the web-based user interface. Through this interface, users can select a drug category, choose a forecasting model, enter a prediction date, and view the forecast result. The frontend also contains pages for analytics, explainability, advanced AI functions, causal analysis, and branch/location visualization.

The API gateway acts as the main communication layer between the frontend and backend services. It receives requests from the frontend and routes them to the correct backend functionality. The API gateway provides endpoints for forecasting, explainability, meta-learning, neural architecture search, federated learning, causal inference, and system status. This gives the frontend a single access point instead of directly calling many different services.

The forecast service is responsible for generating drug sales predictions. It loads the relevant category-wise dataset, identifies the selected forecasting model, loads the trained model artifact, prepares the input features, and returns the predicted sales value. This service handles the main forecasting logic of the system.

The training service is responsible for training forecasting models. It supports training models for selected categories and model types. After training, the generated model artifacts and scaler files are saved in the relevant model folders. This service allows the system to retrain models when new data becomes available.

The explainability service provides interpretation for forecasting outputs. It uses SHAP-based analysis where possible and fallback feature importance methods when SHAP is unavailable. This service helps users understand which lag features or historical sales values influenced the prediction.

The advanced AI service contains the research-oriented AI modules of the system. It supports neural architecture search, federated learning, and causal inference operations. These modules extend the system beyond basic forecasting and provide future-ready capabilities for more intelligent pharmaceutical demand prediction.

Overall, the system architecture supports a complete workflow from user input to forecasting, explanation, model training, advanced analysis, and deployment. By separating these responsibilities into microservices, the project becomes more organized, scalable, and suitable for future real-world implementation.

3.13 API DESIGN

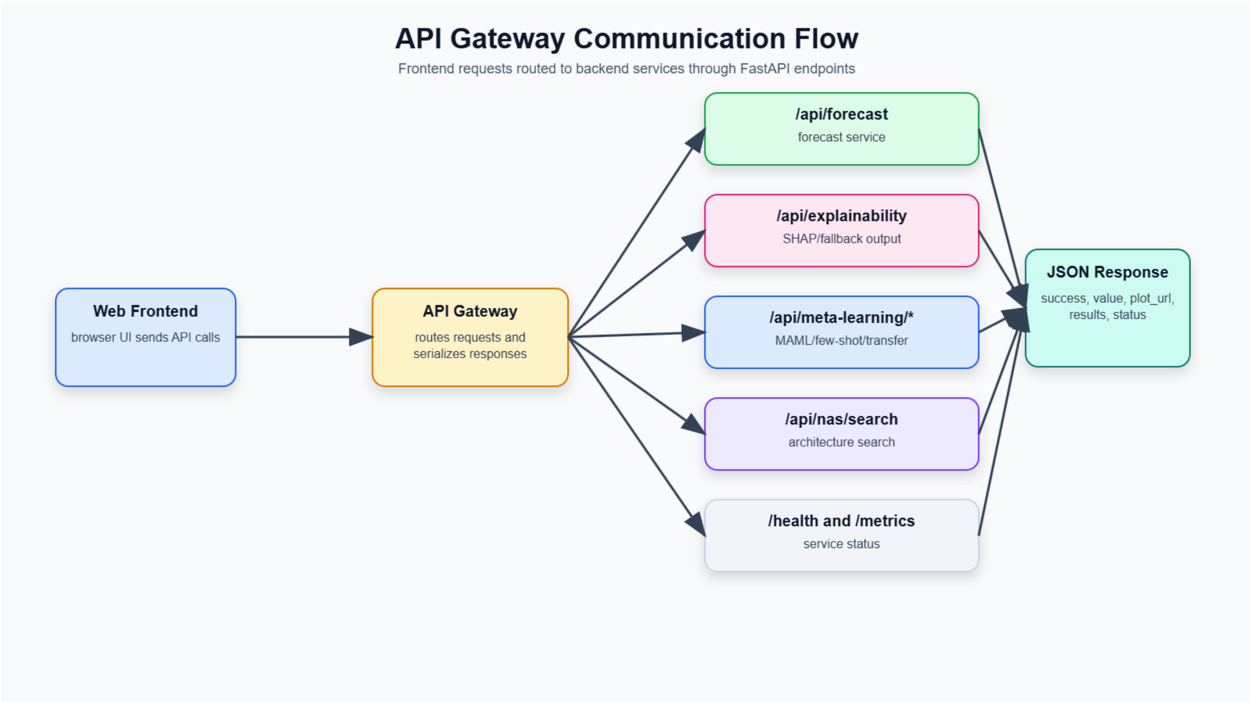

Figure 3:11: API Gateway Communication Flow

The system provides REST API endpoints to allow communication between the frontend interface and backend services. The API layer is important because it makes the forecasting system accessible through both the web application and programmatic requests. The API gateway acts as the main access point and routes requests to the relevant backend functionality.

The forecast endpoint is used to generate drug sales predictions. It accepts inputs such as drug category, prediction date, and selected model type. After receiving the request, the backend loads the required category data and trained model, performs forecasting, and returns the predicted sales value, model used, input date, reference date, and plot information.

The training endpoint is used to train forecasting models. It allows the system to train selected model types for selected drug categories. After training, the generated model files and scaler files are saved in the correct artifact folders. This endpoint supports future retraining when new sales data becomes available.

The explainability endpoint provides interpretation for prediction results. It uses SHAP-based analysis where possible and fallback feature importance methods when SHAP is unavailable. This endpoint returns information about important lag features and their contribution to the forecast.

The advanced AI endpoints are used for research-oriented functions such as meta-learning, neural architecture search, federated learning, and causal inference. These endpoints allow the system to run advanced model adaptation, automated architecture exploration, privacy-preserving training, and causal analysis.

The system also includes health and metrics endpoints. Health endpoints are used to check whether each service is running correctly. Metrics endpoints expose service-level monitoring information such as request counts and uptime. These endpoints are useful for debugging, testing, and monitoring the system during deployment.

3.14 WEB APPLICATION DESIGN

The web application provides a user-friendly interface for interacting with the drug sales prediction system. Instead of using only command-line scripts or API calls, users can access the system through a browser-based frontend. This makes the system easier to use for non-technical users such as pharmacy staff, healthcare planners, and decision-makers.

The dashboard acts as the main entry point of the system. It provides navigation to the major functions of the application, including forecasting, analytics, explainability, advanced AI features, causal analysis, and branch/location-related views.

The forecast page allows users to generate predictions. On this page, the user selects a drug category from C1 to C8, chooses a forecasting model, and enters the prediction date. After submitting the request, the system displays the predicted sales value, selected model, forecast date, and a chart showing the forecast result.

The model selection feature allows users to choose between different forecasting approaches such as SARIMAX, Prophet, XGBoost, LightGBM, LSTM, GRU, Transformer, and ensemble-based forecasting. This makes it possible to compare how different models behave for the same drug category and prediction date.

The explainability page provides interpretation outputs for forecasting results. It helps users understand which previous sales values or lag features influenced the prediction. This supports better trust and decision-making because users can see the reasoning behind the forecast.

The advanced AI pages provide access to research-oriented modules such as meta-learning, neural architecture search, federated learning, and causal inference. These pages allow users to interact with advanced analysis functions and explore deeper AI-based forecasting support.

3.15 DEPLOYMENT AND MONITORING

The project includes deployment and monitoring support to make the system more practical and scalable. Since the system is designed using a microservice architecture, deployment tools are used to run multiple services consistently.

Docker is used to containerize the application. Containerization allows the system and its dependencies to be packaged together so that the services can run in a consistent environment. This reduces issues caused by differences between development and deployment machines.

Docker Compose is used to run multiple services together during local deployment. It defines the services, ports, environment variables, and service connections required for the system. This makes it easier to start the API gateway, frontend service, forecasting service, training service, explainability service, advanced AI service, and supporting services together.

Kubernetes configuration is also included to support future cloud or production deployment. Kubernetes helps manage multiple containers, service scaling, networking, and deployment configuration. The project includes Kubernetes manifests for the different microservices, which makes the system more suitable for scalable deployment.

Prometheus metrics are included for monitoring service behaviour. Each service exposes a metrics endpoint that can provide information such as service uptime, request count, request duration, and service health. These metrics are useful for monitoring the reliability and performance of the system.

Health checks are also included for each service. A health endpoint returns whether the service is running correctly. This is useful during testing and deployment because it allows developers or deployment platforms to confirm that each service is active and responsive.

3.16 EVALUATION METHOD

The evaluation method of this project includes both forecasting evaluation and system-level validation. Since the project is not only a model experiment but also a complete web-based microservice system, both prediction performance and endpoint reliability are considered.

For forecasting evaluation, common error metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and Mean Absolute Percentage Error (MAPE) are used. MAE measures the average absolute difference between actual and predicted sales values. RMSE gives more weight to larger errors and is useful for identifying models that make large prediction mistakes. MAPE shows the prediction error as a percentage, making it easier to understand the relative forecasting accuracy.

These metrics are useful for comparing different models such as SARIMAX, Prophet, XGBoost, LightGBM, LSTM, GRU, Transformer, and ensemble models. Since different drug categories may show different behaviour, evaluation is performed category-wise where possible. This helps identify which models are more suitable for different categories of pharmaceutical sales.

In addition to model performance, endpoint testing was used for system validation. The backend services were tested by sending requests to the health, forecasting, training, explainability, advanced AI, and API gateway endpoints. This confirms that the services respond correctly and that the system workflow functions as expected. During final testing, all tested endpoints successfully passed, showing that the API layer and service communication were working properly.

System validation also includes checking whether the frontend can communicate with the backend, whether forecast results are returned correctly, whether trained models can be loaded, whether explanation outputs are generated, and whether monitoring endpoints are available. This combined evaluation method ensures that the project is assessed both as a forecasting model system and as a practical software implementation.

Chapter 4: Results and Analysis

This chapter presents the experimental results of the implemented drug sales prediction system. The experiments have been carried out in order to examine the forecasting ability of the various model groups applied within this project. Since the dataset is organized into eight pharmaceutical drug categories, C1 to C8, each category was treated as an independent time-series forecasting problem. This chapter aims at exploring the various behaviours of forecasting models in certain conditions and determining whether the chosen forecasting models are suitable for forecasting sales in pharmaceuticals.

The experiments were evaluated using three main forecasting error metrics: namely, Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and Mean Absolute Percentage Error (MAPE). MAE measures the average absolute difference between actual and predicted sales values. RMSE gives higher penalty to larger forecasting errors. MAPE expresses the error as a percentage, which helps compare relative forecasting performance. In addition to these metrics, inference time was also considered because the final system is designed as a web-based forecasting application where users expect fast prediction responses.

4.1 STATISTICAL FORECASTING APPROACH

Statistical forecasting models were used as the first major group of experiments in this research. These models are commonly used for time-series forecasting because they are designed to learn patterns from historical time-ordered data. For this particular project, statistical models were used mainly to learn about the trends, seasonality, and baseline forecasting of pharmaceutical drug sales data.

The statistical forecasting approach included two models: SARIMAX and Prophet.The SARIMAX model was chosen because of its ability to account for the autoregressive component, moving average, and seasonal time-series. Prophet was chosen because of its simplicity in handling the time-series data with trend and seasonality using a date value combination. These models were run on the C1 to C8 drug sales categories using the same forecasting evaluation metrics as the others.

4.1.1 Experimental Setup for Drug Sales Time-Series Forecasting

The statistical forecasting experiments were conducted using the category-wise drug sales dataset. Each drug category was stored as a separate time-series dataset, where the date column represented the time period and the sales column represented the recorded sales value. The models were trained and tested separately for each category so that the unique sales behaviour of each drug group could be analysed.

For this experiment, the chronological order of the dataset was preserved. This is important because time-series forecasting must use past data to predict future values. The data was not randomly shuffled because doing so would create data leakage and produce unrealistic evaluation results. Earlier records were used for model learning, while later records were used to evaluate forecasting performance.

Statistical models were assessed in relation to each of the eight drug categories. The purpose was to understand if the traditional approaches of time series forecasting can be applied to predicting pharmaceutical sales. The results were compared by means of MAE, RMSE, MAPE and average inference time.

Table 4-1: Category-Wise Performance of SARIMAX and Prophet

Category	SARIMAX MAE	SARIMAX RMSE	SARIMAX MAPE (%)	Prophet MAE	Prophet RMSE	Prophet MAPE (%)
C1	19.1074	22.5140	40.5356	10.5368	13.5162	21.1628
C2	18.5494	20.9198	51.5160	13.1088	14.6696	56.3744
C3	17.6812	22.8587	44.2430	10.2570	12.2604	41.5235
C4	96.2937	132.9443	29.0270	143.8519	165.7533	68.3645
C5	29.1626	32.3775	40.3108	32.7109	35.2496	52.8368
C6	5.3813	6.7422	67.5257	4.8752	5.6446	107.3593
C7	40.0671	46.6166	113.3336	31.1265	38.4060	143.5574
C8	17.7142	22.3134	62.0024	14.0789	15.9109	213.0493

4.1.1.1 SARIMAX Forecasting for Category-Wise Drug Sales

SARIMAX was implemented as one of the main statistical forecasting models in this project. SARIMAX is an extension of ARIMA that supports seasonal components. This makes it useful for datasets where sales values may repeat in weekly, monthly, or seasonal patterns. In pharmaceutical sales forecasting, such patterns may occur due to seasonal illnesses, changes in patient demand, availability of medicines, or repeated purchasing behaviour.

In this project, SARIMAX was applied separately to each drug category from C1 to C8. The model attempted to learn the relationship between previous sales values and future sales values using autoregressive and moving average components. Since SARIMAX is a statistical model, it does not require lag features in the same way as XGBoost or LightGBM. Instead, it models the time-series structure directly.

The SARIMAX model produced mixed results across the different categories. It performed reasonably well for some categories, especially where the data had more structured time-series behaviour. However, it was less effective for categories with stronger non-linear or irregular patterns. Based on the final results, SARIMAX achieved an average MAE of 30.4946, average RMSE of 38.4108, and average MAPE of 56.0618 across all categories. The average inference time was 0.1498 seconds, showing that SARIMAX was relatively fast compared with Prophet.

The results show that SARIMAX is useful as a baseline statistical forecasting model. It provides a traditional comparison point for evaluating whether more advanced models improve forecasting accuracy. However, SARIMAX alone was not the best overall model in the project because machine learning and deep learning models achieved better results in several categories.

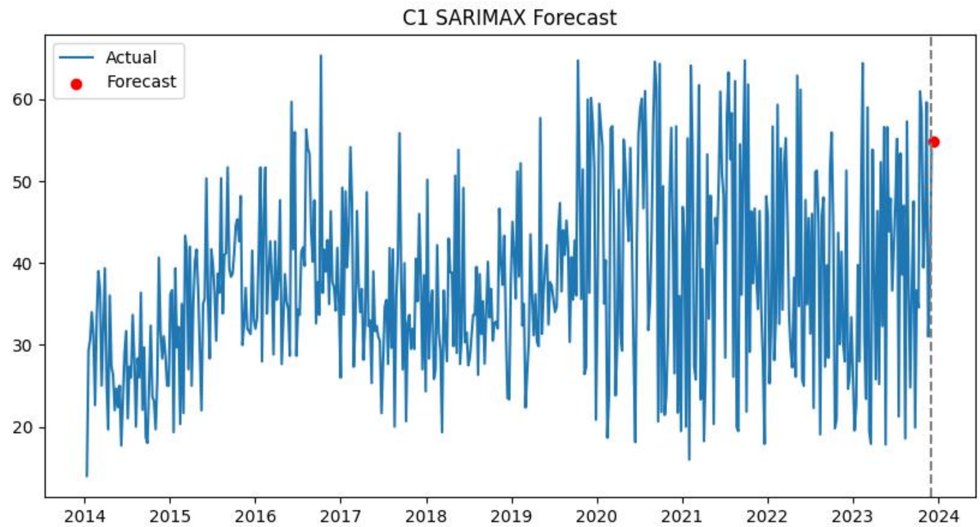

Figure 4:1: C1 Drug Sales Forecast Using the SARIMAX Model

4.1.1.2 Prophet Forecasting with Trend and Seasonality

Prophet was included as one of the statistical forecasting models in this research. Prophet models time-series data by identifying trend and seasonal patterns. Before training the model, the drug-sales data was converted into Prophet's required input format. The date column was renamed ds, representing the timestamp of each observation, while the sales column was renamed y, representing the drug-sales value to be forecast. Prophet then used these historical observations to estimate future drug-sales patterns.

Prophet was effective in this study because pharmaceutical sales may exhibit seasonality and trend-based characteristics. For instance, there may be a higher sales demand for certain types of drugs based on factors like respiratory diseases, allergy seasons, rainy seasons, or frequent customer purchase patterns. Prophet is designed to identify such trends and incorporate them into forecasting models.

However, in the experiment, Prophet performed effectively only in some categories individually. Prophet provided better results in terms of MAE when compared to SARIMAX in some categories such as C1, C2, C3, C6, C7, and C8. It is evident that Prophet was successful in identifying important trends and seasonality in many of the drug categories, especially C4, where the error was much higher than SARIMAX. Because of this, Prophet's overall average MAE and MAPE were weaker than SARIMAX.

Across all categories, Prophet achieved an average MAE of 32.5682, average RMSE of 37.6763, and average MAPE of 88.0285. Its average inference time was 0.8725 seconds, making it slower than SARIMAX and most other tested models. These results show that Prophet is useful for trend and seasonality analysis, but it may not always be the most efficient or accurate model for pharmaceutical sales forecasting.

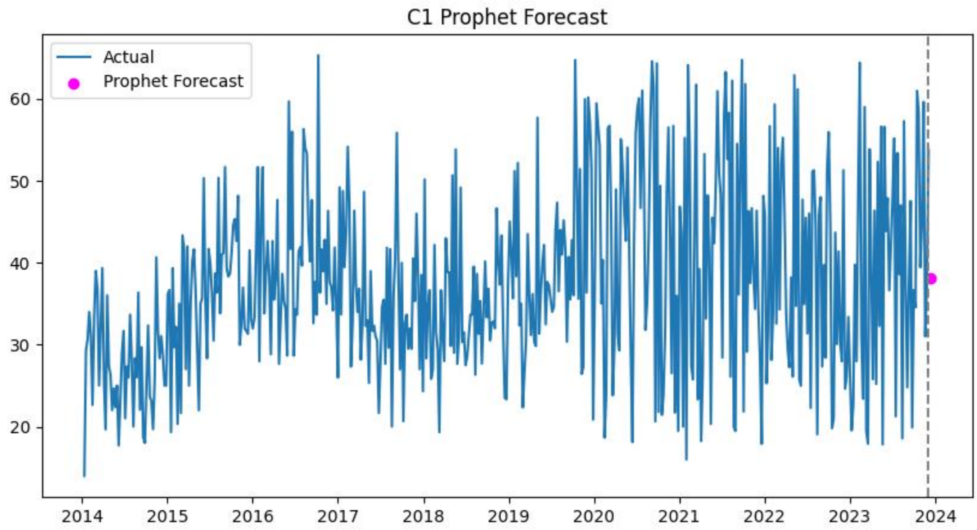

Figure 4:2: C1 Drug Sales Forecast Using the Prophet Model

4.1.1.3 Historical Sales Forecasting Baseline

In addition to future forecasting, the system also includes a historical sales baseline behaviour. If the user selects a date which already exists in the historical data set, then instead of generating a future forecast, the forecasting service will return the closest available historical sales value from the data set. This behaviour is important because the system supports both historical lookup and future forecasting.

For example, if a user selects a past date for category C1, the backend identifies the closest matching date from the dataset and returns the actual recorded sales value. In this case, the model used is identified as Historical Data instead of a forecasting model. This feature prevents unnecessary prediction when actual data is already available.

Such a baseline is important in ensuring the proper working of the system. First, it proves that the backend is able to load data set by category and parse dates and determine whether a given date is historical or future.

4.1.1.4 Statistical Model Comparison

The comparison between SARIMAX and Prophet shows that both models have strengths and limitations. SARIMAX performed better overall in terms of average MAE, MAPE, and inference time. Prophet achieved slightly better average RMSE, but its MAPE and inference time were weaker. Prophet also performed better than SARIMAX in several individual categories, but its poor performance in some categories increased its overall average error.

A summary of the statistical model performance is shown below.

Table 4-2: Average Performance Comparison of Statistical Forecasting Models

Model	Average MAE	Average RMSE	Average MAPE (%)	Average Inference Time (s)
SARIMAX	30.4946	38.4108	56.0618	0.1498
Prophet	32.5682	37.6763	88.0285	0.8725

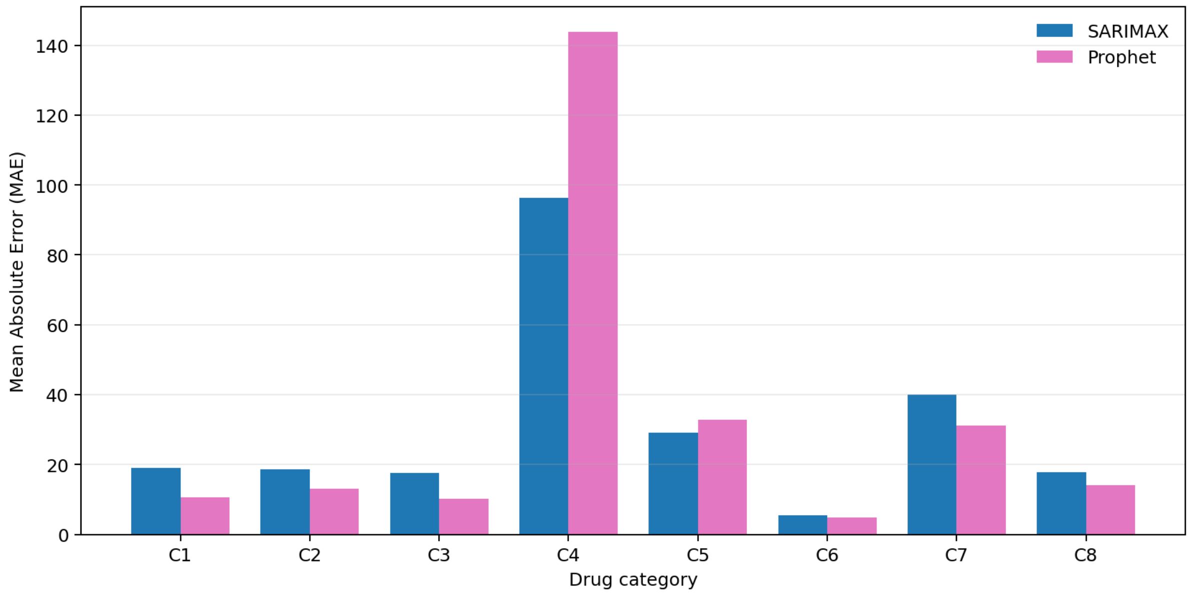

Figure 4:3: Category-Wise MAE Comparison of SARIMAX and Prophet

Based on these results, SARIMAX can be considered the better statistical model for this project overall because it had lower MAE, lower MAPE, and faster inference time. Prophet was still valuable because it performed well in several individual categories and provided trend-seasonality-based forecasting. However, neither SARIMAX nor Prophet achieved the best overall performance when compared with the full set of machine learning and deep learning models.

Therefore, the statistical forecasting experiments show that traditional time-series models are useful as baseline models and for understanding trend and seasonality. However, for category-wise pharmaceutical drug sales prediction, more flexible models such as LightGBM, LSTM, GRU, and ensemble-based methods are required to improve forecasting accuracy across different sales patterns.

4.2 MACHINE LEARNING APPROACH

The second group of experiments focused on machine learning-based forecasting models. In this project, machine learning models were used to learn the relationship between previous drug sales values and future sales demand. Unlike traditional statistical models, machine learning models do not require strict assumptions about the structure of the time-series data. Instead, they learn patterns directly from engineered input features.

In relation to the machine learning algorithm, time-series sales data have been pre-processed in order to transform it into a form for supervised learning task. In this case, lag features which correspond to previous sales values and can be considered input values for predicting the next value of sales were created. This method is suitable for models such as XGBoost and LightGBM because they work effectively with structured tabular data. The main idea behind this approach is that future drug sales are strongly influenced by recent historical sales behaviour.

Two machine learning models were evaluated in this research: XGBoost and LightGBM. Both models are gradient boosting-based algorithms, but they use different training strategies and optimization methods. These models were tested across all drug categories from C1 to C8 and evaluated using MAE, RMSE, MAPE, and inference time.

4.2.1 Experiments with Lag-Based Machine Learning Models

The lag-based machine learning experiments were conducted by transforming each category-wise time-series dataset into input-output pairs. A fixed number of previous sales values were used as input features, and the next sales value was used as the prediction target. This allowed the models to learn how recent sales behaviour affects future demand.

For example, when predicting the next sales value for a category such as C1, the model uses previous sales values from earlier time steps as input. These lag values help the model identify whether the sales trend is increasing, decreasing, or remaining stable. The same process was applied separately for each drug category so that the models could learn category-specific demand behaviour.

The machine learning models were trained and saved as model artifacts. During forecasting, the backend loads the selected model and generates the prediction using the most recent sales values from the dataset. This makes the machine learning approach practical for real-time prediction through the web interface and API.

Table 4-3: Category-Wise Performance of XGBoost and LightGBM

Category	XGBoost MAE	XGBoost RMSE	XGBoost MAPE (%)	LightGBM MAE	LightGBM RMSE	LightGBM MAPE (%)
C1	10.9012	12.3420	23.2977	10.2061	13.5706	19.6222
C2	13.8605	15.2164	44.9367	14.0368	15.9747	57.4618
C3	16.9471	18.7485	60.3954	13.0881	15.3859	44.3339
C4	85.4242	97.5089	35.7980	72.6065	92.3468	29.3020
C5	29.0716	34.7092	44.6566	26.9909	32.4470	38.3756
C6	4.3514	5.2600	78.6966	3.4860	4.2950	68.6662
C7	26.0462	33.0924	106.2338	32.4237	39.8748	90.2034
C8	17.1414	22.8859	297.2098	14.1443	17.4359	243.4210

4.2.1.1 XGBoost Forecasting Using Lag Features

XGBoost was implemented as one of the main machine learning models in this project. XGBoost is a gradient boosting algorithm that builds multiple decision trees sequentially. Each new tree attempts to correct the errors made by the previous trees. This makes XGBoost powerful for regression tasks such as sales forecasting.

In this project, XGBoost was trained using lag features generated from historical drug sales values. The model used previous sales records as input and learned to predict the next sales value. Since XGBoost can capture non-linear relationships, it is useful for pharmaceutical sales data where demand may not follow a simple linear pattern.

The experimental results show that XGBoost performed better than the statistical models in terms of average MAE and RMSE. Across all categories, XGBoost achieved an average MAE of 25.4679, average RMSE of 29.9704, and average MAPE of 86.4031. Although its MAPE was relatively high due to category-level percentage error variation, it achieved strong absolute error performance.

One of the main advantages of XGBoost in this project was its speed. It recorded an average inference time of 0.0343 seconds, making it the fastest model among the tested forecasting models. This is important for a web-based forecasting system because users expect quick responses after submitting prediction requests. Therefore, XGBoost is useful when fast prediction is required.

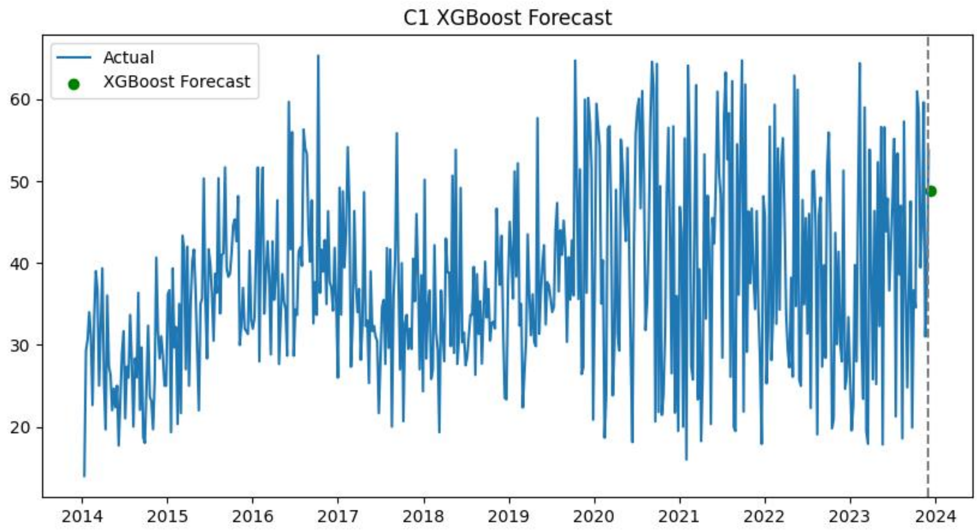

Figure 4:4: C1 Drug Sales Forecast Using the XGBoost Model

4.2.1.2 LightGBM Forecasting Using Normalized Lag Features

LightGBM was also implemented as a machine learning forecasting model. Similar to XGBoost, LightGBM is a gradient boosting framework that uses decision trees. However, LightGBM is designed to be efficient and fast, especially for structured data. In this project, LightGBM used lag-based features created from previous sales values.

Before training LightGBM, the sales values were scaled using MinMaxScaler. This helped normalize the numerical range of the data and allowed the model to train more consistently. After prediction, the forecasted values were converted back to the original sales scale using inverse transformation. This ensured that the final output shown to the user was meaningful and understandable.

LightGBM produced the best overall MAE and RMSE results among all evaluated models. Across all categories, LightGBM achieved an average MAE of 23.3728, average RMSE of 28.9163, and average MAPE of 73.9233. It also maintained a good average inference time of 0.1074 seconds. These results show that LightGBM was highly effective for category-wise drug sales forecasting in this project.

LightGBM performed especially well because it could learn non-linear relationships from lag features while maintaining efficient computation. Since drug sales patterns can differ across categories, LightGBM's ability to model structured historical features made it suitable for this forecasting task.

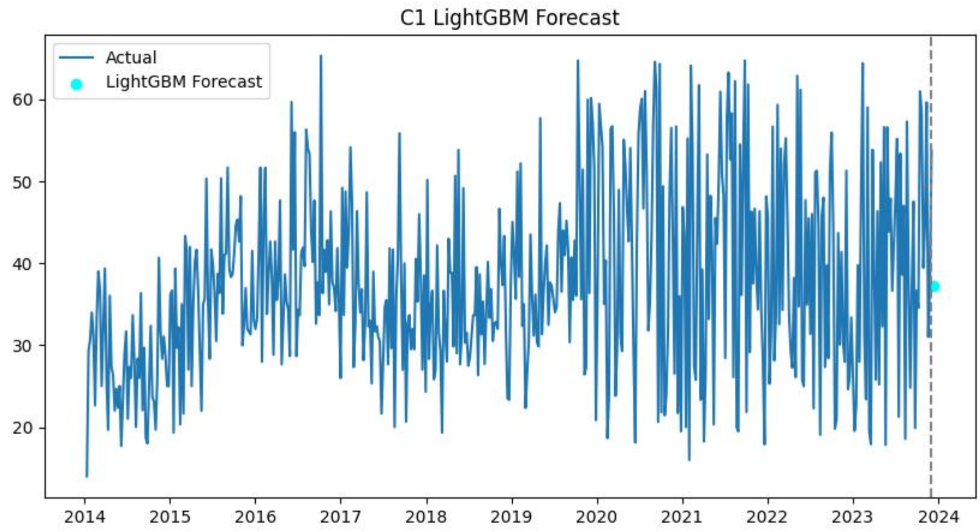

Figure 4:5: C1 Drug Sales Forecast Using the LightGBM Model

4.2.1.3 Feature Importance Analysis for Lag-Based Models

Feature importance analysis was used to understand which lag features contributed most to the predictions made by machine learning models. Since XGBoost and LightGBM use previous sales values as input features, feature importance can show which past sales periods had the strongest influence on the predicted future sales value.

This is important in pharmaceutical sales forecasting because decision-makers need to understand the reason behind a forecast. If the model gives high importance to recent lag values, it means the prediction is strongly influenced by recent demand changes. If older lag values have higher importance, it may indicate a longer repeating pattern in the sales data.

The project includes explainability support through SHAP-based methods and fallback feature importance analysis. When SHAP is available, it can provide more detailed explanations of how each feature affects the model output. When SHAP is not available, the fallback method uses model feature importance or correlation-based importance. This ensures that the system can still provide explanation output for lag-based models.

Feature importance analysis helps improve trust in the forecasting system. Instead of only showing a numerical prediction, the system can show which historical sales values influenced the prediction. This is useful for pharmacists and distributors when making stock-related decisions.

4.2.1.4 Machine Learning Model Performance Comparison

The machine learning experiments showed that both XGBoost and LightGBM were effective for drug sales forecasting. Compared with the statistical models, both machine learning models achieved better average MAE and RMSE. This shows that lag-based machine learning models were better at capturing non-linear patterns in the category-wise sales data.

A summary of the machine learning model performance is shown below.

Table 4-4: Average Performance Comparison of Machine Learning Models

Model	Average MAE	Average RMSE	Average MAPE (%)	Average Inference Time (s)
XGBoost	25.4679	29.9704	86.4031	0.0343
LightGBM	23.3728	28.9163	73.9233	0.1074

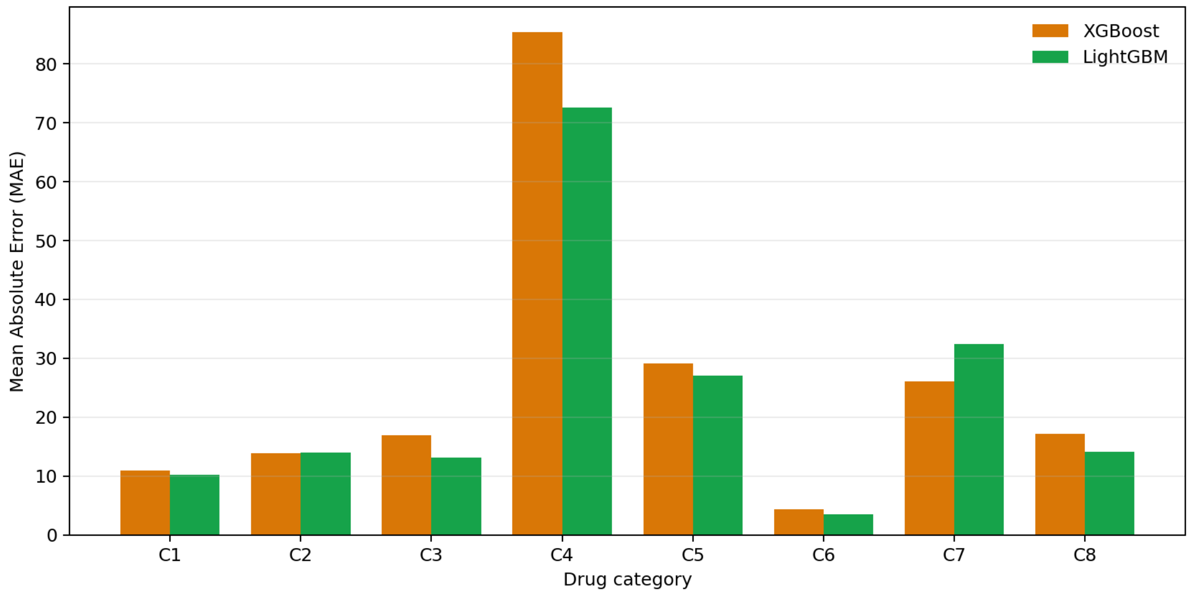

Figure 4:6: Category-Wise MAE Comparison of XGBoost and LightGBM

Based on these results, LightGBM achieved the best overall forecasting accuracy among the machine learning models. It had the lowest MAE, RMSE, and MAPE when compared with XGBoost. However, XGBoost was significantly faster in terms of inference time.

Therefore, LightGBM can be considered the best machine learning model when accuracy is the main priority, while XGBoost is more suitable when fast response time is required. Since the final system is designed as a web-based forecasting platform, both models are valuable. LightGBM supports better prediction accuracy, and XGBoost supports faster forecasting performance.

4.3 DEEP LEARNING APPROACH

The third group of experiments focused on deep learning-based forecasting models. Deep learning models were included because pharmaceutical sales data is sequential, and future demand may depend on patterns that appear across multiple previous time steps. Unlike machine learning models such as XGBoost and LightGBM, which use fixed lag features, deep learning models can learn from sequential input windows and identify temporal relationships in the sales data.

In this project, several deep learning models were created and tested, namely LSTM, GRU, Transformer, Temporal Fusion Transformer, N-BEATS, and Informer. They were chosen due to being used widely for time series forecasting as well as learning different types of sequential patterns. LSTM and GRU models are recurrent neural networks, transformer models are based on attention mechanisms, and N-BEATS and Informer are specifically designed architectures for time series forecasting.

Before training the deep learning models, the sales data was normalized or standardized depending on the model. The dataset was converted into sequential windows, where a sequence of past sales values was used as input to predict a future sales value. This approach allowed the models to learn sales behaviour over time rather than depending only on individual lag features.

4.3.1 Experiments with Sequential Deep Learning Models

Sequential Deep Learning experiments were conducted separately for each drug category C1 to C8. Each category was considered as separate time-series forecasting problem. The input was created as a sequence of previous sales figures, and the deep learning model tried to forecast the next one.

This method is suitable for pharmaceutical sales forecasting because medicine demand may depend on short-term and long-term sales behaviour. For example, if a category shows gradually increasing demand over several weeks, recurrent and attention-based models can learn this trend from the sequence. Similarly, if a category has repeated demand behaviour, deep learning models can capture these temporal dependencies.

The deep learning models were evaluated using MAE, RMSE, MAPE, and inference time. Their results were compared with statistical and machine learning models to identify whether sequential learning improved forecasting performance.

Table 4-5: Category-Wise MAE of the Main Deep Learning Models

Category	LSTM MAE	GRU MAE	Transformer MAE
C1	10.0733	9.8758	11.6626
C2	13.4311	13.4509	14.6117
C3	12.1506	12.5937	11.9895
C4	85.9345	85.5098	113.5993
C5	24.8986	24.9071	24.9922
C6	3.1634	3.1920	3.3981
C7	28.8227	29.1761	35.8801
C8	13.8154	14.1779	14.9933

4.3.1.1 LSTM Forecasting with Sequential Sales Windows

LSTM was implemented as one of the main deep learning models in this project. LSTM is a recurrent neural network architecture designed to learn long-term dependencies from sequential data. It uses memory cells and gates to decide which information should be retained or forgotten across time steps. This makes it suitable for time-series forecasting problems where previous sales behaviour can influence future demand.

In this case, LSTM was trained on sales windows obtained from drug sales per category. The values of sales were normalized using MinMaxScaler and the trained model was stored along with the corresponding scaler file. While doing the forecasting, the scaler file was used to normalize the input data as well as to denormalize the predicted value.

The results of LSTM were excellent in forecasting. For all the drugs, it provided an average MAE of 24.0362, average RMSE of 29.5830, and average MAPE of 62.6915. It also showed an average inference time of 0.1260 seconds. These results indicate that LSTM is one of the best deep learning models in the project.

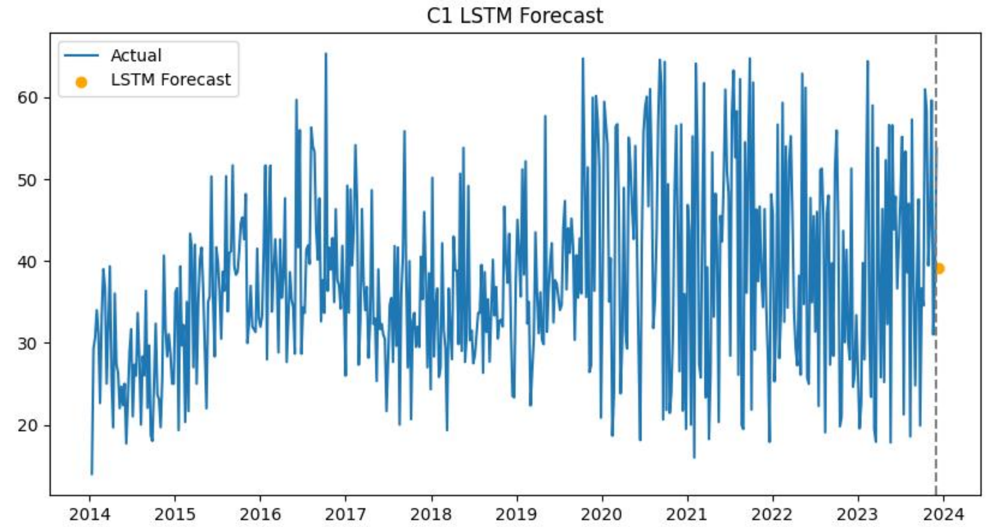

Figure 4:7: C1 Drug Sales Forecast Using the LSTM Model

4.3.1.2 GRU Forecasting with Sequential Sales Windows

GRU was also implemented as a recurrent neural network model. GRU is similar to LSTM but has a simpler structure with fewer gates. Because of this, GRU can sometimes train faster while still learning important temporal dependencies. It is useful for comparing whether a lighter recurrent model can achieve performance similar to LSTM.

In this project, GRU used sequential sales windows as input. The data was normalized using MinMaxScaler, and the trained GRU models were saved with their corresponding scaler files. During forecasting, the backend loads the trained model and scaler to generate future sales predictions.

The GRU model achieved an average MAE of 24.1104, average RMSE of 29.7559, and average MAPE of 63.9476. Its average inference time was 0.1067 seconds, which was slightly faster than LSTM. These results show that GRU performed very close to LSTM while being faster, making it a practical deep learning option for the forecasting system.

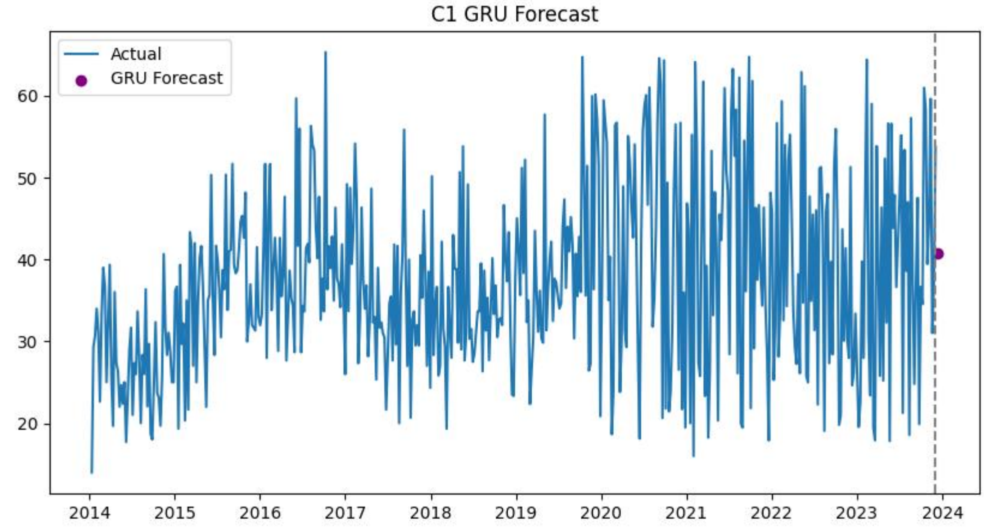

Figure 4:8: C1 Drug Sales Forecast Using the GRU Model

4.3.1.3 Transformer-Based Forecasting

The Transformer model was included to evaluate attention-based forecasting. Unlike LSTM and GRU, which process sequences recurrently, Transformer models use attention mechanisms to identify important relationships between different time steps. This allows the model to focus on the most relevant parts of the historical sequence when making a prediction.

In this project, the Transformer model was trained using normalized sequential sales data. The model attempted to learn which previous sales values were most important for predicting future sales. This is useful for pharmaceutical sales data because some demand patterns may depend not only on the most recent sales values but also on earlier historical behaviour.

The Transformer model achieved an average MAE of 28.8908, average RMSE of 36.5221, and average MAPE of 53.6691. Although its MAE and RMSE were higher than LSTM and GRU, it achieved the best MAPE among the tested models. This indicates that Transformer-based forecasting handled relative percentage error better than other approaches in this experiment.

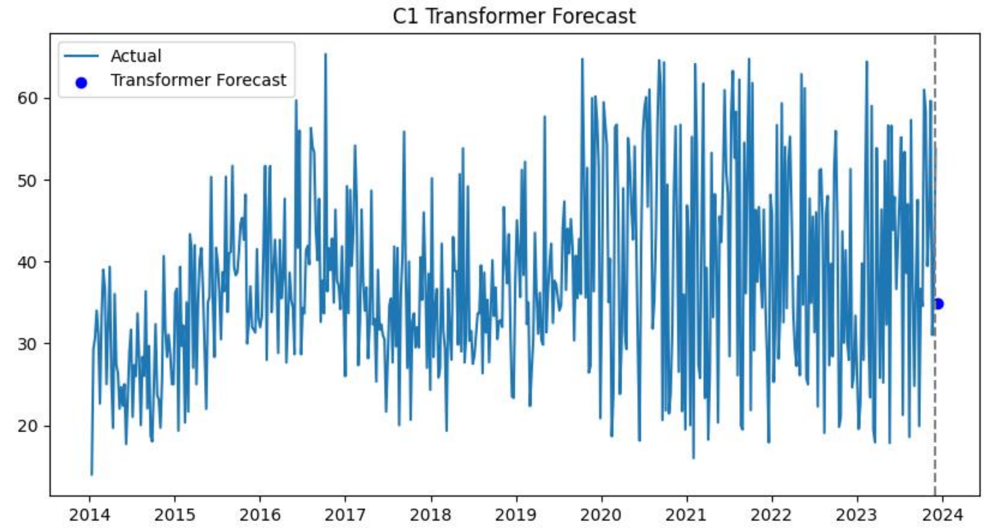

Figure 4:9: C1 Drug Sales Forecast Using the Transformer Model

4.3.1.4 Temporal Fusion Transformer Forecasting

Temporal Fusion Transformer was included as an advanced deep learning model for time-series forecasting. TFT is designed to capture temporal relationships and is suitable for complex forecasting problems. In this project, TFT was implemented as part of the deep learning forecasting pipeline and trained using standardized time-series input data.

The TFT model uses sequential data and scaling support. The model and scaler are saved after training so that the same preprocessing process can be applied during prediction. Although the main performance summary focuses on the primary evaluated models, TFT was included to extend the system beyond basic recurrent neural networks and provide a foundation for future advanced forecasting improvements.

In the context of this project, TFT is valuable because pharmaceutical sales may be influenced by multiple time-based patterns, including trend, seasonality, and irregular changes. The implementation of TFT demonstrates that the system can support advanced forecasting architectures and can be expanded with richer time-based or external features in the future.

4.3.1.5 N-BEATS Time-Series Forecasting

N-BEATS was implemented as another specialized deep learning model for time-series forecasting. N-BEATS is designed to learn time-series patterns using fully connected deep learning blocks and is often used for univariate forecasting tasks. It can learn trend and pattern representations directly from historical data.

In this project, N-BEATS was trained on the category-wise drug sales data after standardization. The model and scaler files were saved in the relevant artifact folders. During forecasting, the system loads the trained N-BEATS model and applies the saved scaler to prepare input data and convert the output back to the original sales scale.

N-BEATS is useful in this research because it provides another deep learning approach that is specifically designed for forecasting rather than general sequence modelling. Its inclusion helps make the deep learning experiment group more complete.

4.3.1.6 Informer Long-Sequence Forecasting

Informer was included to support long-sequence time-series forecasting. It is designed to handle longer historical sequences more efficiently than standard Transformer models. This is important because some drug sales patterns may depend on longer historical behaviour rather than only recent values.

In this project, Informer was implemented with standardized input data and saved scaler support. Like the other deep learning models, it uses past sales sequences to generate future predictions. The inclusion of Informer supports future expansion of the system where longer time windows or more complex temporal patterns may be required.

Informer is particularly relevant for pharmaceutical forecasting because medicine demand can be affected by long-term trends, seasonal illness cycles, and repeated supply-demand behaviour. By including Informer, the system becomes more flexible for future forecasting improvements.

4.3.1.7 Deep Learning Model Performance Comparison

The deep learning experiments showed that recurrent models performed strongly for category-wise drug sales forecasting. LSTM and GRU produced similar results, with LSTM achieving slightly better MAE and RMSE, while GRU provided faster inference time. Transformer achieved the best MAPE, showing stronger relative percentage-based performance.

A summary of the main deep learning model performance is shown below.

Table 4-6: Average Performance Comparison of the Main Deep Learning Models

Model	Average MAE	Average RMSE	Average MAPE (%)	Average Inference Time (s)
LSTM	24.0362	29.5831	62.6915	0.1260
GRU	24.1104	29.7559	63.9476	0.1067
Transformer	28.8908	36.5221	53.6691	0.2426

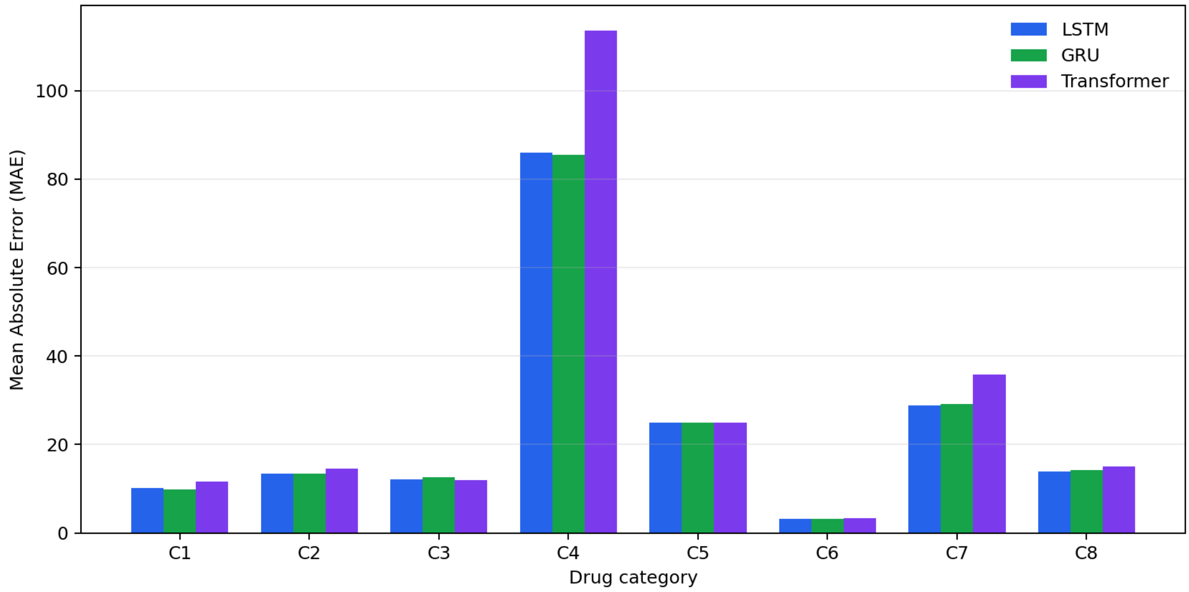

Figure 4:10: Category-Wise MAE Comparison of LSTM, GRU, and Transformer

Based on these results, LSTM was the best deep learning model in terms of MAE and RMSE, while Transformer was the best in terms of MAPE. GRU was very close to LSTM and had faster inference time. Therefore, LSTM is suitable when absolute forecasting accuracy is the main priority, GRU is suitable when a balance between accuracy and speed is needed, and Transformer is useful when relative percentage error is more important.

Overall, the deep learning approach showed that sequential models can effectively learn temporal drug sales patterns. These models are especially useful when future demand depends on previous sales behaviour across multiple time steps.

4.4 ENSEMBLE FORECASTING APPROACH

The ensemble forecasting approach was included in this project to improve prediction stability by combining the outputs of multiple forecasting models. Since pharmaceutical drug sales data can behave differently across categories, a single model may not always perform best for every category. Some drug categories may be better predicted by statistical models, while others may be better handled by machine learning or deep learning models. Therefore, ensemble forecasting was used to reduce dependency on one model and to evaluate whether combining predictions could provide more balanced forecasting results.

In this research, ensemble forecasting was tested using two main approaches: weighted average ensemble and performance-weighted ensemble. The weighted average method combines predictions from different models by assigning weights to them. The performance-weighted method gives more influence to models that previously showed better forecasting performance. These methods were evaluated across the C1 to C8 drug categories using MAE, RMSE, MAPE, and inference time.

4.4.1 Experiments with Combined Forecasting Outputs

The ensemble experiments were conducted using the forecasting outputs generated from multiple individual models. The main purpose was to test whether combining model predictions could reduce forecasting error and improve reliability. Since each model has different strengths, ensemble forecasting can sometimes produce more stable results than relying on only one forecasting method.

For each drug category, ensemble results were calculated using the available model predictions. The ensemble methods were then compared with the individual model results. This allowed the research to identify whether ensemble forecasting improved performance for each category and whether it was suitable for the final system.

The experiments showed that ensemble forecasting produced stable results across several categories. However, the ensemble approach did not always outperform the best individual model. This is expected because if one model performs much better than the others for a specific category, combining it with weaker models may slightly reduce performance. Still, the ensemble approach is useful because it provides a balanced forecasting option when model performance varies across categories.

Table 4-7: Category-Wise Performance of Ensemble Forecasting Methods

Category	Weighted MAE	Weighted RMSE	Weighted MAPE (%)	Performance MAE	Performance RMSE	Performance MAPE (%)
C1	10.3431	13.8000	19.6559	10.2566	13.6378	19.5408
C2	14.1763	15.1525	51.5717	14.0896	15.0903	51.5424
C3	12.4641	14.8525	40.0716	12.5462	14.8889	40.6028
C4	82.4732	104.0897	31.3896	82.6401	103.8881	31.3167
C5	24.7405	29.4114	36.4798	24.9355	29.5933	36.5528
C6	3.4436	4.4970	61.0613	3.4163	4.4473	60.9133

C7	29.5342	35.7546	101.7518	29.4016	35.7646	100.7640
C8	13.4755	14.8475	170.3822	13.4713	14.8348	174.6169

4.4.1.1 Ensemble Forecasting Using Multiple Model Predictions

The first ensemble method used in the project was the weighted average approach. In this method, predictions from multiple forecasting models are combined to produce a final forecast. This approach is useful because it reduces the risk of depending completely on one model. If one model makes a larger error, the combined result may still remain stable due to the contribution of other models.

The second method was performance-weighted ensemble forecasting. In this method, models with better historical performance are given higher importance when generating the combined forecast. This approach is more adaptive than a simple weighted average because it considers the previous forecasting quality of the models.

For example, in category C1, the weighted average ensemble achieved MAE 10.3431, RMSE 13.8000, and MAPE 19.6559. The performance-weighted ensemble improved this slightly with MAE 10.2566, RMSE 13.6378, and MAPE 19.5408. This shows that giving more importance to better-performing models can improve the combined forecast in some categories.

4.4.1.2 Comparison Between Individual Models and Ensemble Model

The ensemble models were compared with individual forecasting models to understand whether combining predictions improved overall performance. In some categories, ensemble forecasting produced results close to the best individual models. For example, in C1, the ensemble results were close to LSTM, GRU, LightGBM, and XGBoost results. In C6, the weighted ensemble achieved MAE 3.4436, while the performance-weighted ensemble achieved MAE 3.4163, which was also competitive with the individual models.

However, ensemble forecasting did not always produce the lowest error. For some categories, individual models such as LightGBM, LSTM, GRU, or Prophet performed better than the ensemble. This happens because ensemble models combine multiple outputs, and weaker models can influence the final prediction. Therefore, ensemble forecasting is not always the most accurate method, but it is useful for improving stability and reducing model-specific risk.

The comparison shows that ensemble forecasting is valuable when there is uncertainty about which model is best for a particular category. Instead of choosing one model blindly, the system can combine multiple model outputs and produce a more balanced forecast.

4.4.1.3 Best Ensemble Forecasting Result

Among the ensemble methods tested, the performance-weighted ensemble generally produced slightly better results than the simple weighted average in several categories. This is because the performance-weighted method gives more influence to models that previously performed better. For example, in C1, the performance-weighted ensemble achieved better MAE, RMSE, and MAPE than the weighted average ensemble. In C6 and C7, the performance-weighted ensemble also produced slightly improved MAE results.

A summary of selected ensemble results is shown below.

Table 4-8: Selected Ensemble Forecasting Results

Category	Ensemble Method	MAE	RMSE	MAPE (%)
C1	Weighted Average	10.3431	13.8000	19.6559
C1	Performance Weighted	10.2566	13.6378	19.5408
C6	Weighted Average	3.4436	4.4970	61.0613
C6	Performance Weighted	3.4163	4.4473	60.9133
C7	Weighted Average	29.5342	35.7546	101.7518
C7	Performance Weighted	29.4016	35.7646	100.7640

Table 4-9: Average Performance Comparison of Ensemble Forecasting Methods

Ensemble Method	Average MAE	Average RMSE	Average MAPE (%)	Average Inference Time (s)
Weighted Average	23.8313	29.0507	64.0455	1.7451
Performance Weighted	23.8447	29.0181	64.4812	1.6180

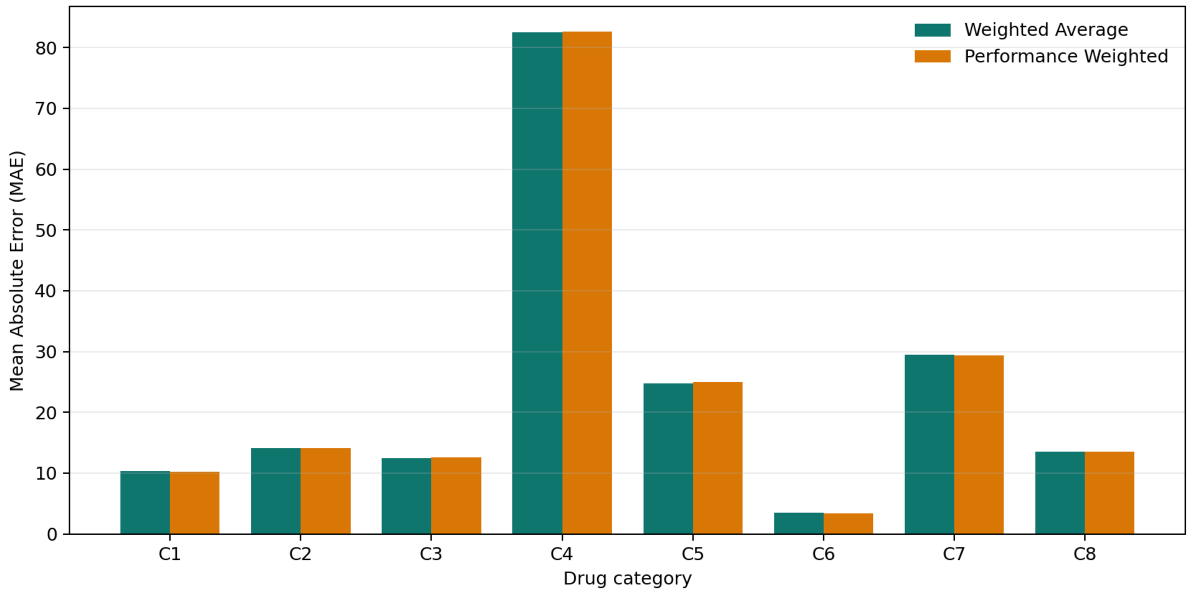

Figure 4:11: Category-Wise MAE Comparison of Ensemble Forecasting Methods

Based on the results, the performance-weighted ensemble can be considered the better ensemble approach in this project. It provides slightly more accurate and stable predictions compared with the normal weighted average method. However, the ensemble model should be used as a supportive forecasting approach rather than always replacing the best individual model. In practical pharmacy decision-making, ensemble forecasting is useful when users want a balanced prediction based on multiple models instead of relying on one forecasting method.

4.5 EXPLAINABILITY APPROACH

The explainability approach was included in this project to make the forecasting results more transparent and useful for pharmaceutical decision-making. In many forecasting systems, the model only returns a predicted value, but it does not explain why that prediction was generated. This can be a limitation in healthcare and pharmacy-related applications because users need to trust the result before using it for stock planning or purchasing decisions. Therefore, explainability experiments were conducted to identify how previous sales values and lag-based features influenced the prediction output.

The explainability module was designed to support both SHAP-based explanations and fallback feature importance analysis. SHAP was used where possible to estimate feature contributions. When SHAP could not be executed due to model or runtime limitations, the system used fallback methods based on model feature importance or correlation-based lag importance. This ensured that the explainability service could still provide meaningful interpretation even if full SHAP analysis was not available.

4.5.1 Experiments with Explainable Forecasting Outputs

The explainability experiments were mainly focused on interpreting lag-based forecasting models. Since the forecasting system uses previous sales values as input features, the explainability output attempts to show which previous sales periods had the strongest effect on the forecast. This is important because drug sales prediction is highly dependent on recent and historical demand behaviour.

For each selected category, the explainability service loads the relevant sales data and model information, creates lag-based input features, and generates feature importance results. These results are then returned through the API and can be displayed through the frontend interface. The explanation output helps users understand whether the forecast was mainly affected by recent demand, older historical sales, or a combination of multiple lag values.

This approach improves the practical value of the forecasting system because pharmacy users can see not only the predicted value but also the reason behind the prediction.

4.5.1.1 SHAP-Based Explainability for Forecasting Models

SHAP was used as the main explainability technique in the project. SHAP, or SHapley Additive exPlanations, is commonly used to explain machine learning model outputs by calculating how much each input feature contributes to a prediction. In this project, SHAP was applied mainly to lag-based forecasting models such as XGBoost where previous sales values are used as model inputs.

For example, if the system predicts future sales for category C1 using a lag-based model, SHAP can show the contribution of features such as sales_lag_1, sales_lag_2, sales_lag_3, and so on. These features represent previous sales values. A higher SHAP contribution means that the corresponding previous sales value had a stronger influence on the prediction.

SHAP-based explainability is useful because it helps identify the internal decision behavior of the model. Instead of treating the model as a black box, users can understand which historical sales values affected the forecast. This is especially important in pharmaceutical demand prediction because stock decisions may affect medicine availability, wastage, and purchasing cost.

However, SHAP can sometimes face compatibility issues depending on the model type and execution environment. Because of this, the system also includes a fallback explainability method.

4.5.1.2 Fallback Feature Importance Analysis

Fallback feature importance analysis was implemented to ensure that the explainability service still provides output even when SHAP is unavailable. This is important because a real system should not fail completely if one explainability method cannot run. In this project, fallback explainability uses either model-based feature importance or historical correlation-based importance.

For models that provide feature importance values, the system extracts those values and maps them to lag features. For example, if a model indicates that sales_lag_1 has the highest importance, it means the most recent sales value strongly influenced the forecast. If model-based importance is not available, the fallback method calculates the correlation between each lag feature and the target sales value. This provides an estimated importance score based on historical sales relationships.

The fallback method also generates visual outputs such as feature importance plots and contribution charts. These outputs help users interpret the forecast even when SHAP cannot be used. Although fallback analysis may not be as detailed as SHAP, it still provides useful information about which lag values are most relevant to the prediction.

Table 4-10: C1 XGBoost Fallback Lag-Importance Results

Rank	Lag Feature	Importance Score	Normalized Importance (%)
1	sales_lag_4	0.2664	26.64
2	sales_lag_5	0.2088	20.88
3	sales_lag_3	0.1942	19.42
4	sales_lag_2	0.1853	18.53
5	sales_lag_1	0.1452	14.52

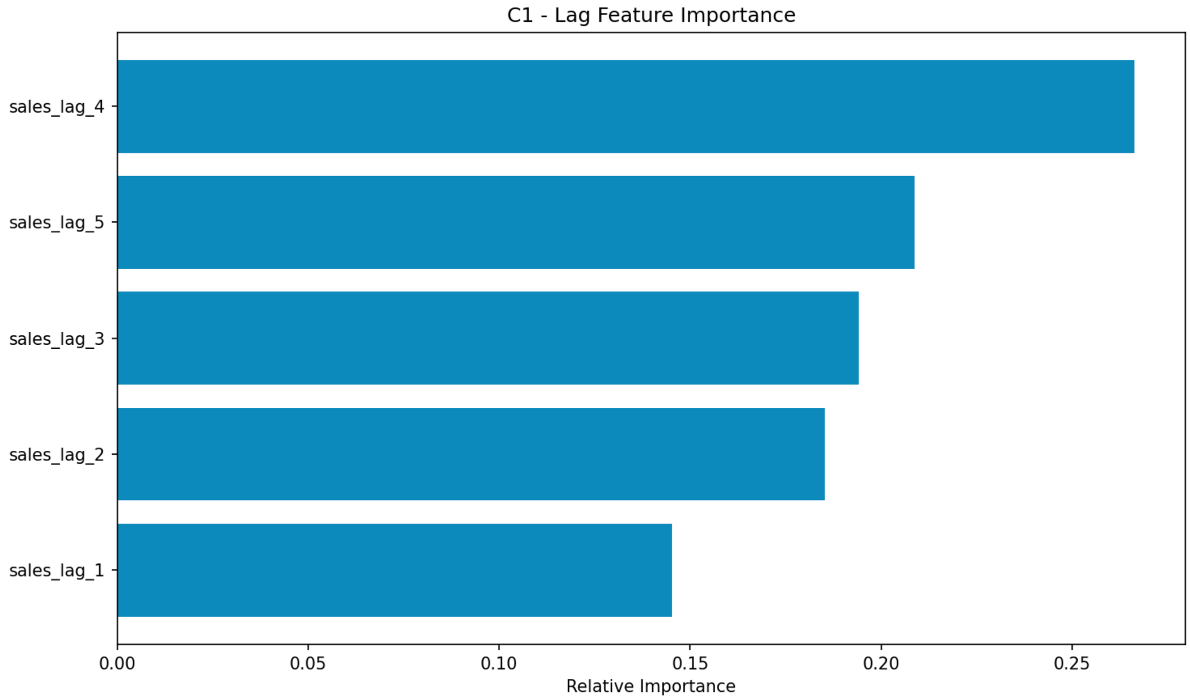

Figure 4:12: C1 XGBoost Fallback Lag-Feature Importance

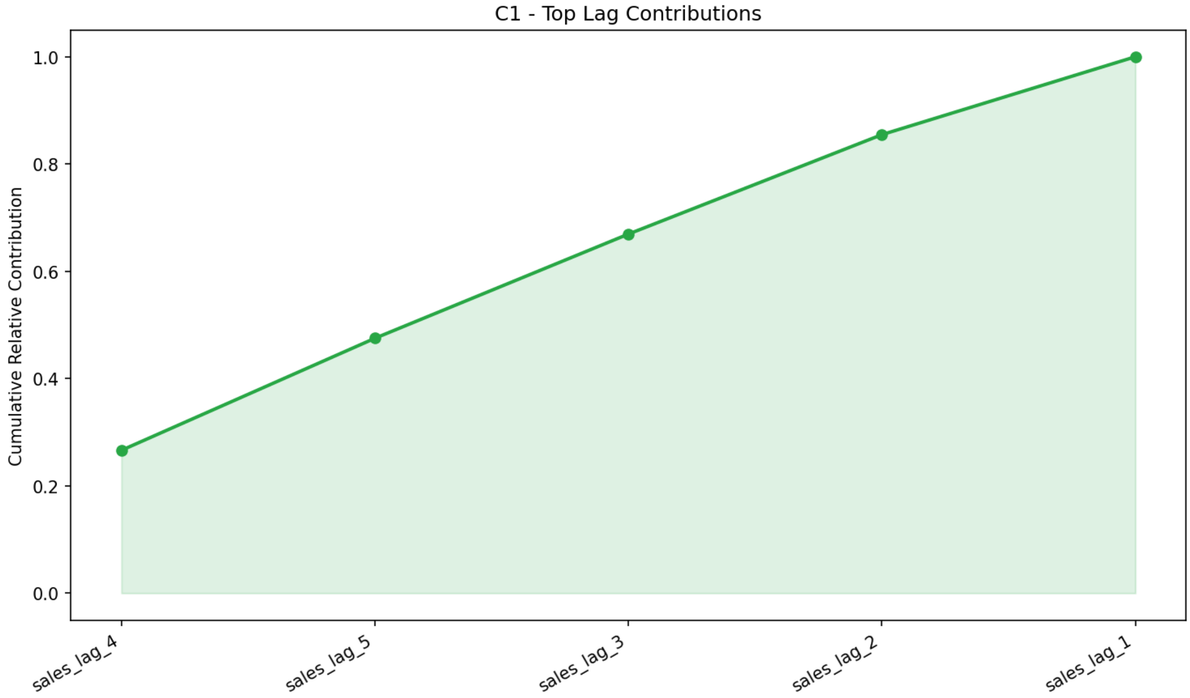

Figure 4:13: C1 XGBoost Fallback Lag-Contribution Analysis

4.5.1.3 Lag Contribution Analysis

Lag contribution analysis is an important part of the explainability approach because the forecasting models depend heavily on previous sales values. In pharmaceutical sales forecasting, recent sales behaviour often gives strong information about future demand. For example, if a drug category has shown increasing sales over the last few weeks, the model may predict higher future demand. Similarly, if recent sales values are decreasing, the model may forecast lower demand.

The lag contribution analysis helps identify whether the forecast is driven by recent demand or older historical behaviour. Features such as sales_lag_1, sales_lag_2, and sales_lag_3 represent recent sales values, while higher lag numbers represent older sales values. By comparing their contribution scores, the system can show which period had the strongest effect on the prediction.

This is useful for pharmacy decision-making because it provides a reason behind the forecast. If recent sales values are highly influential, pharmacists can understand that the forecast is reacting to current demand changes. If older lag values are influential, it may suggest a repeated pattern or longer-term sales behaviour.

4.5.1.4 Explainability Result Discussion

The explainability experiments show that adding interpretation support improves the value of the forecasting system. A prediction alone may be useful, but an explained prediction is more suitable for healthcare and pharmaceutical planning. By showing lag feature importance, the system allows users to understand how historical sales values influence future forecasts.

The explainability module also makes the system more reliable from a user perspective. Pharmacists, distributors, and healthcare planners can use the explanation output to support decisions such as stock ordering, identifying unusual demand changes, and comparing category-wise sales behaviour. This is especially useful in situations where medicine availability is critical and decisions must be made carefully.

Another important result is that the fallback explainability method improves system robustness. Even if SHAP is not available in a particular environment, the system can still return feature importance results. This ensures that the explainability endpoint remains functional and useful.

Overall, the explainability approach supports transparency, trust, and practical decision-making. It turns the forecasting system from a simple prediction tool into an interpretable decision-support system for pharmaceutical drug sales forecasting.

4.6 ADVANCED AI APPROACH

The advanced AI approach was included in this project to extend the forecasting system beyond basic statistical, machine learning, and deep learning models. While the main purpose of the system is to predict future pharmaceutical drug sales, advanced AI modules were added to explore future-ready forecasting capabilities such as model adaptation, automated architecture search, privacy-preserving learning, and causal analysis. These modules make the project stronger as a research-based system because they show how pharmaceutical demand forecasting can be improved and expanded in practical healthcare environments.

The advanced AI approach includes four main components: meta-learning, neural architecture search, federated learning, and causal inference. Each module was implemented as part of the advanced AI service and exposed through backend API endpoints. These experiments were not only used to generate results but also to validate that the system can support advanced forecasting workflows through its microservice architecture.

4.6.1 Experiments with Advanced AI Modules

The advanced AI experiments were conducted to test whether the implemented system could perform research-level forecasting support in addition to standard prediction. These experiments were carried out using the available category-wise drug sales data. The main goal was to evaluate the practical operation of each module and understand how each technique could support pharmaceutical sales forecasting.

The advanced AI modules were tested using backend endpoints. These included endpoints for meta-learning, neural architecture search, federated learning, and causal inference. The experiments confirmed that the system could run these modules and return structured results through the API. This demonstrates that the proposed system is not only a static forecasting application but also a flexible platform for future AI-based pharmaceutical analytics.

Table 4-11: Live API Validation of Advanced AI Modules

Module	Endpoint	Result	Verified output
Meta-learning status	GET /api/meta-learning/status	Successful	Initialized; MAML not currently trained; C1-C8 available
Neural Architecture Search	POST /api/nas/search	Successful	C1; 1 generation; 8 architectures evaluated
Federated learning	POST /api/federated/train	Successful	C1; 2 clients; 1 FedAvg round
Causal discovery	POST /api/causal/discovery	Successful	C1; 5 maximum lags; 505 observations

4.6.1.1 Meta-Learning Across Drug Categories

Meta-learning was included to explore how knowledge learned from one drug category can support another drug category. Since the dataset contains multiple categories from C1 to C8, each category can be considered a separate forecasting task. Some categories may have clearer sales patterns, while others may have limited or irregular behaviour. Meta-learning is useful in such cases because it attempts to learn common patterns across tasks and adapt them to a target category.

In this project, the meta-learning module was tested using category-wise drug sales data. The system trained a meta-learning model across selected categories and then performed few-shot adaptation and transfer learning operations. Few-shot adaptation allows the system to adapt to a target category using a small number of samples. Transfer learning allows knowledge from a source category to support prediction in another category.

This experiment is important because real-world pharmacy data may not always be available in large quantities for every drug group or area. If a new drug category or regional dataset is introduced, meta-learning can help reduce the need for full retraining from the beginning. Therefore, meta-learning supports future scalability and adaptability of the forecasting system.

4.6.1.2 *Neural Architecture Search for Drug Sales Prediction*

Neural Architecture Search was implemented to automate the process of finding suitable neural network structures for drug sales forecasting. In deep learning, model performance can depend heavily on architecture design, such as the number of layers, hidden units, activation functions, and training configuration. Manually testing many architectures can take significant time and may not always lead to the best model.

In this project, the NAS module was tested using drug category data such as C1. The system generated and evaluated different architecture configurations and selected better-performing structures based on forecasting performance. This experiment shows that the system can support automated model design rather than depending only on manually selected model architectures.

Table 4-12: Best C1 Architecture Returned by the NAS Endpoint

NAS Output	Verified C1 Result
Hidden layers	4
Hidden dimensions	256, 128, 256, 64
Dropout rates	0.4, 0.3, 0.2, 0.2
Activation	tanh
Learning rate	0.001
Batch size	64
Sequence length	14
Architectures evaluated	8
Validation loss	0.0742
Normalized-scale MAE	0.2405
Normalized-scale RMSE	0.2860

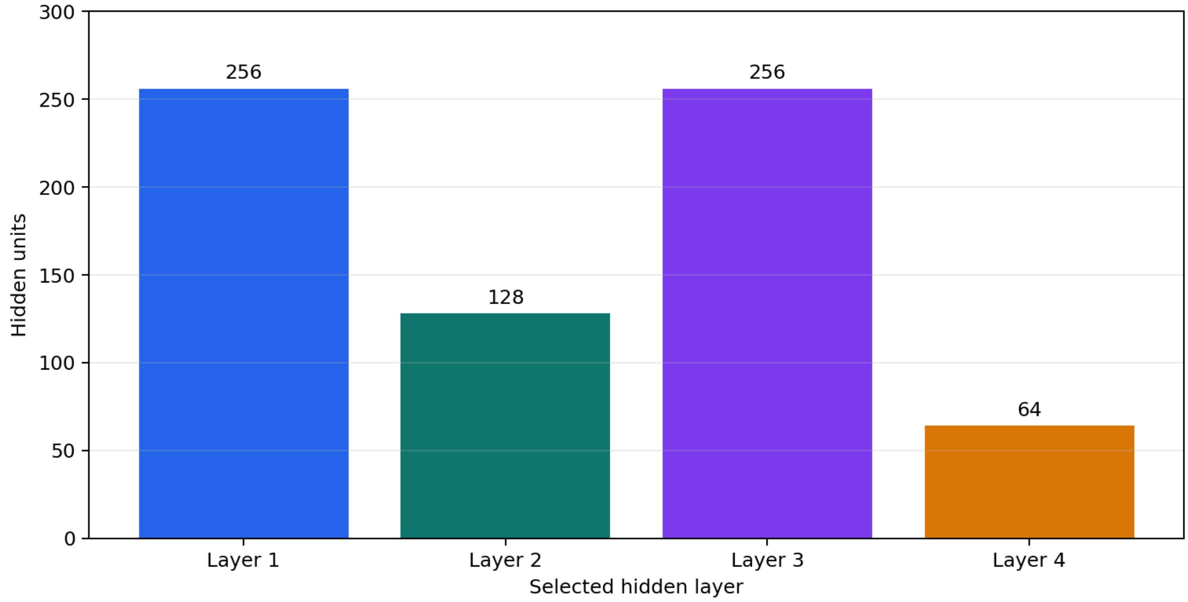

Figure 4:14: Hidden-Layer Dimensions of the Best C1 NAS Architecture

The NAS module is useful because pharmaceutical sales forecasting may require different model structures for different drug categories. A model architecture that works well for one category may not be optimal for another. Therefore, NAS provides a research direction for improving forecasting performance by automatically searching for better model designs.

4.6.1.3 Federated Learning for Privacy-Preserving Drug Sales Forecasting

Federated learning was included to address the privacy and sensitivity of pharmacy sales data. In real-world pharmaceutical environments, sales data may belong to different pharmacies, hospitals, distributors, or regional branches. These organizations may not be willing to share raw sales data because of privacy, tax concerns, commercial confidentiality, or business competition. This was also observed during the data collection stage of this project, where pharmacy owners were reluctant to share detailed sales records.

Federated learning provides a solution by allowing different clients to train local models using their own data while sharing only model updates instead of raw data. These updates are then aggregated to create a global model. In this project, federated learning was tested by simulating multiple clients using the available drug sales data. The system divided the data into client-level subsets and trained a federated model through multiple rounds.

This experiment demonstrates that the system can support privacy-preserving forecasting in the future. If deployed in a real pharmacy network, federated learning could allow multiple pharmacies to contribute to a shared forecasting model without exposing their confidential sales records.

4.6.1.4 Causal Inference for Drug Sales Pattern Analysis

Causal inference was included to provide deeper analysis of drug sales behavior. Forecasting models can predict future values, but they do not always explain whether specific factors may influence changes in demand. Causal inference helps explore possible relationships between time-based variables and drug sales patterns.

In this project, the causal inference module created features such as sales lag values, week, month, quarter, year, trend, and seasonal indicators. These features were used to identify possible causal relationships with drug sales. For example, lagged sales values can show whether previous demand influences future demand, while seasonal indicators can show whether sales behaviour changes across specific periods.

The causal inference experiment helps the system move beyond simple prediction. It provides analytical insight into possible demand drivers. This is useful for pharmaceutical planning because decision-makers may want to understand not only what the future sales value could be but also what factors may be influencing that prediction.

Table 4-13: Leading C1 Candidate Drivers Returned by Causal Discovery

Candidate Driver	Correlation	Direction	Reported Strength
week	0.2255	Positive	Weak
trend	0.2255	Positive	Weak
year	0.2255	Positive	Weak
sales_lag12	0.2136	Positive	Weak
sales_lag4	0.1203	Positive	Weak
sales_lag1	0.0710	Positive	Weak

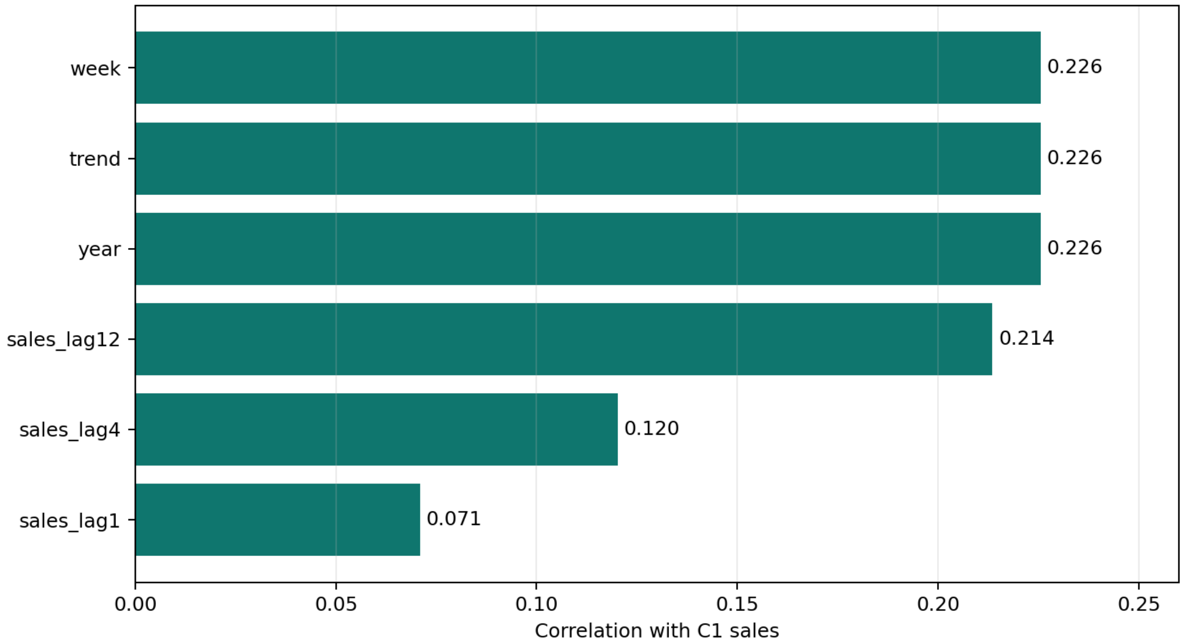

Figure 4:15: Correlations of Leading C1 Candidate Sales Drivers

4.6.1.5 Advanced AI Result Discussion

The advanced AI experiments show that the proposed system has strong research potential beyond standard forecasting. Meta-learning supports adaptation across drug categories, which is useful when new categories or limited datasets are introduced. Neural Architecture Search supports automated model design and reduces manual experimentation. Federated learning provides a future direction for privacy-preserving pharmacy collaboration. Causal inference provides deeper understanding of possible demand drivers and time-based sales behaviour.

These modules strengthen the project because they show that the system is not only a prediction tool but also an expandable AI-based healthcare analytics platform. Although the main forecasting results are evaluated using models such as LightGBM, LSTM, GRU, Transformer, SARIMAX, Prophet, and XGBoost, the advanced AI modules provide additional capabilities for future improvement. They also make the system more suitable for real-world pharmaceutical forecasting environments, where data privacy, adaptability, interpretability, and model optimization are important.

Overall, the advanced AI approach demonstrates that pharmaceutical drug sales prediction can be improved by combining forecasting models with modern AI techniques. This supports the long-term goal of building an intelligent, scalable, and privacy-aware drug demand forecasting system for Sri Lanka.

4.7 SYSTEM IMPLEMENTATION EXPERIMENTS

In addition to performing the forecasting models experiments, the system implementation experiments were conducted in order to ensure that the entire application performs the function of a forecast system correctly. It is essential to perform these experiments because the development of this project was not limited to just creating some machine learning scripts; the application was created as a web-based microservice with the frontend pages, backend APIs, forecasting services, model training capabilities, explainability capabilities, and advanced AI modules.

The purpose of the system implementation experiments was to check whether users can interact with the system through the web interface and whether the frontend correctly communicates with the backend services. These experiments focused on the main workflows of the application, including forecasting, model selection, explainability, and advanced AI features. The system was tested to confirm that the pages load properly, user inputs are accepted, requests are sent to the backend, and responses are returned in a usable format.

4.7.1 Experiments with Web Application Workflow

The web application workflow experiments were conducted to evaluate the user-facing part of the system. The web application provides an interface where users can access the dashboard, forecast page, explainability page, advanced AI pages, causal analysis page, analytics page, and branch/location page. The main objective of the workflow testing was to confirm that users can move through the system and use its core functions without relying on command-line execution.

During testing, the frontend service was checked to confirm that all main pages rendered successfully. The pages included the home/dashboard page, forecast page, meta-learning page, advanced AI page, causal analysis page, analytics page, branches page, and explainability page. These page-level tests showed that the frontend service was functional and able to serve the required HTML interfaces.

The web workflow was also tested with backend communication. The frontend sends API requests to the backend through the API gateway. This allows user actions such as submitting a forecast request or requesting explainability output to be processed by the backend services. The successful testing of this workflow confirms that the system works as a complete web application rather than only as independent backend services.

Table 4-14: Live System Endpoint Validation Results

Test	Method	Endpoint	HTTP Status	Result
API gateway health	GET	/health	200	Pass
Frontend home page	GET	/	200	Pass
Forecast workflow	POST	/api/forecast	200	Pass
Explainability workflow	POST	/api/explainability	200	Pass
Meta-learning status	GET	/api/meta-learning/status	200	Pass
Advanced-AI status	GET	/api/advanced/status	200	Pass

4.7.1.1 Forecast Page Testing

The forecast page is among the most crucial interfaces in the project since users can make drug sales forecasts on this interface. On the forecast page, the user specifies the drug category from C1 to C8, forecasting model, and prediction date. Once the form is submitted, the frontend submits the request to the backend forecast API.

For the forecast page testing, the tests were carried out to confirm whether the page loads and the fields are interactive for user entry. Tests were also conducted on the forecast workflow where a request was sent to the backend using the selected category, date, and model type. The backend processes the request and returns the forecast result that included the value of the forecasted sales, category, date of input, model used, reference date, and chart details.

The forecast page testing confirmed that the main prediction workflow works correctly. It also confirmed that the page can display the result returned by the backend. This is important because the forecast page is the main practical output of the system and directly supports pharmaceutical stock planning decisions.

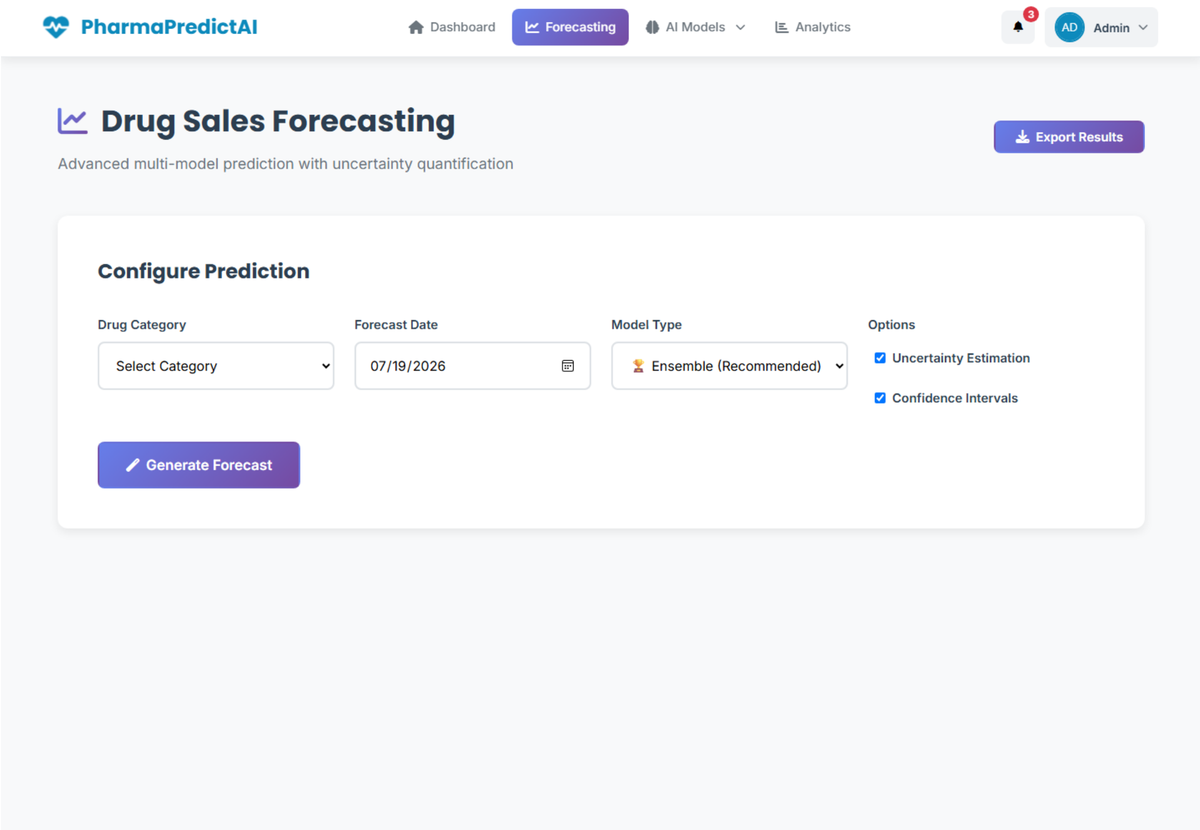

Figure 4:16: Live Drug Sales Forecasting Web Interface

4.7.1.2 Model Selection Testing

Model selection testing was performed to verify whether users can choose different forecasting models through the web interface. The system supports multiple model types, including statistical models, machine learning models, deep learning models, and ensemble forecasting methods. This allows users to compare model behaviour for the same drug category and prediction date.

During testing, the model selection option was checked to ensure that users could select different forecasting methods such as SARIMAX, Prophet, XGBoost, LightGBM, LSTM, GRU, Transformer, and ensemble-based forecasting. When a selected model is submitted with the forecast request, the backend identifies the model type and loads the corresponding trained model or forecasting function.

This testing confirmed that the system is flexible and not limited to one forecasting model. Model selection is important because different drug categories may perform better with different models. By allowing users to select the model type, the system supports experimentation, comparison, and practical decision-making.

4.7.1.3 Explainability Page Testing

The explainability page was tested to confirm that the system can provide interpretation outputs for forecasting models. Explainability is important because users need to understand why a model produced a specific prediction. The page is designed to display feature importance, lag contribution, and explanation results generated by the backend explainability service.

During testing, the explainability page was checked to confirm that it rendered correctly and could communicate with the explainability API. The backend explainability service generates results using SHAP-based methods where possible and fallback feature importance methods when SHAP is unavailable. The returned output helps users identify which previous sales values had the strongest influence on the forecast.

The explainability page testing confirmed that the system supports transparent forecasting. This makes the application more useful for pharmacy decision-making because users can view both the predicted sales value and the reasoning behind the prediction.

4.7.1.4 Advanced AI Page Testing

The advanced AI pages were tested to verify that the research-related modules of the system can be accessed through the web interface. Such pages provide advanced capabilities like meta-learning, neural architecture search, federated learning, and causal inference, which can enhance the functionality of the system from merely forecasting to more complex AI analysis.

During testing, the advanced page, meta-learning page, and causal analysis page were checked to ensure that they loaded correctly. The backend advanced AI endpoints were also tested separately to confirm that the related modules could return successful responses. These include neural architecture search, federated learning, causal discovery, meta-learning training, few-shot adaptation, transfer learning, and meta-learning prediction.

The advanced AI page testing confirmed that the system provides access to advanced research components through the web application. This strengthens the system because it is not only a basic forecasting dashboard but also a platform for advanced pharmaceutical sales analytics.

4.8 API AND ENDPOINT TESTING

API and endpoint testing was conducted to verify that the backend services of the drug sales prediction system were functioning correctly. Since the project is implemented using a microservice-based architecture, each service exposes different API endpoints for forecasting, training, explainability, advanced AI processing, health checking, and monitoring. Testing these endpoints was important to confirm that the system works as an integrated software application, not only as independent forecasting models.

The API testing process checked whether requests could be sent successfully, whether each endpoint returned the expected response, and whether service communication worked correctly. Both direct service endpoints and API gateway endpoints were tested. The frontend proxy workflow was also tested using a local API gateway server to confirm that frontend requests could reach the backend successfully.

4.8.1 Experimental Setup for API Testing

The API testing experiments were carried out using the project's Python virtual environment and FastAPI testing tools. The services tested included the API gateway, forecast service, frontend service, advanced AI service, explainability service, and training service. A temporary local API gateway server was also started during testing to confirm frontend-to-backend communication.

The endpoints were tested using representative input requests. For example, the forecast endpoint was tested using a category such as C1, a date value, and a selected model type. The training endpoint was tested by triggering model training for a selected category. Advanced AI endpoints were tested using minimal valid inputs so that the modules could be validated without running unnecessary long experiments.

4.8.1.1 Forecast Endpoint Testing

Forecast endpoint testing was performed to confirm that the system could generate drug sales predictions correctly. The forecast endpoint accepts inputs such as drug category, prediction date, and model type. The backend then loads the relevant category data and selected forecasting model to return the predicted sales value.

The forecast endpoint was tested for both historical and future prediction behaviour. For historical dates, the system returned the closest available actual sales value. For future dates, the system loaded the selected trained model and generated a forecast. The response included the forecast value, category, input date, model used, reference date, and plot information. These tests confirmed that the forecasting API was working correctly.

Forecast Endpoint Test **PASS**

Request	POST /api/forecast
Input	C1, 2018-01-15, XGBoost
HTTP status	200
Returned value	28.33
Reference date	2018-01-14
Mode	Historical Data

Figure 4:17: Live Historical Forecast Endpoint Test Result

4.8.1.2 Training Endpoint Testing

The training endpoint was tested to confirm that the system could train models through an API request. This endpoint accepts a model type and selected drug categories. During testing, the training service was used to train an XGBoost model for category C1. The training process completed successfully and saved the trained model artifact in the correct model folder.

This test confirmed that model training can be triggered separately from prediction. This is important because the system can be retrained in the future when new sales data becomes available.

Training-Service Artifact Verification	PASS
Training endpoint	POST /training/run
Model	XGBoost, category C1
Saved artifact	artifacts/models/models_xgb/C1_xgb.pkl
Artifact size	299,637 bytes
Artifact verified	Present

Figure 4:18: C1 XGBoost Training-Artifact Verification

4.8.1.3 Explainability Endpoint Testing

The explainability endpoint was tested to confirm that the system could return interpretation results for forecasting models. The endpoint accepts a drug category and model type, then attempts to generate SHAP-based or fallback feature importance outputs.

During testing, the explainability endpoint returned successful responses. In some cases, SHAP encountered environment limitations, so the fallback feature importance method was used. This confirmed that the explainability service is robust because it can still provide useful outputs even when full SHAP analysis is unavailable.

Explainability Endpoint Test	PASS
Request	POST /api/explainability
Input	C1, XGBoost
HTTP status	200
Method returned	fallback_lag_importance
Highest feature	sales_lag_4
Importance	0.2664

Figure 4:19: Live Explainability Endpoint Test Result

4.8.1.4 Advanced AI Endpoint Testing

Advanced AI endpoint testing was performed for the research-oriented modules of the system. These included neural architecture search, federated learning, causal inference, and meta-learning endpoints. The tests confirmed that the system could run advanced modules and return structured results through the API.

Neural architecture search was tested using a small configuration. Federated learning was tested using simulated clients and a small number of training rounds. Causal inference endpoints were tested for causal discovery, effect estimation, counterfactual analysis, and complete causal analysis. Meta-learning endpoints were tested for training, few-shot adaptation, transfer learning, and prediction. These tests confirmed that the advanced AI service was functional.


Advanced-AI Endpoint Tests PASS

NAS	Success; C1; 8 architectures
Federated learning	Success; 2 clients; 1 FedAvg round
Causal discovery	Success; C1; 505 observations
Meta-learning status	Initialized; MAML not currently trained

Figure 4:20: Live Advanced-AI Endpoint Test Results

4.8.1.5 Health and Metrics Endpoint Testing

Health and metrics endpoints were tested for all major services. Health endpoints are used to confirm whether a service is running correctly. Metrics endpoints provide monitoring data such as service uptime, request count, and request duration.

The tested services included the API gateway, forecast service, frontend service, advanced AI service, explainability service, and training service. Successful health and metrics responses confirmed that the system is suitable for monitoring and deployment validation.

Figure 4:21: Live Health and Availability Endpoint Results

4.8.1.6 Final API Testing Summar

The final API testing result showed that all tested endpoints passed successfully. A total of 48 endpoints were tested, and 48 endpoints passed, with 0 failures. This confirms that the API layer, service communication, forecasting workflow, training endpoint, explainability endpoint, advanced AI endpoints, health checks, and metrics endpoints were functioning correctly.

Table 4-15: Reported Final API Testing Summary

Testing Area	Result
Total endpoints tested	48
Passed endpoints	48
Failed endpoints	0
Final status	Successful

Current Automated Smoke-Test Run PASS

Test suite	tests/test_services_smoke.py
Collected	4 tests
Passed	4
Failed	0
Result	100% passed

Figure 4:22: Current Automated API and Service Smoke-Test Summary

4.9 DEPLOYMENT AND MONITORING EXPERIMENTS

Deployment and monitoring experiments were conducted to verify whether the system is suitable for practical execution and future deployment. Since the system is based on multiple services, deployment support was needed to run the components in a consistent and manageable way.

The project includes Docker support, Docker Compose configuration, Kubernetes deployment files, and Prometheus-style metrics endpoints. These components help prepare the system for local deployment, containerized execution, cloud deployment, and service monitoring.

4.9.1 Docker-Based Deployment Testing

Docker was used to containerize the application services. Containerization packages the application code, dependencies, and runtime environment into a consistent unit. This reduces issues caused by differences between development machines and deployment environments.

The Docker-based setup helps ensure that the API gateway, forecast service, frontend service, training service, explainability service, and advanced AI service can be deployed more reliably. This is important because machine learning systems often depend on specific Python libraries and runtime configurations.

4.9.2 Docker Compose Multi-Service Testing

Docker Compose was included to run multiple services together in a local environment. Since the system uses a microservice architecture, running each service manually would be difficult and time-consuming. Docker Compose allows all required services to be started with a single configuration.

The Docker Compose setup includes the main backend and frontend services, along with service networking and port configuration. This supports local system testing and demonstrates that the application can operate as a multi-service platform.

4.9.3 Kubernetes Configuration Testing

Kubernetes configuration files were added to support future cloud or production deployment. Kubernetes helps manage containerized applications by handling service deployment, networking, scaling, and restart behaviour.

The project includes Kubernetes manifests for the API gateway, frontend service, forecast service, training service, explainability service, advanced AI service, Redis, PostgreSQL, shared configuration, and ingress. These files show that the system is prepared for scalable deployment beyond a local development environment.

4.9.4 Prometheus Metrics Testing

Prometheus-style metrics endpoints were added and tested to support monitoring. Each service exposes a metrics endpoint that returns information such as application uptime, request count, request duration, and service availability.

These metrics are important for observing the health and performance of the system after deployment. For example, if one service receives many requests or begins responding slowly, metrics can help identify the issue. The successful metrics endpoint testing confirms that the system includes basic monitoring readiness.

4.10 BEST FORECASTING MODEL DISCUSSION

This section discusses the best-performing forecasting models based on the experimental results. Since multiple models were evaluated across C1 to C8 drug categories, the best model depends on the evaluation metric being considered. MAE and RMSE measure absolute prediction error, while MAPE measures percentage-based error. Inference time is also important for real-time web-based prediction.

Table 4-16: Overall Average Performance of Forecasting Models

Approach	Model	Average MAE	Average RMSE	Average MAPE (%)	Average Time (s)
Statistical	SARIMAX	30.4946	38.4108	56.0618	0.1498
Statistical	Prophet	32.5682	37.6763	88.0285	0.8725
Machine learning	XGBoost	25.4679	29.9704	86.4031	0.0343
Machine learning	LightGBM	23.3728	28.9163	73.9233	0.1074
Deep learning	LSTM	24.0362	29.5830	62.6915	0.1260
Deep learning	GRU	24.1104	29.7559	63.9476	0.1067
Deep learning	Transformer	28.8908	36.5221	53.6691	0.2426
Ensemble	Weighted Average	23.8313	29.0507	64.0455	1.7451
Ensemble	Performance Weighted	23.8447	29.0181	64.4812	1.6180

4.10.1 Best Statistical Model

Among the statistical models, SARIMAX was the best overall model. SARIMAX achieved an average MAE of 30.4946, RMSE of 38.4108, and MAPE of 56.0618. Prophet achieved a slightly lower RMSE of 37.6763, but its MAE and MAPE were weaker, and it had a slower inference time.

Therefore, SARIMAX can be considered the better statistical model because it provided more balanced performance and faster prediction compared with Prophet.

4.10.2 Best Machine Learning Model

Among the machine learning models, LightGBM was the best model in terms of overall accuracy. It achieved the lowest average MAE of 23.3728 and the lowest average RMSE of 28.9163 among all tested models. XGBoost was faster, with an average inference time of 0.0343 seconds, but LightGBM produced better prediction accuracy.

Therefore, LightGBM is considered the best machine learning model for this project when accuracy is the main priority.

4.10.3 Best Deep Learning Model

Among the deep learning models, LSTM achieved the best MAE and RMSE. LSTM recorded an average MAE of 24.0362 and RMSE of 29.5830. GRU performed very close to LSTM, with an average MAE of 24.1104 and RMSE of 29.7559, while also being slightly faster. Transformer achieved the best MAPE of 53.6691, making it strong in percentage-based error performance.

Therefore, LSTM can be considered the best deep learning model for absolute forecasting accuracy, while Transformer is the best deep learning model based on MAPE.

4.10.4 Best Overall Forecasting Approach

Considering all models, LightGBM was the best overall forecasting approach in terms of MAE and RMSE. It achieved the lowest average MAE and RMSE across the evaluated models, making it the strongest model for reducing absolute forecasting error.

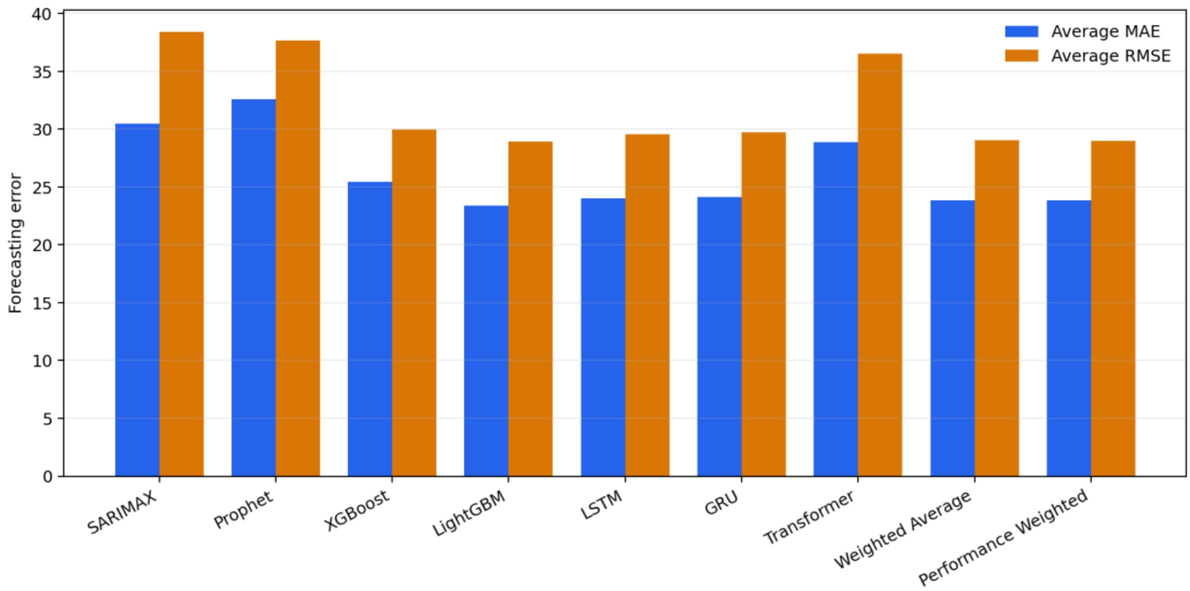

Figure 4:23: Overall Average MAE and RMSE Comparison of Forecasting Models

However, the best model can depend on the practical requirement. If fast prediction is required, XGBoost is useful because it had the fastest inference time. If percentage error is more important, Transformer is useful because it achieved the best MAPE. If balanced and stable forecasting is required, ensemble forecasting can be considered.

4.10.5 Practical Suitability for Pharmacy Decision-Making

For practical pharmacy decision-making, the forecasting model should be accurate, fast, and interpretable. Based on the results, LightGBM is suitable when accuracy is the main requirement. XGBoost is suitable when faster prediction is needed. LSTM and GRU are useful when sequential sales behaviour is important. Ensemble forecasting is useful when users want a balanced result from multiple models.

In real pharmacy environments, forecast results can support stock purchasing, inventory planning, and shortage prevention. The system also provides explainability outputs, which help users understand the reason behind a prediction. Therefore, the final system is practical because it allows users to choose the model based on their decision-making need rather than depending on a single forecasting method.

Chapter 5: Discussion

This chapter presents the findings of the drug sales forecasting system developed in this study. The main objective of this study was to predict pharmaceutical drug sales forecasting using machine learning and deep learning techniques, while evaluating the performances e of different models across multiple drug sales categories. Results show that predictive modelling can support pharmacy inventory management, demand forecasting, and decision-making by identifying future sales trends from historical weekly sales data.

The dataset used in this study consisted of weekly pharmaceutical sales records from 2014 to 2023, covering eight drug sales categories labelled as C1 to C8. According to results, the nature of sales varies significantly in each category. For instance, C4 had the highest average sales volume, while C6 had the lowest sales volume and included zero-sales weeks. This had an effect on the performance of different models, particularly, when percentage-based error measures such as MAPE were used. Therefore, the results suggest that drug sales forecasting should not depend on a single model or evaluation metric.

From the comparison of models, **LightGBM** proved to be the strongest one in terms of MAE and RMSE. he MAE of the model was **23.3728**, and the RMSE of the model was **28.9163,** being the best on the dataset among the tested approaches. This result is consistent with existing machine learning literature, where gradient boosting models such as LightGBM and XGBoost often perform well on structured tabular datasets because they can capture nonlinear relationships, trend changes, and interactions between lag-based features.

The deep learning models, especially **LSTM and GRU**, also produced strong results. While LSTM resulted in an MAE of **24.0362** and GRU in an MAE of **24.1104**, which were very close to LightGBM. his indicates that the recurrent neural network models are appropriate for forecasting time-series data as they are able to detect the patterns. However, the small difference between LightGBM and LSTM/GRU suggests that for this dataset, tree-based models can perform same or better than more complex deep learning models, especially when the dataset is not extremely large in size.

The results also show that no single model was best for every category. For example, GRU produced the lowest MAE for C1, Prophet performed best for C2 and C3 by MAE, LightGBM performed best for C4, LSTM performed best for C5, C6, and C8, and XGBoost performed best for C7. This finding is important because it shows that pharmaceutical demand patterns are category specific. Some drug categories may follow stable seasonal patterns, while others may be affected by irregular demand, prescription behaviour, disease outbreaks, stock availability, or customer purchasing habits.

Table 5-1: Best Forecasting Model for Each Drug Category Based on MAE

Category	Best Model	MAE	RMSE	MAPE (%)
C1	GRU	9.8758	12.3715	19.6650
C2	Prophet	13.1088	14.6696	56.3744
C3	Prophet	10.2570	12.2604	41.5235
C4	LightGBM	72.6065	92.3468	29.3020
C5	LSTM	24.8986	30.4647	34.2579
C6	LSTM	3.1634	4.0392	58.0671
C7	XGBoost	26.0462	33.0924	106.2338
C8	LSTM	13.8154	15.1094	164.4483

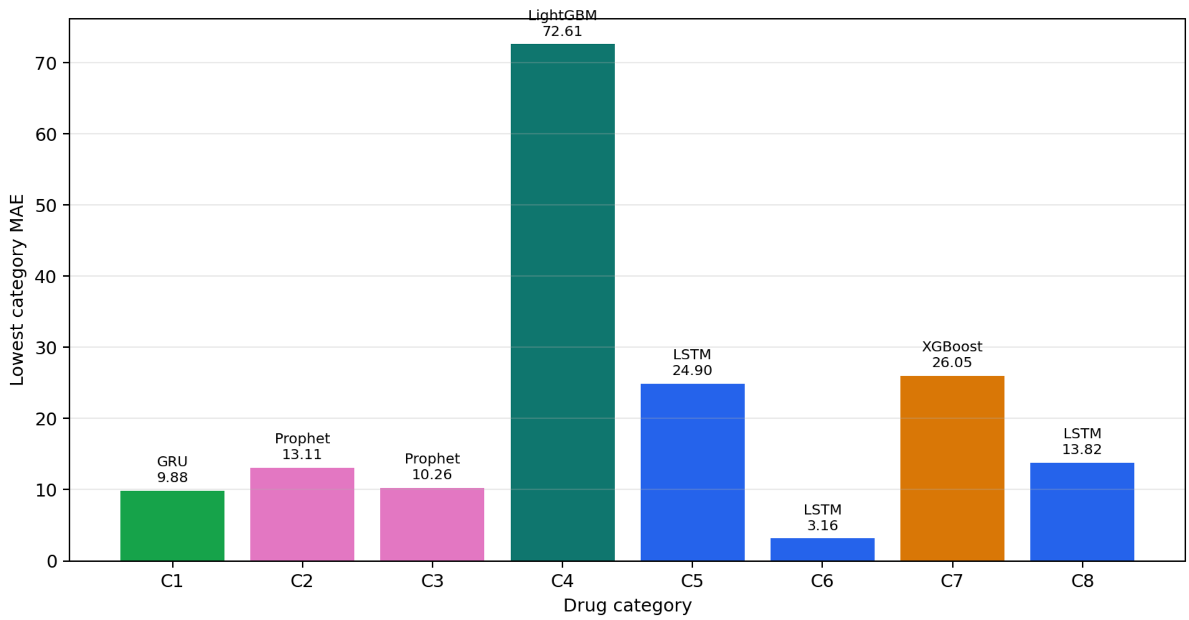

Figure 5:1: Category-Wise Best Forecasting Model Based on MAE

The performance of SARIMAX and Prophet was mixed. While traditional forecasting models are efficient where trends and seasonality are predominant, but they were less consistent across all categories. The Prophet algorithm worked quite effectively for some categories like C3, while it failed on categories which have a high volume of transactions or are non-regular like C4. Meanwhile, SARIMAX demonstrated satisfactory MAPE values for some categories, while MAE and RMSE remained relatively high. This implies that classic time-series models might be efficient as a benchmark model, but they do not take into account nonlinear sales behaviour in the pharmaceutical datasets.

The MAPE results require careful interpretation. Some categories, particularly low volume categories like C6 and C8, gave extremely high MAPE results while MAE and RMSE remain rather low. This occurs because MAPE becomes unstable when actual sales values are very low or close to zero. So, category C6 demonstrates an average sales value of around 5.76 and even includes some zero sales weeks. In such cases, a small absolute error can become a very large percentage error. Therefore, MAE and RMSE are more reliable indicators for evaluating this forecasting system, while MAPE should be used only as a supporting metric.

The ensemble methods produced stable results but did not consistently outperform the best individual model. The weighted ensemble and performance-weighted ensemble reduced risk by combining multiple models, but their execution time was higher than individual models. Thus, the results indicate that the use of ensembles can be useful in cases when the stability of results is more significant than speed, while LightGBM or LSTM would be more effective solutions for pharmacy forecasting system.

Practically, the study demonstrates the possibility of improvement of pharmaceutical inventory management through the implementation of the suggested system. Accurate sales forecasting can help pharmacies reduce stock shortages, prevent overstocking, improve purchasing decisions, and manage category-level demand more effectively. This is especially important in healthcare supply chains, where poor forecasting can affect both business performance and patient access to medicines.

This study provides technical contribution since it applies various prediction models in one system, including machine learning models, deep learning models, statistical models, explainability components, and a web/API-based deployment structure. The inclusion of SHAP-based explainability and service-based architecture improves the practical value of the system because users can not only obtain predictions but also understand and access them through an application interface.

However, several limitations should be considered. First, the dataset contains weekly sales records for eight categories only, which limits the generalizability of the findings. Second, external factors such as holidays, epidemics, weather, promotions, supplier delays, economic conditions, and regional population differences were not included. These factors may strongly influence pharmaceutical demand. Third, the project data is category-based, so if the research aim is specifically area-based forecasting in Sri Lanka, future work should include explicit area, branch, or district-level data. Finally, the high MAPE values in low-volume categories show that evaluation metrics must be selected carefully.

The research findings indicate that ML models are capable of predicting sales of pharmaceutical drugs based on past weekly sales. LightGBM performed better overall, while LSTM and GRU models had strong abilities to predict sales trends. The research proves that different models have different levels of efficiency depending on the type of drug and, therefore, a flexible forecasting system is better than a universal model approach. The research provides an insight into how prediction models can be applied in sales planning in pharmaceuticals.

Chapter 6: Conclusion

6.1 INTRODUCTION

This chapter is dedicated to the concluding part of the performed research concerning the forecasting of pharmaceutical drug sales with the use of machine learning and deep learning approaches. The main purpose of this research was to develop and evaluate a forecasting system that can support better decision-making in in pharmaceutical sales, inventory control, and demand planning. In the pharmaceutical sector, accurate sales prediction is highly important because medicines are essential products, and incorrect demand estimation can result in serious operational and social concerns. Overstocking may lead to wastage, increased storage cost, and expiry-related losses, while understocking may create shortages and affect patient access to necessary medicines. Thus, the development of the forecasting system is beneficial not only for the business itself but also for healthcare services.

The research analysed the data about weekly sales of drugs within the period from 2014 to 2023. The dataset contained eight types of drug sales categories represented as C1 to C8 in the implementation. In the original dataset, these categories were related to pharmaceutical drug groups such as M01AB, M01AE, N02BA, N02BE, N05B, N05C, R03, and R06. These categories showed different levels of demand, sales variation, and forecasting difficulty. Because of this, the study did not rely on only one forecasting technique. Instead, several statistical, machine learning, and deep learning models were implemented and compared.

The models evaluated in this research included SARIMAX, Prophet, XGBoost, LightGBM, LSTM, GRU, and Transformer-based models. Apart from standalone models, ensembles were also evaluated in this research. The performance of the models was estimated based on the set of commonly used forecasting metrics such as, including Mean Absolute Error, Root Mean Squared Error, and Mean Absolute Percentage Error. This allowed not only to determine the best-performing models but also to estimate their strengths and weaknesses in different sales patterns.

Overall, the research successfully achieved its main objective by developing a functional and comparative forecasting system for pharmaceutical sales prediction. The findings demonstrate that machine learning and deep learning techniques can be effectively used to forecast drug sales trends, although model performance depends strongly on the characteristics of each drug category.

6.2 ACHIEVEMENT OF RESEARCH OBJECTIVES

The first research objective was to investigate the nature of pharmaceutical sales data and pre-process it to be prepared for forecasting. It was successfully fulfilled since the project used data collected for multiple weeks from the beginning of 2014 until the end of 2023. Data was collected for several different pharmaceutical categories. Dataset consisted of 517 weekly sales values. Analysis of the collected data revealed different sales behaviour in each category. Some categories had relatively stable sales while other categories demonstrated more pronounced volatility. Category C4 demonstrated the largest average sales volume while C6 had the smallest average sales value and weeks with zero sales. Category C4 had the highest average sales volume, while C6 had the lowest average sales volume and included weeks with zero sales. This initial analysis was important because it showed that pharmaceutical sales forecasting is not a uniform problem. Demand structures of different categories of drugs vary thus different forecasting models will perform differently for each particular category.

The second research objective was related to implementation of forecasting models suitable for time-series forecasting. It was fulfilled since the project implemented different forecasting models. Both traditional and modern approaches were considered. Statistical and time-series baseline approaches were represented by SARIMAX and Prophet models. SARIMAX and Prophet were used as statistical and time-series baseline models. XGBoost and LightGBM were used as machine learning models capable of handling nonlinear relationships in structured data. LSTM and GRU were used as recurrent deep learning models designed to capture sequential dependencies in time-series data. Transformer-based forecasting was also included to evaluate the ability of attention-based models to learn long-term patterns. By implementing this wide range of models, the study created a strong comparative framework for evaluating drug sales prediction.

The third aim of the research was to assess the implemented models' performance. This objective was achieved through the calculations of MAE, RMSE, and MAPE of the models for all drug categories. The results indicate that LightGBM has the best performance among all the models with the final MAE of 23.3728 and final RMSE of 28.9163. Besides, LSTM and GRU have performed well with the final MAE of 24.0362 and 24.1104 respectively. These results show that both gradient boosting models and recurrent neural networks are suitable for pharmaceutical sales forecasting. However, the study has shown that there is no universal model that can give the best performance for every drug category. For example, GRU has the best performance for C1 using MAE criterion, Prophet has the best performance for C2 and C3, LightGBM for C4, LSTM for C5, C6, and C8, and XGBoost for C7. Thus, different drug categories require different approaches in terms of modelling. This confirms that different drug categories require different modelling approaches.

The fourth goal was to develop a realistic forecasting system, not just conduct the theoretical comparison. This objective was achieved through the implementation of a complete software system. This project has included the trained models, their evaluation, forecasting utilities, web interface, API services, explainability modules, and deployment materials. Thus, it is more practical and usable. A pharmacy, a distributor or any other company can use it for sales forecasts, inventory management and making decisions. A pharmacy, distributor, or healthcare organization could use such a system to support sales forecasting, inventory planning, and decision-making.

The final objective was to find out the limitations of the proposed system and suggest ways of improvement for future research. This objective was addressed through the analysis of model performance, metric behaviour, data constraints, and practical deployment considerations. The findings show that the current system provides a strong foundation, but further improvements are possible by including external demand factors, expanding the dataset, improving area-level data representation, and integrating real-time inventory information.

6.3 SUMMARY OF KEY FINDINGS

The most important finding of this research is the ability of the models to predict pharmaceutical sales based on the historical weekly sales. From the models tested, LightGBM performed best with the lowest MAE and RMSE scores. The results point out that gradient boosting models work best in solving structural sales forecasting problems. It is capable of handling nonlinearity and interaction between different variables while being computationally fast. Compared to the deep learning model, it is easier and faster to train and make predictions with it.

Another major finding is that recurrent neural networks, especially LSTM and GRU, are also highly effective for drug sales forecasting. These models are designed to learn temporal dependencies, which is important when past sales influence future sales. The results showed that LSTM and GRU performed very close to LightGBM overall. LSTM achieved the best MAE in several categories, including C5, C6, and C8, while GRU performed best for C1. This indicates that deep learning models can capture sequential sales behaviour well, especially where demand patterns depend on previous time periods.

In addition, it was revealed that the application of traditional models like SARIMAX and Prophet models was helpful but inconsistent. SARIMAX model was able to give satisfactory results for specific categories but still showed lower values for MAE and RMSE compared to LightGBM, LSTM and GRU. Prophet models performed well in certain categories like C2 and C3 but at the same time poor results for other categories like C4. It can be concluded that traditional methods will perform well only if there is enough clear trend and seasonality, otherwise will be unable to catch the nonlinear demand patterns in pharmaceutical sales.

One more significant result of the research was related to evaluation metrics application. MAE and RMSE gave more accurate interpretation of models' results in different categories while MAPE was sometimes misleading. If there were low volume categories, like C6, then MAPE values became extremely high although the actual error rate was insignificant. This can happen due to the way of calculation, which was based on a percentage from the actual value. When actual sales values are close to zero, even a small forecasting error can produce a very large percentage error. Therefore, this study concludes that MAPE should be interpreted carefully in pharmaceutical sales forecasting, especially for low-demand or irregularly sold drug categories.

The study also found that ensemble methods can provide stable predictions but may not always outperform the best individual model. Weighted average and performance-weighted ensemble approaches were tested. These methods reduced dependency on a single model, but they required more processing time because they combined outputs from several models. In practical environments, ensemble models may be useful when reliability and stability are more important than speed. However, when efficiency is important, LightGBM or LSTM may be better choices.

6.4 RESEARCH CONTRIBUTIONS

This research makes several contributions to the field of pharmaceutical sales forecasting and healthcare analytics. First contribution that is made is related to the comparative machine learning approach of forecast development. Instead of relying solely on one kind of forecasts, the research compared different types of models belonging to various classes such as statistical models, machine learning models, deep learning models, and ensemble methods. In this way, it gives an overview of models that are appropriate to use for various sales dynamics in pharmaceuticals.

The second contribution is the practical implementation of a forecasting system. The research does not limit at mere development and evaluation of models. The system developed includes a web interface, forecasting services, API endpoints, trained model files, and deployment. The actual practical implementation adds value to the research and demonstrates how forecasting models can be used in practice. In this way, it can be useful for pharmacy managers and healthcare suppliers.

Third, the contribution of category-wise analysis of models' performance. Results clearly indicate that the performance of models is different by drug categories. The value of this contribution is evident as it proves that pharmaceutical demand is not homogeneous. Depending on the medicine category or product group, demand could be either stable or unstable/seasonal/very volatile, and hence, the best way of forecasting is different.

Fourth, the contribution related to the recognition of certain limitations in percentage-based evaluation metrics. The study revealed the unreliability of MAPE when used in low-volume drug categories. This information will be valuable for future researchers as it will make them pay attention to the choice of metrics for the evaluation of pharmaceutical sales forecasting models.

Fifth, emphasizing the inclusion of such aspects as explainability and advanced system components. The system has such explainability features as SHAP-based interpretation, as well as advanced concepts like meta-learning, uncertainty quantification, and service-oriented architecture. It should be noted that all these components could be developed even further in future, but their presence proves the practical nature of the system design

6.5 PRACTICAL IMPLICATIONS

The findings of this research have several practical implications for pharmaceutical sales and inventory management. A forecasting system such as the one developed in this study will enable pharmacies and pharmaceutical distributors to forecast their future demands and thereby avoid the risk of stockouts and stock buildup. It is particularly relevant to be able to forecast future demand in the case of pharmaceuticals since they come with expiry periods and storage issues. Excess inventory leads to financial loss and shortages impact negatively on the patients.

The forecasting system can also be useful for category-based planning. Different types of drugs behave differently in terms of demand and therefore, it would be easier for pharmacists to plan based on categories of drugs.

Another practical implication is that the system can support digital transformation in pharmacy operations. Many pharmacies still depend on manual experience or simple historical averages for stock planning. A machine learning-based forecasting system can provide more systematic and data-driven decision support. This can improve operational efficiency and help organizations respond better to changing demand patterns.

6.6 LIMITATIONS OF THE STUDY

Although the study achieved its objectives, there are still some limitations should be acknowledged. First, the size of the dataset utilized for this research. As mentioned above, the dataset includes the information about the weekly sales between 2014 and 2023 in eight categories of drugs; however, to make the results more generalizable, the dataset needs to be larger with more categories of products, brands, pharmacies and different regions involved.

Secondly, the datasets do not contain any external variables that can impact the pharmaceutical sales. For example, there are numerous external factors which might affect the demand on drugs, including seasonal illnesses, epidemics, holidays, weather conditions, various promotions, doctors' prescriptions, delays in suppliers, price changes, and economic conditions among others. Consequently, all these variables are missing, and the models work with the sales history data.

The third limitation pertains to area-specific forecasting. The title of the project involves the forecasting of drug sales in specific areas in Sri Lanka. However, the existing implementation dataset consists primarily of data categorized according to drug type. If the research objective is area-level prediction, future work should include explicit district, city, branch, or pharmacy-level sales data. This would allow the system to predict demand for specific geographical locations more accurately.

The fourth limitation concerns the instability of MAPE for low volume categories. As it has been stated above, low volume categories tend to result in extremely high MAPE values despite the fact that the errors themselves might not be that high. Thus, MAPE cannot be solely used as a measure of model accuracy in such cases.

The fifth limitation concerns the method used for testing the effectiveness of the developed models. In fact, the systems were tested against historical test data which is sufficient for academic testing purposes. However, real-world implementation would require constant monitoring because the pattern of sales might change with time due to changing market conditions or changing medical needs of the population.

6.7 RECOMMENDATIONS FOR FUTURE WORK

There are several ways that future research can enhance the results of this study. Firstly, the database can be enlarged. More drug types, historical periods, and pharmacies/regions can be used to make the model learn more realistic patterns of demand.

Moreover, external variables can be added to future research. Holiday season, illness statistics, weather, epidemics, promotional activities, price and population data for each region can increase the effectiveness of predictions. In case of pharmaceuticals, the external variables can play an important role since medicine purchases depend on people's health.

Third, further research should pay more attention to the area-based forecasting. Since the forecasting system is supposed to forecast drug sales on the regional market of Sri Lanka, the data should be obtained from districts, branches or pharmacies. In this way, the system would take into account differences in demands and provide more relevant forecasts.

Fourth, model selection could be automated. As no one model was the best for each category of drug, the system could automatically select the most suitable model depending on the previous experience. For instance, LightGBM could be used for some categories while LSTM, for others and so forth.

Fifth, further research might focus on the improvement of ensemble learning. It is possible to conduct tests of more complex methods of ensemble learning, for instance, stacking with a meta-model, dynamic weighting and error-based model selection.

Sixth, the module that explains predictions could be improved. The explainability of the model is crucial in any decision-making system. Thus, the analysis of SHAP values and feature importance could be helpful for users.

Finally, the system can be integrated with real pharmacy inventory management software. This would allow automatic data collection, real-time forecasting, reorder alerts, stock monitoring, and decision support. Such integration would make the research more useful in practical healthcare supply chain environments.

6.8 FINAL CONCLUSION

In conclusion, this study successfully developed and evaluated a machine learning-based pharmaceutical drug sales forecasting system. The research demonstrated that historical weekly sales data can be used to predict future drug sales with reasonable accuracy. By evaluating several models, the research revealed that LightGBM provided the highest performance with the minimum MAE and RMSE values. At the same time, LSTM and GRU have also provided high-quality forecasting results. It has been proven that machine learning and deep learning approaches are well suited for forecasting pharmaceutical sales.

The study also showed that drug sales patterns differ across categories. this research has shown that there are differences between drugs' sales patterns depending on their categories. This means that one model of prediction will not fit for all drug types. Therefore, a flexible forecasting system that can choose different models depending on the category would be more effective. Another important point is related to the need for careful selection of metrics for forecasting.

The research further highlighted that evaluation metrics must be selected carefully, especially when dealing with low-volume sales categories where MAPE can become misleading.

The proposed system has practical value for pharmaceutical inventory management. It can help reduce stock shortages, avoid overstocking, improve purchasing decisions, and support better planning in pharmacy operations. Although the study has limitations, especially regarding dataset size, external factors, and area-level data availability, it provides a strong foundation for future research and system development.

Overall, this research contributes to the application of predictive analytics in pharmaceutical sales forecasting. It shows that data-driven forecasting methods can support better healthcare supply chain decisions and improve the efficiency of medicine availability. With further improvements, such as real-time data integration, area-specific forecasting, external variable inclusion, and automated model selection, the proposed system can become a more powerful and practical tool for pharmaceutical demand prediction in Sri Lanka and similar healthcare environments.

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